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Development of a Transferable Predictive Modelling Framework for Mammalian Cell Culture Scale-Up Using CHO and iPSC Data

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A dissertation submitted in partial fulfilment of the requirements for the award of the degree of
Master of Science in Data Science.

Abstract

Scale-up remains one of the major challenges in modern biomanufacturing of therapeutic products. There is currently no standardised, data-driven framework to support efficient scale-up of cell-based processes, as existing datasets are typically small and research remains costly and resource-intensive. This study therefore aimed to develop a data pipeline for a predictive model capable of transferring knowledge between two distinct mammalian cell lines. A comprehensive dataset was first constructed using the widely recognised CHO cell line and subsequently adapted to the iPSC system. Using CHO scale-up data ranging from 15 mL to 5000 L, an LSTM-based model was developed to predict the full trajectory of viable cell density over time. The model effectively captured cell growth dynamics, accurately representing transitions between growth phases across different reactor scales. Furthermore, the trained model was successfully transferred and fine-tuned to the iPSC dataset to predict cell growth for this target cell line. These findings demonstrate that a bridge can be established between distinct mammalian systems through the adaptation of learned temporal features, confirming the feasibility of knowledge transfer across biological domains. The proposed approach establishes a foundation for scalable, data-driven process modelling in biomanufacturing.

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1. Introduction.

Background

The large-scale production of modern medicines, such as therapeutic proteins, vaccines, and regenerative therapies, relies heavily on biotechnology and the biopharmaceutical industry's shift toward the Pharma 4.0 paradigm. Bioprocess development, as a key driver of this transformation, is being continuously reshaped by data-driven and computational technologies including automation, machine learning, and advanced analytics, aimed at improving product quality, robustness, and efficiency. A central technical hurdle in this transition is the scale-up of stirred-tank reactors, which involves translating cell culture conditions that perform well in small laboratory vessels of a few millilitres to reactors of several thousand litres while maintaining process performance and product quality consistency.

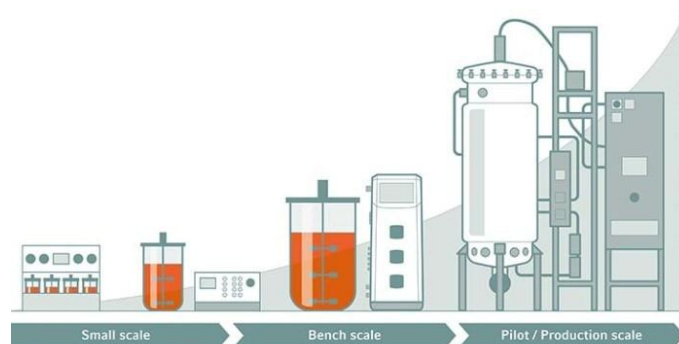


Figure 1: Bioprocessing Scale Up (adapted from Eppendorf, n.d.)

The overall yield of a successful bioprocess scale-up depends strongly on a thorough understanding of key operational parameters, such as mixing strategy, nutrient delivery, oxygen transfer, pH and temperature control, cultivation time, and media composition, and their influence on cellular physiology and proliferation. Adjusting these parameters across increasing bioreactor scales requires extensive optimisation and is widely recognised as a time- and resource-intensive process.

The multidimensional datasets generated in bioprocess development provide a valuable foundation for data-driven strategies, particularly those based on machine learning, which can

efficiently manage and interpret such complex information. Several machine learning frameworks have already shown considerable promise. However, one major limitation remains. Most existing models are based on a single mammalian cell line, Chinese Hamster Ovary (CHO) cells, which are well established and widely used in modern biomanufacturing. Other models have been developed for bacterial systems, which differ markedly from mammalian cultures in their environmental responsiveness, growth kinetics, nutrient requirements, and scale-up behaviour. Because CHO cells remain the most commercially utilised platform in industrial bioprocessing, the abundance of well-curated datasets makes them a logical and accessible basis for model development and training.

However, the biotechnology industry is increasingly shifting its focus toward mammalian induced pluripotent stem cells (iPSCs) because of their broad applications in cell-based therapies, regenerative medicine, cultured meat production, and emerging food technologies. Compared with other mammalian systems such as CHO cells, iPSCs present a far greater technical challenge due to their fragility and complex growth behaviour, including the tendency to form aggregates. These characteristics make iPSC-based research considerably more demanding, requiring greater experimental effort and higher resource investment. Nevertheless, research dedicated specifically to iPSC bioprocesses remains limited, and the application of machine learning in this area is still in its early stages.

In a field where conventional experimental design and execution remain costly and resource-intensive, data-driven modelling offers a promising means of reducing costs and improving process optimisation through computational innovation. Such approaches respond to the growing need for robust frameworks capable of supporting the scale-up of iPSC bioprocesses.

In living biological systems, where even minor parameter variations can lead to substantial differences in outcomes, the complexity and multidimensionality of cell culture conditions significantly limit the predictability of scale-up performance. Computational models therefore present a valuable opportunity to forecast cell growth across different bioreactor scales. Machine learning models can identify and learn complex relationships between process inputs (such as mixing time and mass transfer) and key outputs like VCD, reducing reliance on traditional trial-and-error experimentation.

Aim and Objectives

By employing structured datasets for both, target domain iPSC and source domain CHO cell lines, this project aims to develop a predictive model for bioreactor scale-up through the application of transfer learning. Positioned at the intersection of bioprocess engineering and

The modelling phase commences with the development of a supervised machine learning model trained on CHO datasets to predict VCD over the cultivation period, an essential indicator of culture health and productivity, before being optimised and adapted for cross-system application. To assess the predictive accuracy and robustness of the transfer learning approach, the model's performance is evaluated on large-scale iPSC bioprocesses by comparing predicted and observed VCD values using metrics such as RMSE. Prior to this validation, the model is fine-tuned with small-scale iPSC data to enable it to capture the distinct physiological and process characteristics of iPSC cultures.

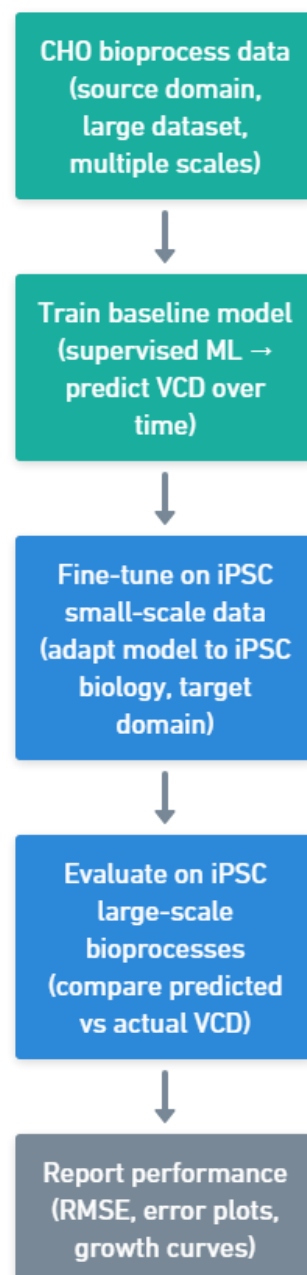


Figure 3: TL workflow for VCD prediction.

This study serves as a proof of concept for transfer learning between two distinct yet related mammalian cell systems, demonstrating that CHO bioprocess datasets can provide a foundation for predictive model development applicable to iPSC processes. The approach aims to confirm that transferring knowledge to systems with limited experimental data can yield models that improve process scale-up strategies, making them more time- and cost-efficient, while supporting early-stage iPSC process development and data-driven innovation in biomanufacturing. Supporting outputs include visual representations of model performance and validation results.

Significance

By leveraging industrial CHO datasets, this project applies a transfer learning approach within bioprocess modelling to advance early-stage iPSC process development. The proposed framework has the potential to significantly lower optimisation costs and reduce the experimental burden associated with iPSC research. In data-limited environments, the model is expected to enable a more data-efficient scale-up strategy, supporting predictive, knowledge-driven biomanufacturing. The application of such modelling approaches in bioprocessing is anticipated to enhance efficiency, minimise resource use, and reduce experimental waste, thereby contributing to sustainable development objectives.

While the model's limitations are explicitly acknowledged to avoid overreliance on computational outcomes, the responsible interpretation of predictions remains critical. Data-driven insights are intended to complement, rather than replace, experimental validation, ensuring that model-based findings support, refine, and strengthen empirical research practices.

Ethical considerations related to data collection and research integrity are described in Appendix A.

2. Literature Review

The Digital Transformation of Biomanufacturing (Pharma 4.0)

In today's biomanufacturing landscape, the rapid growth of process data is accelerating the transition toward Pharma 4.0, a stage where biotechnology becomes increasingly data driven (Karimi Alavijeh et al., 2022). Scaling up mammalian cell cultures remains one of the central challenges, as results obtained at laboratory scale are difficult to translate directly to larger systems. At the same time, experiments at production scale are expensive and difficult to forecast, while product quality and yield must be consistently maintained. To address this, modern scale-up strategies integrate engineering principles, process analytical technologies, and computational modelling, creating a more reliable and data-informed approach to biomanufacturing.

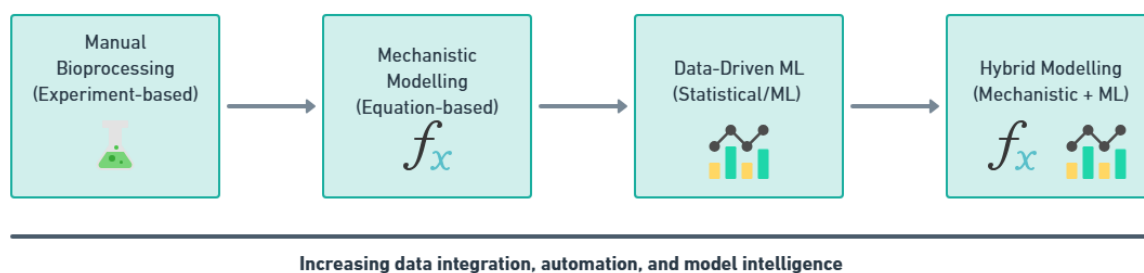


Figure 4: The evolution of modelling approaches in biomanufacturing.

Traditional Engineering Scale-Up Strategies

Engineering similarity criteria remain the basis for bioreactor scale-up. Traditional strategies rely on fixed parameters such as P/V, kLa, tip speed, or Reynolds number to maintain comparable mixing and oxygen transfer across scales (Karimi Alavijeh et al., 2022). However, these approaches have shown clear limitations. Geometric inconsistencies between reactor sizes, differences in gas transfer, and the lack of integration of biological variables make direct translation difficult (Karimi Alavijeh et al., 2022). Since these parameters cannot be matched simultaneously, compromise strategies are often applied, prioritizing the most critical needs of the cells.

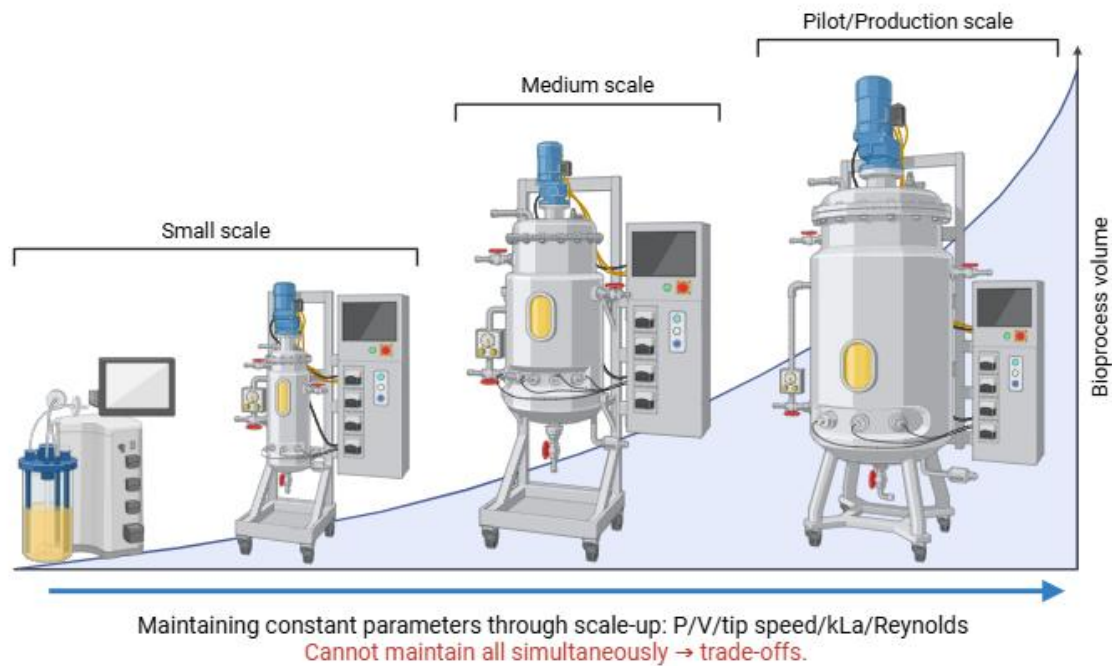


Figure 5: Conventional engineering scale-up based on physical similarity parameters.

However, as bioprocesses became more complex and biologically sensitive, these physically based approaches proved insufficient, giving rise to mechanistic and later hybrid models that aimed to capture both biological and engineering dynamics.

Mechanistic Modelling in Bioprocess Development

Mechanistic models use systems of ordinary differential equations (ODEs) to describe the dynamics of cell culture, including changes in biomass, substrates, and products (Karimi Alavijeh et al., 2022; Mahanty, 2023). Mechanistic models, such as stoichiometric, kinetic, genome-scale, and glycosylation-focused frameworks, provide a causal understanding of CHO metabolism. They have been applied to optimize nutrient utilisation, scale-up strategies, and glycosylation outcomes (Ranpura et al., 2025). Mechanistic modelling of CHO culture processes has been widely investigated as a tool to guide operational decision-making in large-scale protein production. These models usually follow a structured workflow, starting with the definition of metabolic pathways, followed by the identification of activation and inhibition parameters, formulation of reaction rates, and subsequent calibration and validation against experimental data (González-Hernández & Perré, 2024). While mechanistic models provide

valuable insights into CHO metabolism, their complexity often limits practical application. A key challenge highlighted in recent reviews is maintaining the balance between complexity and predictive accuracy. Over-parameterized models often compromise robustness and limit practical usability. Furthermore, CHO culture processes remain inherently difficult to model due to complex and dynamic metabolic shifts over time (Lu et al., 2025).

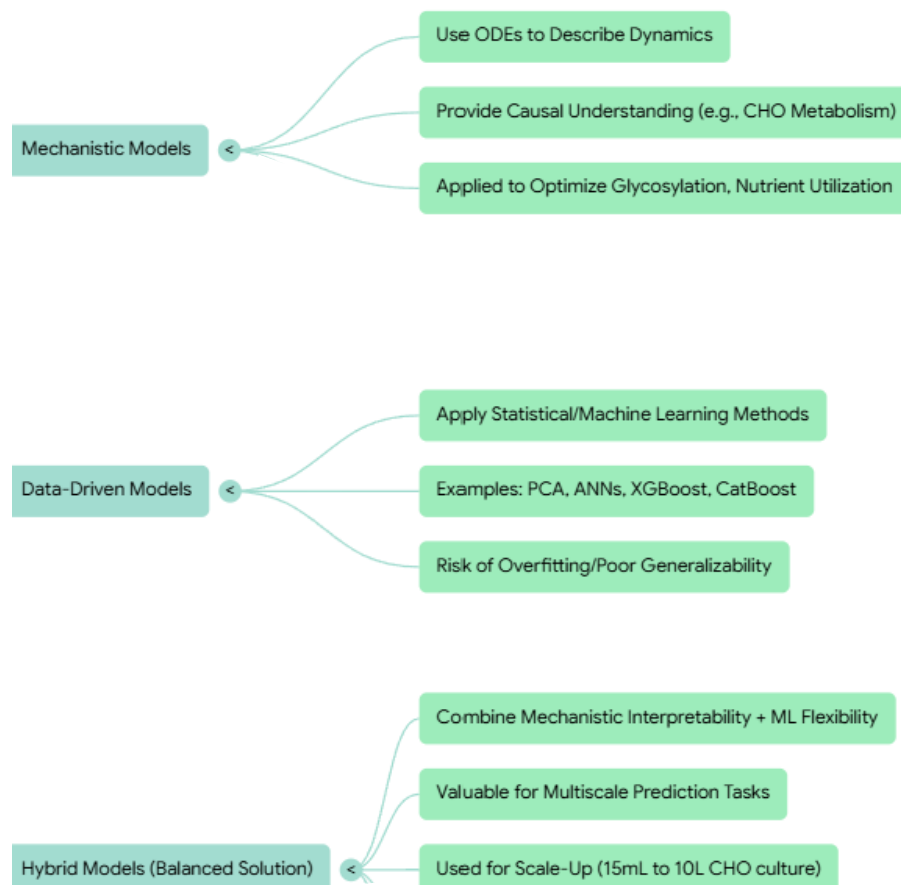


Figure 6: Overview of mechanistic, data-driven, and hybrid modelling approaches.

The Rise of Hybrid and Data-Driven Approaches

To overcome these challenges, recent advances combine mechanistic understanding with data-driven methods, leading to hybrid approaches that support the digitalisation of

biomanufacturing. Because traditional scale-up criteria such as P/V or kLa cannot fully capture biological complexity, the industry is moving toward digitally enhanced approaches.

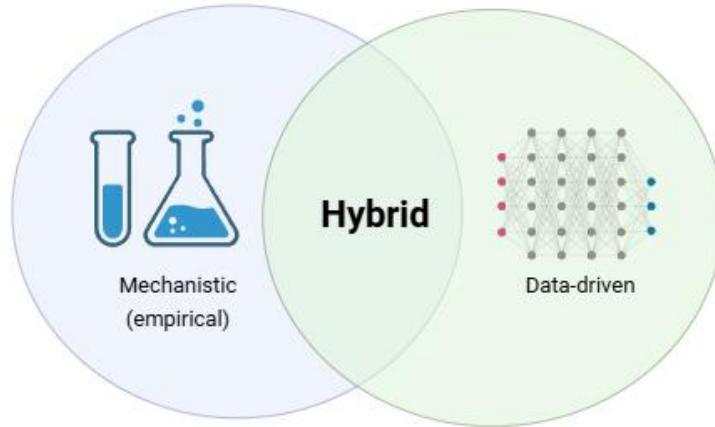


Figure 7: Comparison of mechanistic, data-driven, and hybrid modelling paradigms.

Mechanistic, data-driven, and hybrid modelling methods are increasingly recognized for their potential to improve scalability, robustness, and process understanding, particularly under data-scarce conditions (Karimi Alavijeh et al., 2024; Mahanty, 2023; Rogers et al., 2022).

The Biopharma 4.0 framework brings together advanced modelling strategies to optimise bioprocess development. Mechanistic models describe cellular metabolism and growth dynamics based on first principles, while data-driven models apply statistical and ML methods such as PCA, ANNs, and gradient boosting approaches like XGBoost.

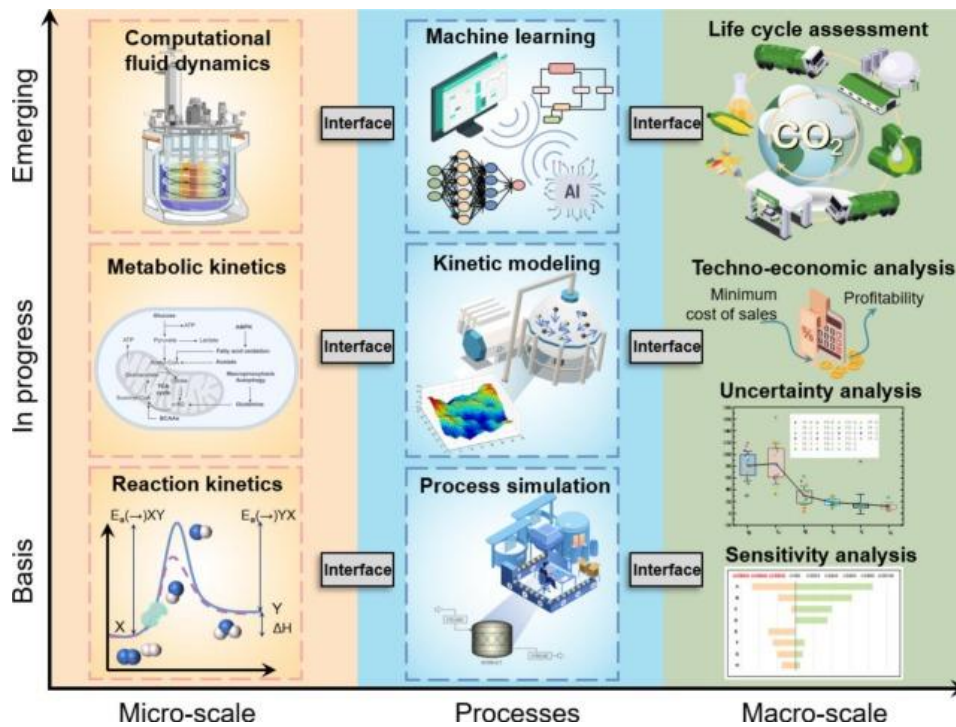


Figure 8: Hierarchical integration of models in bioprocess development (adapted from Chen & Wang, 2024)

Applications of Hybrid Modelling in Mammalian Cell Systems

Hybrid models, combining the interpretability of mechanistic systems with the flexibility of machine learning, have shown particular value for multiscale prediction tasks (Karimi Alavijeh et al., 2022; Karimi Alavijeh 2025, Lee & Gras, 2024). For instance, Bayer et al. Have shown that ANN trained on shake flask data could be successfully transferred to predict viable cell density in 15 L bioreactors, requiring minimal retraining and showing promising results for mammalian cell lines like CHO cells (Bayer et al., 2021). Similarly, models based on XGBoost and CatBoost, tuned via Bayesian optimisation and cross-validation, created by Karimi Alavijeh et al. (2024), have shown high predictive accuracy in scale-up bioprocess of CHO cell culture.

Mowbray et al. (2023) have introduced a RL-based hybrid modelling framework that combines mechanistic kinetics with data-driven learning for model structure identification and parameter estimation. Applied to in-silico fermentation case studies, the approach achieved high accuracy in representing combinatorial, history-dependent, and time-varying kinetics, demonstrating the potential of hybrid modelling for complex bioprocesses.

In another example, Williams et al. (2023) have applied a hybrid modelling strategy that integrated a refractometry-based PAT system with metabolic modelling and machine learning to

optimise lentiviral vector production in HEK293T cells. This study provides further evidence of the applicability of hybrid models in mammalian cell systems to enhance process optimisation.

Deep Learning for Bioprocess Modelling: RNN and LSTM

RNNs have recently gained attention as promising alternatives to traditional parametric and hybrid approaches for upstream bioprocess modelling (Bayer et al., 2021; Karimi Alavijeh et al., 2022). Unlike mechanistic models, which depend on detailed kinetic knowledge and extensive parameter calibration, RNNs learn directly from time-series data and can flexibly integrate process variables such as temperature, pH, DO, or modified feeding strategies (Bayer et al., 2021; Karimi Alavijeh et al., 2022; Tan et al., 2018; Wang et al., 2025). Studies have shown that both product-specific and generic RNN models achieve high predictive accuracy for growth, metabolite concentrations, and product titer, with normalized MAE and RMSE values below unity and performance exceeding experimental variability (González-Hernández & Perré, 2024).

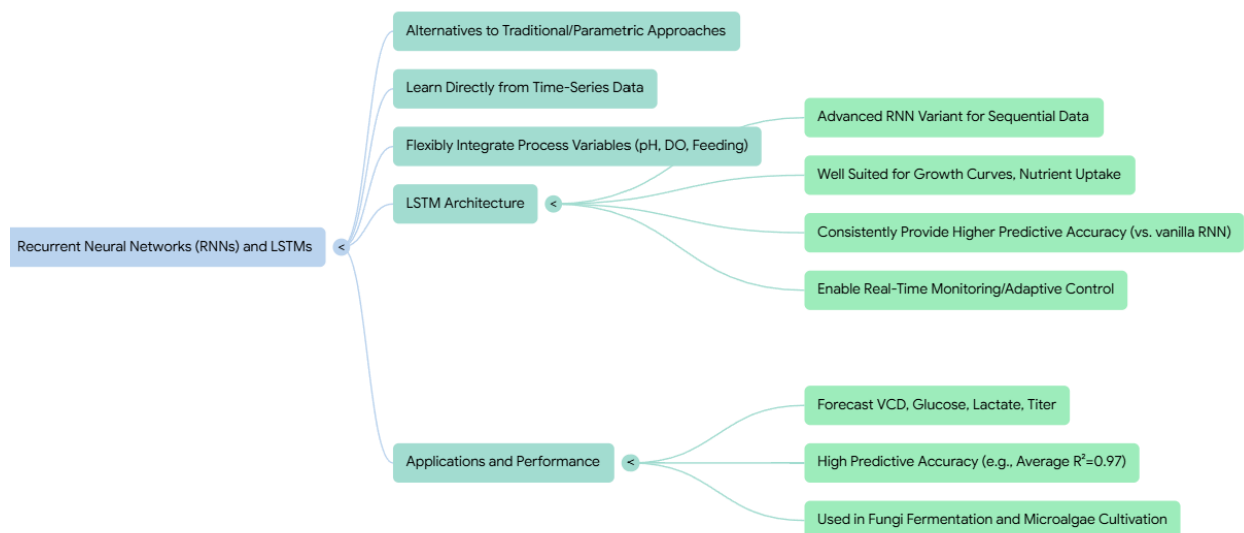


Figure 9: Overview of LSTM based models.

Building on this, RNNs and their advanced variant, the LSTM architecture, have emerged as powerful tools for sequential data, making them particularly well suited to modelling growth curves, nutrient uptake, and DO trajectories (Bayer et al., 2021; Karimi Alavijeh et al., 2022;

Wang et al., 2025). Despite their potential, they remain underexplored in bioprocessing (Syed et al., 2024). A study by Lu et al. (2025) have demonstrated the application of RNNs to forecast key parameters such as glucose uptake, lactate, VCD, and viability in fed-batch cultures, underlining their promise for improving control, optimisation, and scale-up in biomanufacturing. Similarly, Rayes et al. (2025) have combined online bioreactor data with historical offline measurements to predict next-day process outcomes using RNNs. Their model accurately forecasted product titer, VCD, glucose, lactate, and ammonia profiles over a 17-day run, achieving high predictive performance (average $R^2 = 0.97$).

A recent comparative study done by Yunita et al. (2025), evaluated nine recurrent neural network architectures, including RNN, LSTM, GRU, and their hybrid variants, across diverse time-series datasets such as sunspot activity, COVID-19 cases, and DO concentrations. While statistical testing did not reveal significant differences, clear trends were observed: vanilla RNNs were computationally efficient but delivered the weakest forecasts, whereas LSTM-based architectures, particularly LSTM-RNN and LSTM-GRU hybrids, consistently provided higher predictive accuracy. These findings indicate that LSTM-driven models offer robust performance for time-series forecasting in bioprocessing and beyond, although the optimal configuration may still depend on dataset properties and computational resources.

Further evidence is provided by the study of Lee, Jeong & Kim (2025), which demonstrated that LSTM models can accurately predict fermentation dynamics and, when combined with IoT sensors and blockchain-secured data, enable real-time monitoring, adaptive control, and improved product quality in industrial fermentation. More recently, RNN extensions with LSTM architectures have been applied to bioprocess modelling, as shown in the work of Liu & Zhang (2021), reinforcing their applicability to complex biological systems.

Furthermore, recurrent deep learning models are also gaining application outside mammalian systems. For instance, domain adaptation ensemble LSTM models have been successfully applied to fungi fermentation, predicting cell growth and inulinase production by leveraging unlabelled data and cross-batch information to improve accuracy under variable operating conditions (Wang et al., 2025). Similarly, in microalgae cultivation, LSTM-based soft sensors have been developed to forecast biomass concentration in photobioreactors. These models outperformed SVR by capturing nonlinear temporal dynamics, where conventional approaches showed limited generalisation (Syed et al., 2024). Together, these studies highlight the strong potential of RNN and LSTM models as scalable, data-driven tools for process control,

optimisation, and simulation, supporting predictive decision-making across both development-scale and large-scale cell culture processes.

Transfer Learning in Bioprocessing

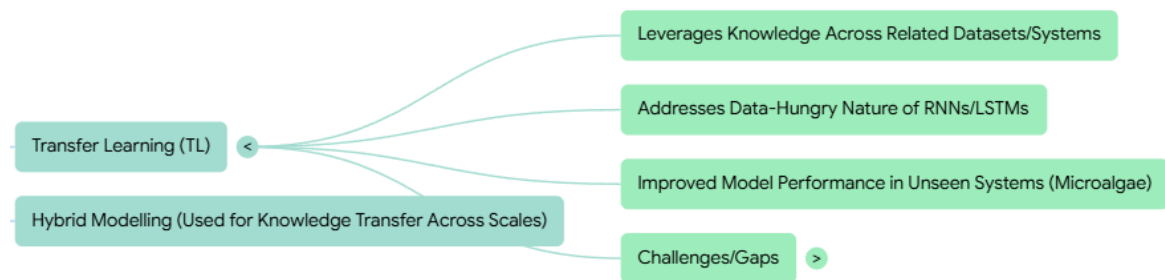


Figure 10: Overview of transfer learning.

Despite their strong predictive power, RNN and LSTM models remain data-hungry, making them difficult to apply under data-scarce conditions. Transfer learning has therefore emerged as a complementary strategy, enabling models to leverage knowledge across related datasets or systems (Helleckes et al., 2023). Hybrid modelling has also been explored as a way to transfer knowledge across related mammalian systems. In a study by Yatipanthawala et al., serial hybrid models were developed to predict CHO cell culture performance, including VCD, product titer, and glucose consumption. These models combined mechanistic mass balances with data-driven algorithms such as RF, XGB, and FFNN, demonstrating robustness against extrapolation errors and adaptability to process variability. Notably, the trained models were successfully applied to a different CHO cell line producing another therapeutic protein, with retraining on a subset of new data further improving predictive accuracy.

A recent industry case study reported by DataHow (nd) described the use of a hybrid mechanistic machine learning model to scale up a CHO cell culture from 15 mL to 10 L. The model, integrating mechanistic mass balances with ML components, was trained on small-scale data together with a limited number of pilot runs. By transferring knowledge from the small scale, it was able to accurately predict large-scale outcomes with minimal experimental input. Furthermore, when trained on both small- and large-scale data, the model successfully recommended and simulated a mitigation strategy to restore final titer after a feeding error, demonstrating its value for process robustness.

Beyond mammalian systems, knowledge transfer has also been applied to microalgae-based processes. Rogers et al. (2022) showed that transfer learning improved model performance in previously unseen microalgal cultivation systems by leveraging shared structural and process attributes. This enabled reliable simulation of new cultivation conditions despite limited data availability. The models consistently outperformed traditional kinetic approaches in cross-domain applications, highlighting the promise of transfer learning under low-data constraints. However, their performance remained strongly dependent on domain similarity, and optimal topology selection is still largely empirical, pointing to the need for more standardised transfer learning protocols in bioprocessing.

Current Limitations and Research Gap

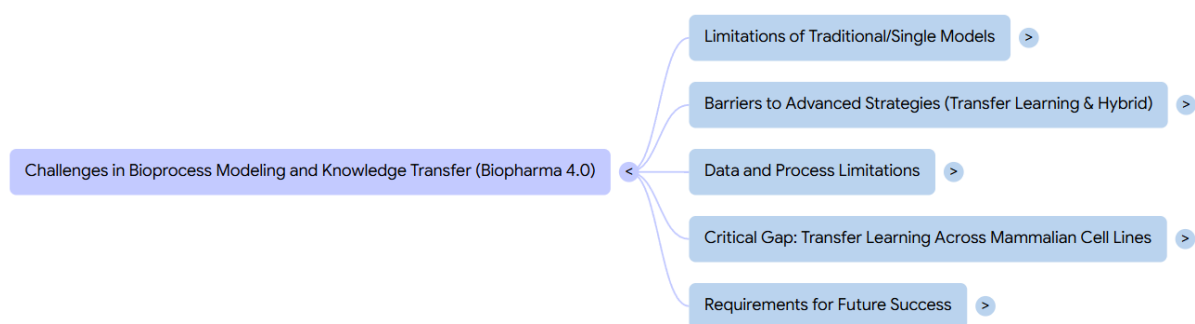


Figure 11: Overview of challenges in scale-up modelling.

Despite the advances brought by Biopharma 4.0, challenges remain. Traditional scale-up metrics such as P/V and kLa lack the flexibility to capture nonlinear biological interactions and are inadequate for modern bioprocess environments (Karimi Alavijeh et al., 2022). While mechanistic models offer biological fidelity, they are often overparameterized and inflexible. Conversely, data-driven models may suffer from overfitting, poor generalisability, and limited interpretability (Rogers et al., 2022; Helleckes et al., 2023; Mahanty, 2023; Williams et al., 2023).

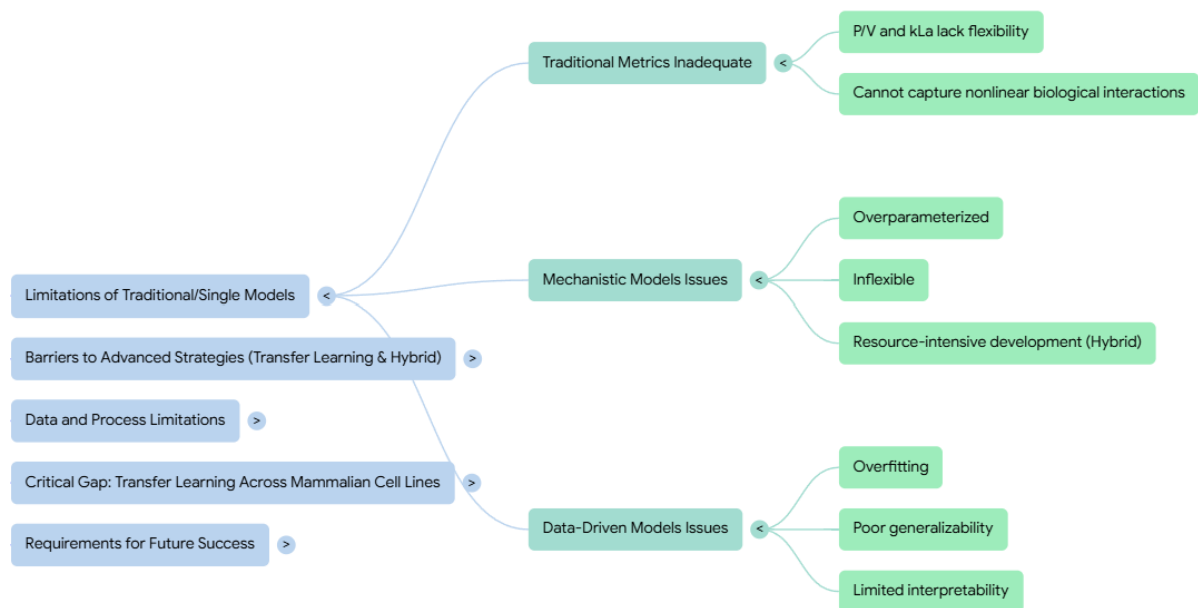


Figure 12: Limitation of traditional and data-driven scale-up approaches.

Hybrid models emerge as a balanced solution, yet their development is resource-intensive and requires cross-disciplinary expertise. To date, most hybrid models in bioprocessing have relied on simple FFNN, with only limited exploration of more advanced or diverse algorithms such as RF, XGBoost, or RNNs within hybrid frameworks (Yatipanthala et al., 2024).

Moreover, advanced strategies such as transfer learning and digital twins remain underutilised in industrial bioprocessing, while challenges related to reproducibility and open access to data and code continue to limit broader adoption (Rogers et al., 2022; Mahanty, 2023). Uncertainty also persists regarding the extent to which hybrid models can generalise across different products, cell lines, or broader process conditions (Rogers et al., 2022; Yatipanthala et al., 2024). Although transfer learning has been widely explored in areas such as image classification, natural language processing, and chemical fault detection, its systematic application in bioprocess engineering remains rare (Rogers et al., 2022).

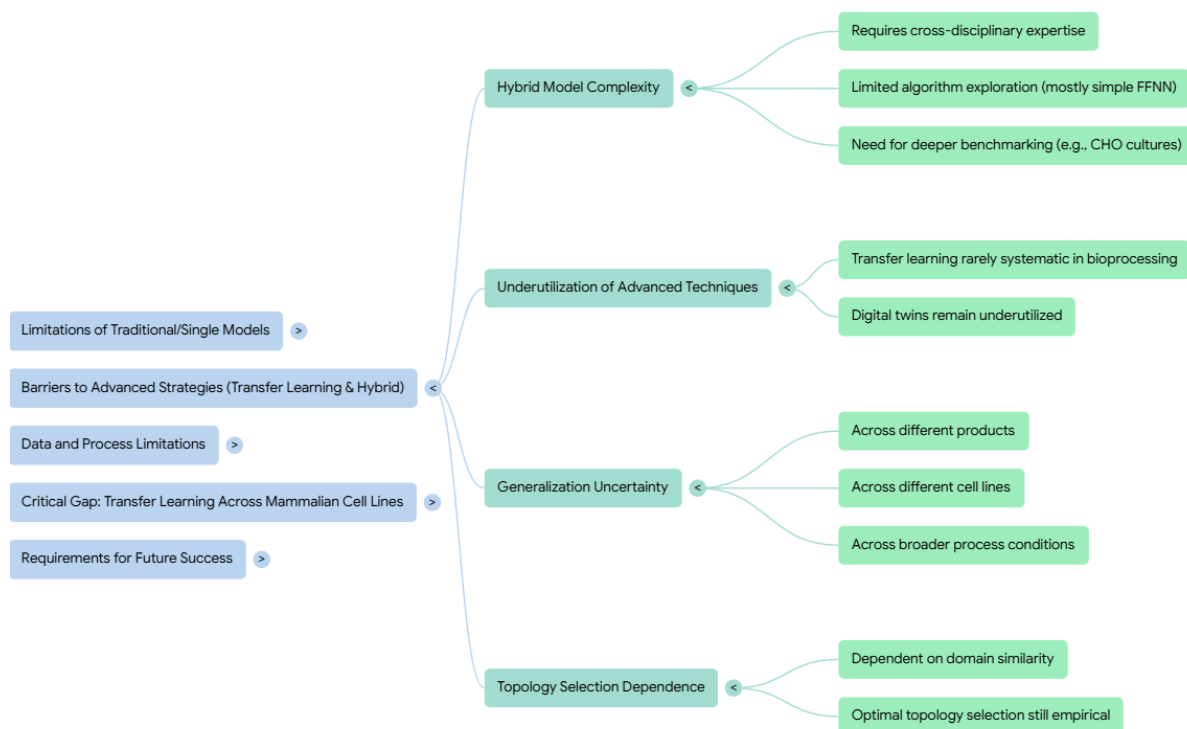


Figure 13: Limitation of advanced scale-up approaches.

Data variability and quality represent additional barriers (Bayer et al., 2021; Karimi Alavijeh, Lee & Gras, 2024; Mahanty, 2023; Rogers et al., 2022; Wang et al., 2025). Limited availability of negative results, which are often underreported compared to successful runs, along with a scarcity of high-quality, well-annotated datasets, continues to restrict model training and validation (Helleckes et al., 2023). Narrow design spaces, such as models trained only on processes with similar seeding densities or glucose conditions, further constrain extrapolation capacity (Yatipanthalawa et al., 2024). The performance of transfer models remained strongly dependent on domain similarity (Rogers et al., 2025) and, comparable to hybrid models, on optimal topology selection, which is still largely empirical (Rogers et al., 2022; Mahanty, 2023).

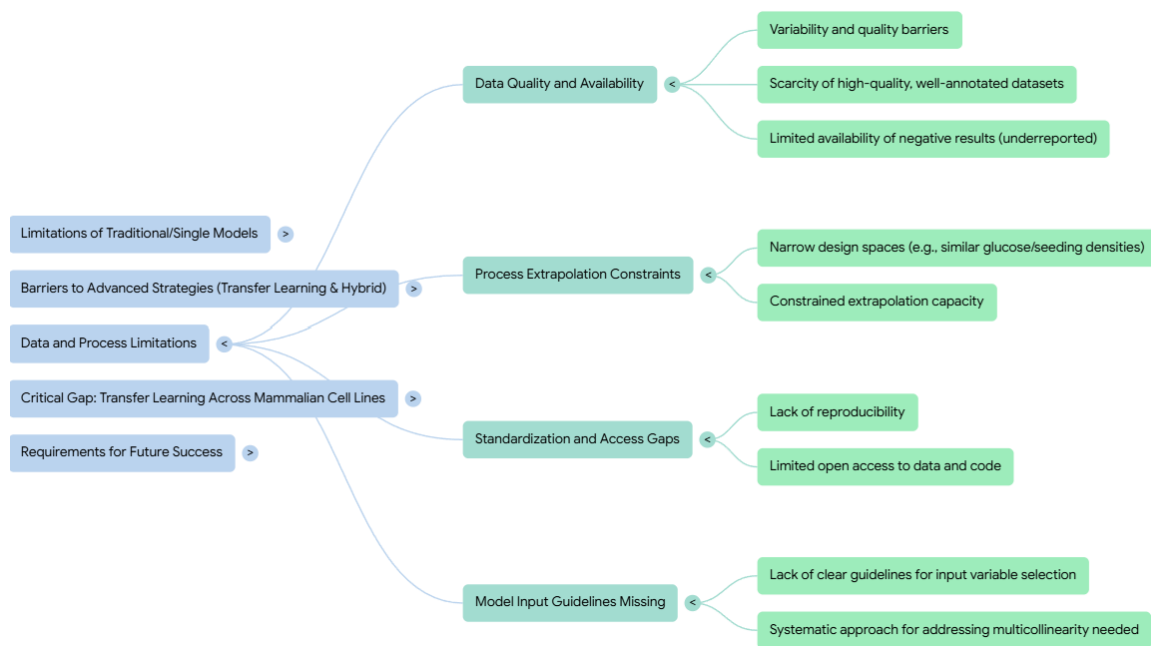


Figure 14: Limitation due to data availability and quality.

Few studies have compared multiple machine learning algorithms within hybrid frameworks for CHO cultures, highlighting the need for deeper benchmarking (Yatipanthalawa et al., 2024). In addition, there is still a lack of clear guidelines for selecting input variables and addressing multicollinearity in a systematic way (Yatipanthalawa et al., 2024). These areas will need to be addressed to ensure broader applicability and reliability of modelling approaches in bioprocessing.

While Biopharma 4.0 represents a paradigm shift in bioprocess modelling, its success depends on the integration of domain knowledge with scalable digital strategies and robust, interpretable modelling frameworks. Overcoming current limitations through systematic validation, standardisation, and interdisciplinary collaboration will be essential to accelerate the adoption of data-centric approaches and support the development of more efficient, resilient, and scalable biomanufacturing systems (Helleckes et al., 2023; Karimi Alavijeh et al., 2024).

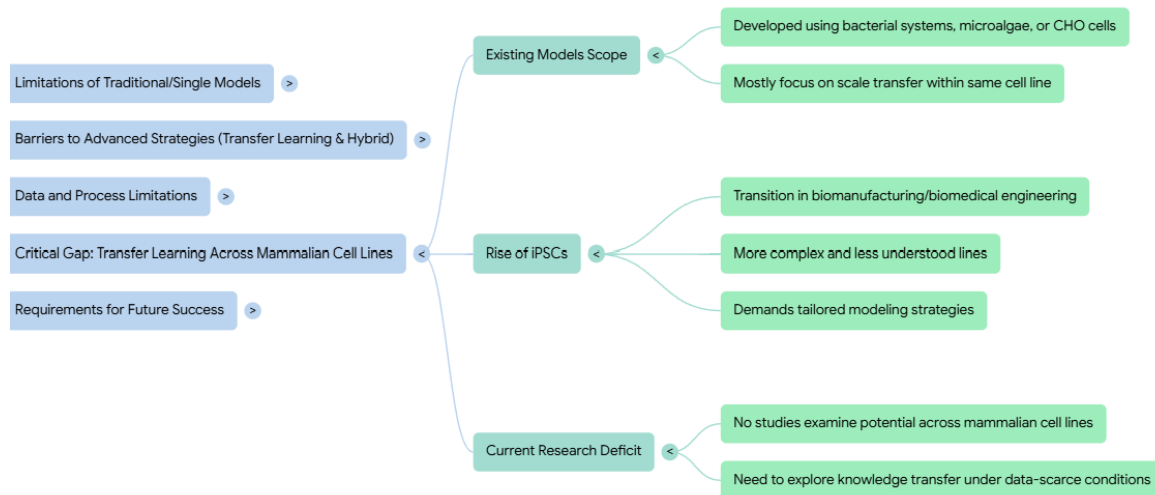


Figure 15: Overview of the research gap.

Furthermore, while the literature review highlights the early stages of a shift toward data-driven approaches in biomanufacturing, most existing models have been developed using bacterial systems, microalgae, or a single mammalian cell line, CHO cells. Owing to their widespread use and well-characterized biology, CHO cells remain the most extensively studied mammalian system (Masuda et al., 2024). However, the growing transition toward iPSCs in both biomanufacturing and biomedical engineering (Fujisawa et al., 2023; Kriedemann, 2024; Sun et al., 2024) introduces a new research frontier: assessing the transferability of scale-up models from established mammalian systems to the more complex and less understood iPSC lines. This shift emphasizes the need to adapt modelling strategies to capture the unique characteristics of iPSCs and to enable their broader use in therapeutic development and regenerative medicine.

Taken together, these limitations highlight a critical gap. While transfer learning has been explored in microbial and algal systems, no studies to date have examined its potential across mammalian cell lines. This gap becomes even more pressing with the rise of iPSCs, which introduce additional complexity and demand tailored modelling strategies. To date, existing studies have either addressed simpler microbial systems, such as the pioneering work of Rogers et al. (2025) on transfer learning in bioprocesses or focused on transferring knowledge across scales within the same cell line rather than across different mammalian systems.

Current evidence shows that scale-up processes for a given cell line can be successfully predicted and are already applied in industry. However, these approaches remain highly dependent on the availability of large datasets, which are costly to generate due to the extensive number of experiments required. By contrast, the concept of transferring knowledge about the impact of scale-dependent bioreactor parameters on cell behaviour from one mammalian cell line to another under data-scarce conditions has not been explored.

Despite these advances, no studies have yet explored transfer learning across mammalian cell lines. This gap is particularly relevant given the shift toward iPSCs, which present unique modelling challenges. Addressing these forms the central focus of the present study.

3. Methodology

The methodological framework aimed to develop and evaluate a data-driven approach for bioprocess scale-up modelling using ML. It involved constructing a predictive model to forecast VCD trajectories across two mammalian cell systems. The workflow included a pilot feasibility study, full-scale model development, and a transfer learning stage adapting the CHO-trained model to iPSC data. Each phase comprised feature selection, data preprocessing, EDA, model design, training, and evaluation, guided by literature insights to ensure reproducibility and biological interpretability.

3.1 Pilot Study

To establish and validate the methodological framework for applying ML to bioprocess scale-up, a pilot study was designed to assess feasibility and identify potential implementation challenges. The study relied on a smaller dataset, enabling the practical implementation of methods informed in the literature review. Within this framework, all essential stages were tested, including feature selection, exploratory data analysis, data preprocessing, baseline model development using CHO cell data, and a transfer to iPSC data.

An initial baseline model was developed as a proof of concept to capture fundamental growth dynamics in the CHO dataset, with success criteria defined as maintaining a global MAPE below 30% across valid timesteps. This staged design minimized implementation risks, clarified methodological requirements, and demonstrated the potential of ML models in bioprocess scale-up.

3.2 Feature selection

Scale-up in bioprocess engineering is defined as a systematic transfer of a process from a small to a larger production-scale system, while maintaining product quality attributes, a deep understanding of the process is essential for the correct selection of scale-up variables (Lawson et al., 2021). In this study, feature selection represented a critical methodological step in dataset development, with the objective of identifying variables relevant to scale-up that are either accessible from published datasets or derivable through feature engineering. The selection process was guided by domain knowledge in bioprocess engineering and supported by an extensive literature review.

Bioprocess parameters were classified as scale-independent or scale-dependent. Scale-independent parameters are generally maintained at constant levels during scale-up (Seidel et al., 2023; Reyes et al., 2024) and were therefore excluded from modelling. In contrast, scale-dependent parameters were prioritized due to their direct dependence on vessel geometry and operating conditions, as well as their influence on hydrodynamics, shear forces, and mass transfer (Seidel et al., 2023; Reyes et al., 2024; Neuss et al., 2025).

The structured review of traditional scale-up strategies, based on fixed engineering parameters, further informed the selection process. Parameters identified as critical to process performance and cell behaviour included direct sensor measurements, as well as engineered features derived from vessel geometry and agitation characteristics.

Certain variables were excluded to maintain model generalisability. Cell line-specific features were not incorporated, as they vary significantly across clones and cell types. Similarly, unreported parameters and qualitative descriptors were omitted unless they could be transformed into quantitative measures.

The final variable set is therefore represented by scale-dependent physical phenomena, which are accessible from standard secondary data sources and independent of specific cell culture systems. This approach ensured that the dataset incorporated features most meaningful for scale-up modelling, while remaining practical within the constraints of publicly available data.

3.3 Data collection

Data for this study were collected through a systematic review of bioreactor scale-up studies on industrial CHO cell lines expressing antibodies, primarily sourced from PubMed and biopharmaceutical reports. These sources provided both process data and bioreactor design specifications. iPSC data were obtained from laboratory runs and used as secondary data. Quantitative values reported in graphical form were digitised to reconstruct time-series trajectories for inclusion in the dataset.

3.4 EDA

EDA was performed to assess the structure and quality of the CHO and iPSC datasets and to support informed decision-making during subsequent modelling. The datasets were scanned for errors, missing values, outliers, and relationships between variables to ensure data integrity and reliability.

Analyses were performed in Python using libraries such as pandas, numpy, matplotlib, and seaborn, with the initial objective of determining the number of processes and scales represented in the study. A key focus was placed on examining statistical distributions, detecting missing values and outliers, and investigating relationships between variables across processes and scales.

Visualisation of data trajectories and correlation analysis supported the identification of key growth patterns and potential multicollinearity among scale-dependent features, guiding feature selection and model design, and ensure that the final input set comprised robust, non-duplicative predictors (Yerdelen et al., 2023; Karimi Alavijeh, Lee & Gras, 2024).

By combining statistical distributions, trajectory visualisation, and correlation analysis, EDA provided a comprehensive understanding of the dataset, informed feature selection, and established a foundation for the machine learning workflow.

3.5 Data Preprocessing

To ensure the reliability and robustness of the predictive model, raw bioprocess data underwent careful preprocessing prior to model training. Preprocessing was essential to address missing values, variable scaling, and the sequential structuring of data for use in a RNN based on LSTM layers.

Datasets were designed with independent training, validation, and test partitions to ensure unbiased evaluation. This design allowed sufficient data for model learning, while reserving independent subsets for hyperparameter optimization and unbiased performance evaluation (scikit-learn developers, 2025).

A systematic evaluation of missing data guided decisions regarding exclusion or imputation. Variables with a high proportion of missing values were excluded to prevent noise and bias (Wells et al., 2013; Cui et al., 2020). For other variables, imputation strategies were guided by both statistical characteristics and established bioprocess engineering principles.

Because RNNs are sensitive to differences in variable scales, all continuous variables were standardised. This step supported model convergence, reduced the risk of vanishing gradients, and ensured that no single feature dominated the learning process due to its unit of measurement (Ahmed et al., 2022).

Categorical variables were numerically encoded to incorporate contextual information. Additional handling was implemented for unseen categories in validation and test sets, thereby preserving model generalisability.

To enable full trajectory prediction, the dataset was structured into sequences corresponding to individual bioprocess runs. Grouping by process- and run-specific identifiers ensured temporal coherence, while sequence padding facilitated the uniform processing of variable-length time series (Yoon & Yu, 2020).

All preprocessing steps were fitted exclusively on the training data and subsequently applied to validation and test sets to avoid data leakage. This ensured that model evaluation remained unbiased and reflective of real-world deployment scenarios (scikit-learn developers, 2025).

3.6 Model development

Model architecture

In the context of cell growth prediction in bioreactors, the selection of neural sequence models such as RNNs and LSTM networks is strongly supported in the literature. Standard FFNN treat each data point independently (Hewamalage, Bergmeir & Bandara, 2021), ignoring sequential structure, whereas RNN-based models are specifically designed to capture temporal dependencies (Mienye & Jere, 2024; Yunita et al., 2025).

In this study, RNNs were selected as the modelling foundation because cell culture processes produce sequential data, where each measurement depends on the system's previous state. Unlike static models such as RF or XGBoost, RNNs are specifically designed to capture temporal dependencies. This makes them well suited for representing time-evolving phenomena such as cell growth in bioreactors. Since VCD trajectories are influenced by earlier conditions, RNNs provide a biologically meaningful way to model these dynamic relationships that cannot be effectively captured by non-sequential algorithms.

Classical RNNs, however, encounter challenges such as vanishing and exploding gradients, which limit their ability to learn long-term dependencies (Mienye & Jere, 2024). LSTMs were introduced to address this issue (Hewamalage, Bergmeir & Bandara, 2021). It mitigates vanishing gradient issues common in long bioprocess time-series and retains information about early culture conditions that influence later growth outcomes. This makes it particularly well suited for modelling extended bioprocess trajectories where long-term temporal dependencies are biologically significant.

Their architecture is based on three gates, forget, input, and output, which regulate information flow, allowing the model to selectively discard irrelevant signals, incorporate new inputs, and retain useful information over time (Mienye & Jere, 2024). In TensorFlow/Keras, the default configuration of LSTM applies the hyperbolic tangent (tanh) activation to the hidden state update and the sigmoid activation to the input, forget, and output gates (figure 16).

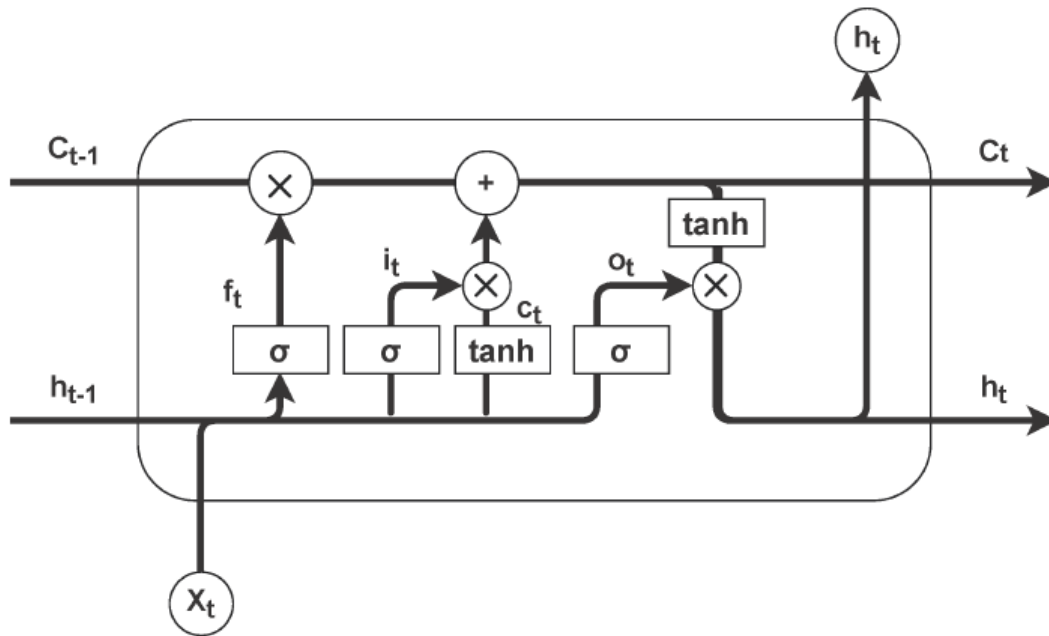


Figure 16: Structure of an LSTM cell (adapted from Mienye & Jere, 2024).

This mechanism enables LSTMs to preserve critical early culture conditions, such as initial seeding density or culture phase, and use them for later predictions. Their gating structure also enhances robustness by filtering out noise while retaining the underlying biological growth signal (Mienye & Jere, 2024).

The study was therefore based on an RNN model with LSTM layers. This choice aligns with the overarching objective of the study to develop a data-driven, generalisable model capable of capturing dynamic cell growth behaviour in CHO systems and transferring this knowledge to iPSC cultures, where experimental data are limited.

Training and Evaluation

The training strategy for the predictive models was designed to ensure stable convergence, reduce overfitting, and provide interpretable evaluation metrics in the context of bioprocess modelling. Methodological choices regarding optimiser, batch size, epoch limits, loss function, and evaluation criteria were carefully considered.

The Adam optimizer was selected due to its dynamic adjustment of learning rates, combining the advantages of AdaGrad and RMSProp (Mishra et al., 2020). This enables efficient and stable convergence (Liu et al., 2022), particularly in cases of noisy or sparse gradients (Kingma & Ba, 2015), making it suitable for training RNN on limited bioprocess dataset.

Model training parameters were selected to ensure stable learning and prevent overfitting. Batch size, epoch limit, and early stopping were optimised to balance convergence speed, generalisation, and computational efficiency in line with best practices for small datasets (Prechelt, 2012; Masters & Luschi, 2018).

MSE loss function was used as it is standard for continuous regression tasks and emphasises larger deviations, supporting accurate modelling of growth trends (Yatipanthalawa et al., 2024; Richter et al., 2025). MAE was additionally monitored to provide an interpretable measure of average prediction deviation in biological terms.

Predictions were designed to be inverse transformed to enable biologically interpretable evaluation metrics. This step ensured that error metrics and forecasts could be directly interpreted in terms relevant to bioprocess development, making the outcomes not only statistically valid but also biologically meaningful.

Model evaluation focused on both statistical and biological interpretability. Performance was assessed using standard regression metrics (MAE, RMSE, and MAPE) to quantify prediction accuracy, while ensuring that only valid timepoints were included in the analysis (Park et al., 2023; Solís-Martín, Galán-Páez & Borrego-Díaz, 2025).

3.7 Transfer Knowledge

In this study, a transfer learning strategy was evaluated, as this approach has already been widely adopted across different domains (Liu et al., 2022), including drug response prediction (Zhu et al., 2020) and more recently bioprocess scale-up (Bayer et al., 2021). Transfer learning enables models trained for one task to be reused for related tasks (Zhu et al., 2020; Park et al., 2022), reducing the need to develop new models entirely from scratch.

Since bioprocess scale-up data is conceptually related for both CHO and iPSC systems, model parameters trained on CHO data may be transferred to a new model targeting iPSC data. This strategy was motivated by the limited availability of iPSC datasets and the assumption that a model does not need to be fully retrained when a sufficiently large CHO dataset is available (Kůrková et al., 2018; Tan et al., 2018).

The rationale is that early recurrent layers capture generalisable temporal patterns of bioprocess dynamics, such as growth phases or responses to agitation and oxygen transfer, while later layers represent more abstract, dataset-specific behaviours. In particular, higher layers are more likely to reflect cell line dependent growth characteristics. Transfer learning therefore enables the reuse of broadly applicable low-level representations, while selectively adapting higher-level abstractions to the target dataset (Zhu et al., 2020).

A network-based deep transfer learning framework was adopted to enable efficient adaptation of the model from CHO to iPSC systems. This methodological choice was informed by evidence from related bioprocess studies (Kůrková et al., 2018; Kingma & Ba, 2015), which demonstrated that retaining pretrained lower layers while retraining only the upper layers effectively reduces prediction error and model uncertainty.

This approach was selected to maximise knowledge reuse from the source system, while minimising data requirements for the target model. iPSC datasets are typically limited and costly to generate, while CHO systems are well-characterised and data rich. Leveraging shared dynamic behaviour between the two systems allows the model to generalise fundamental bioprocess patterns while adapting to the unique physiology of iPSCs. This directly addresses the research aim of developing scalable, data-efficient predictive tools for early-stage iPSC bioprocess development.

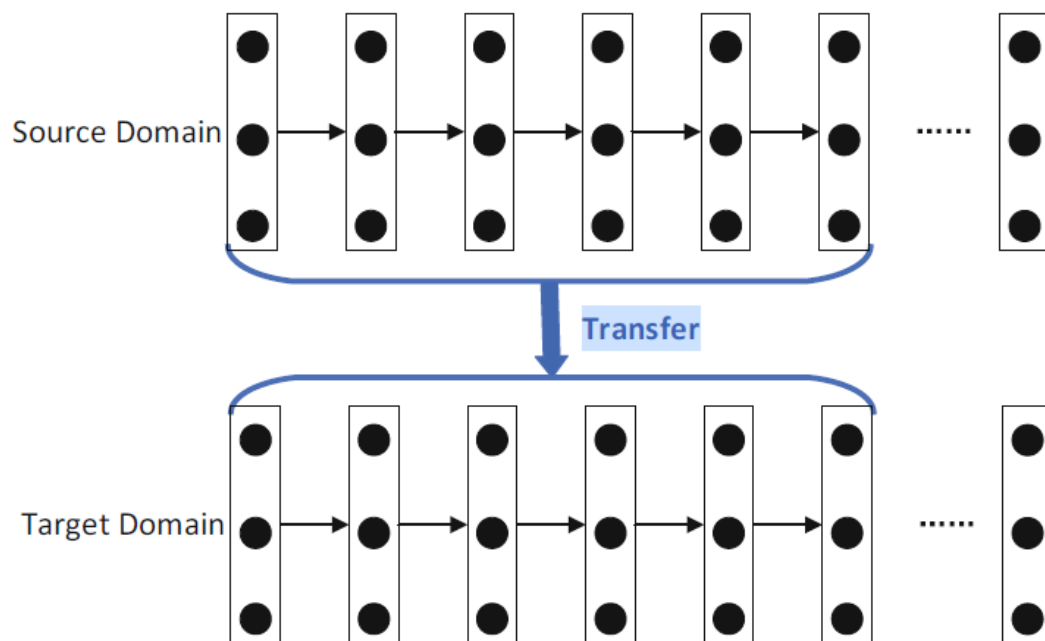


Figure 17: Transfer learning (adapted from Tan et al., 2018).

4. Implementation

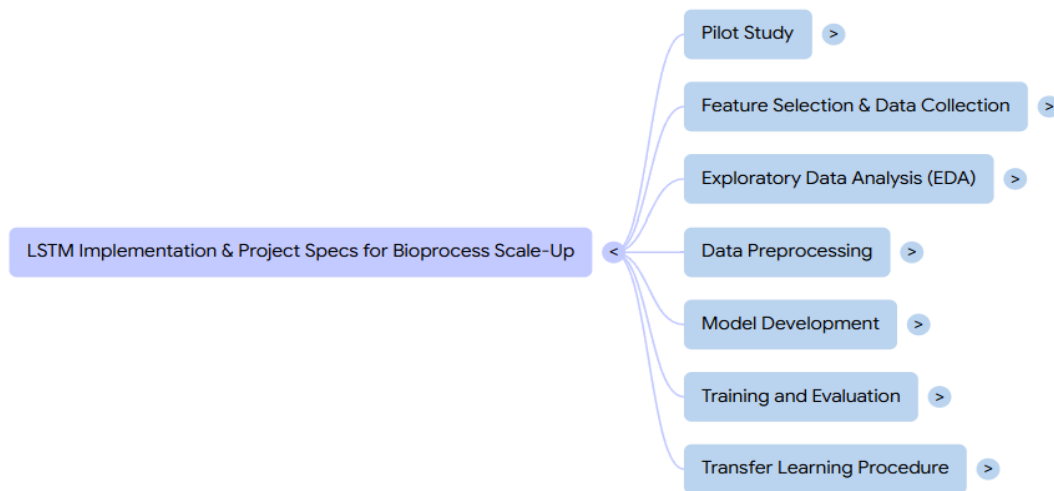


Figure 18: Project framework outlining the LSTM implementation.

The project followed a structured workflow, starting with a pilot study to define objectives and assess data feasibility. Subsequent steps included feature selection, data collection, and EDA to identify key patterns. Clean, normalised, time-aligned datasets were then used to train and tune an LSTM model for predicting VCD, with performance evaluated for accuracy, generalisation, and robustness. Finally, transfer learning was applied to adapt the CHO-trained model to iPSC data (figure 18).

4.1 Pilot Study

The study relied on a reduced CHO cell dataset comprising five scale-up processes and twelve distinct clones, enabling practical testing of methods identified in the literature review.

Insights from the pilot phase informed refinements in preprocessing, feature selection, and model architecture, which were subsequently expanded and formalized in the main study. The overall workflow for both the pilot and main study is illustrated in figure 19.

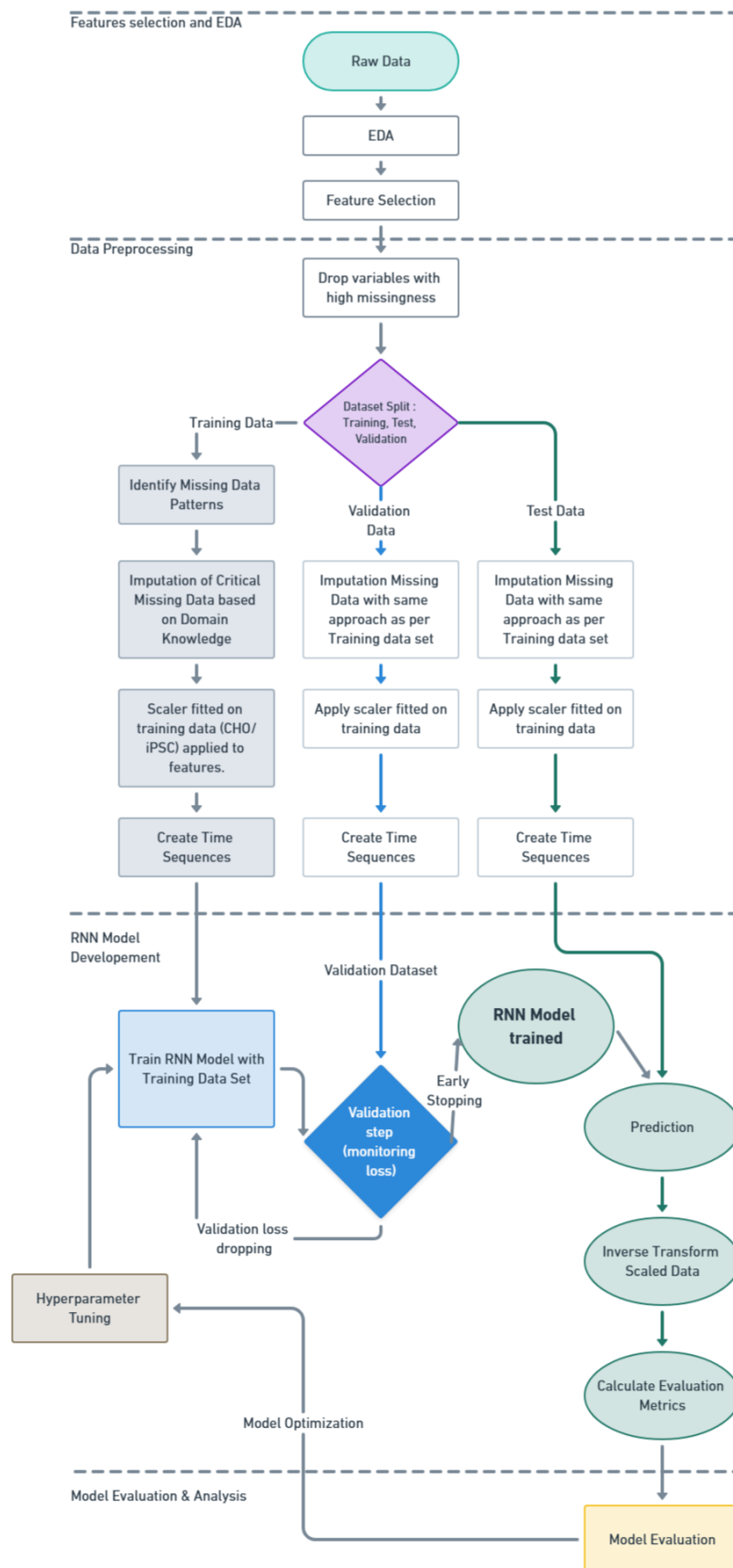


Figure 19: The overall workflow of model development and evaluation.

4.2 Feature selection & Data collection

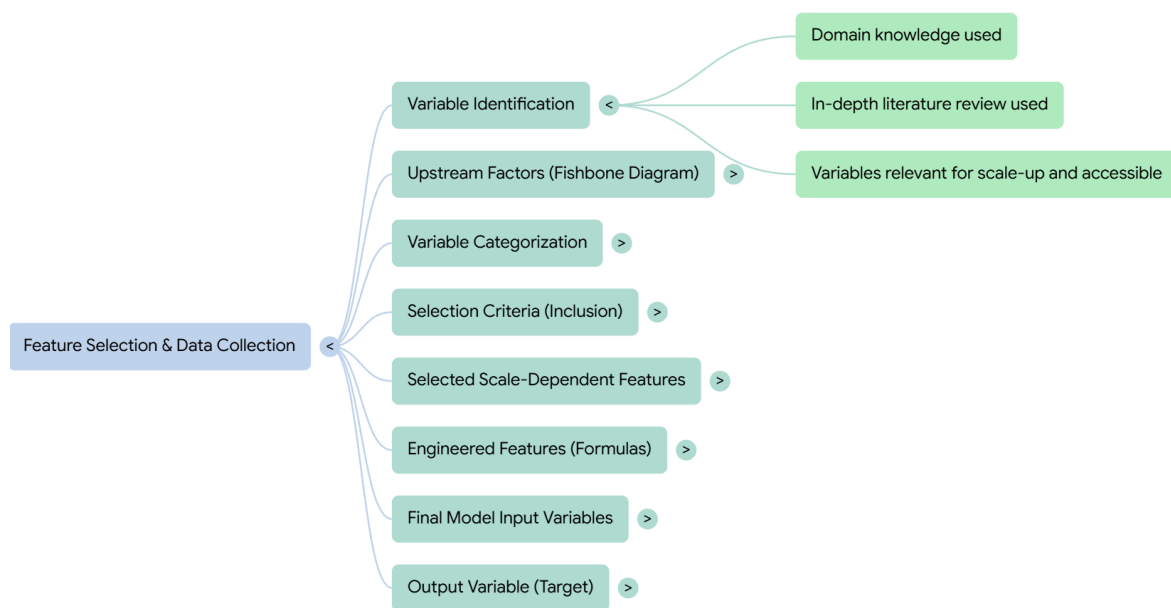


Figure 20: Project framework outlining the variable identification.

Grounded in bioprocess engineering, this study combined domain knowledge and literature insights to select variables relevant to bioprocess scale-up that were available or derivable from published data. The resulting dataset incorporated experimental measurements and engineered features meaningful for modelling mammalian cell culture scale-up. A fishbone diagram (figure 21) summarised six key upstream factors influencing performance: operational conditions, mass transfer, bioreactor design, hydrodynamics, feeding strategy, and cell line, grouped into biological, design, and scale-related categories.

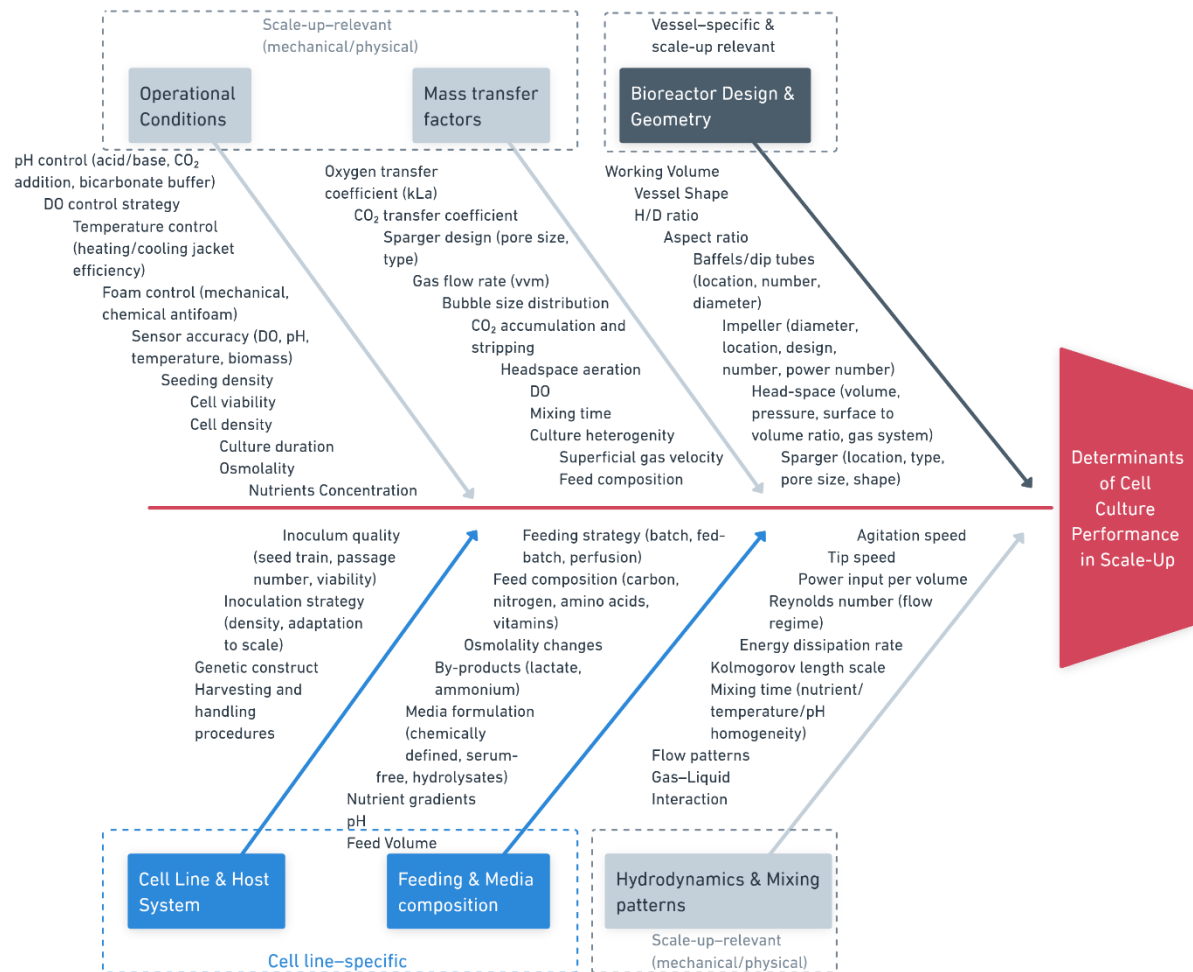


Figure 21: Key determinants influencing cell culture performance during bioreactor scale-up.

These parameters collectively define scale-up strategies. Their strong interconnectivity and nonlinear relationships (figure 22) demonstrate the inherent complexity of scale-up.

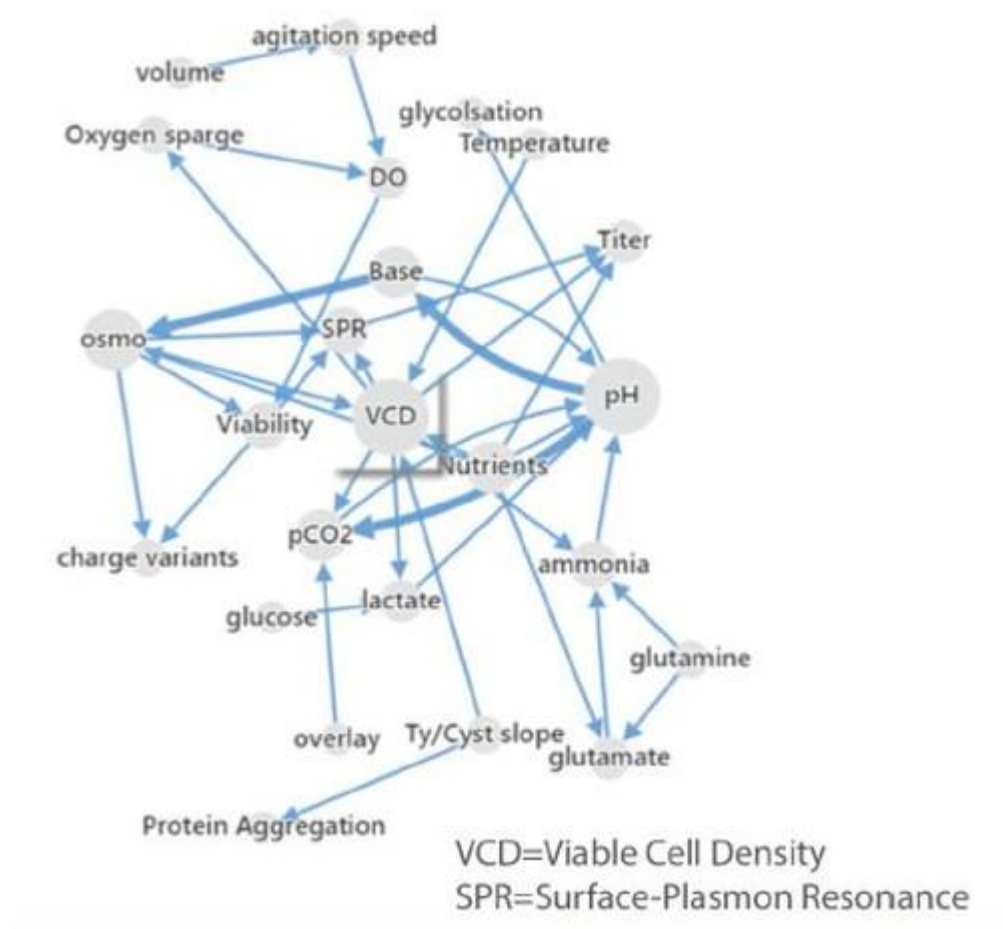


Figure 22: Simplified representation of multicollinearity between bioprocess variables (adapted from Ahmed et al., 2019).

The literature review primarily focused on traditional scale-up approaches based on fixed engineering parameters, which served as an important guide for variable selection in machine learning model development. From this body of work, process- and scale-dependent engineering and hydrodynamic factors with direct impact on culture performance and behaviour were selected (table 1).

Variable	Reference
Oxygen mass transfer coefficient k_La	Xu et al., 2017b; Doi et al., 2020; Kreitmayer et al., 2023; Xing et al., 2024
Gas flow rates (superficial gas velocities) including sparger rate and overlay rate)	Xu et al., 2017a; Xu et al., 2017b; Doi et al., 2020; Xing et al., 2024
pO ₂	Xu et al., 2017a; Xu et al., 2017b; Kreitmayer et al., 2023
pCO ₂	Xu et al., 2017a; Xu et al., 2017b; Doi et al., 2020; Xing et al., 2024
Volumetric power input P/V (W/m ³)	Xu et al., 2017a; Xu et al., 2017b; Doi et al., 2020; Paul & Herwig, 2020; Gaugler et al, 2024; Xing et al., 2024
Mixing time	Van 't Riet & Van der Lans, 2011; Xu et al., 2017a; Xu et al., 2017b; Doi et al., 2020; Paul & Herwig, 2020; Gaugler et al, 2024; Xing et al., 2024
DO heterogeneities between compartments	Xu et al., 2017b; Zakrzewski, Lee & Lye, 2022; Gaugler et al, 2024
Lactate metabolism	Xu et al., 2017b; Zakrzewski, Lee & Lye, 2022
Osmolality	Xu et al., 2017b; Gaugler et al, 2024
Glucose uptake/feeding strategy	Xu et al., 2017b; Gaugler et al, 2024; Yehya et al., 2024
pH	Xu et al., 2017b; Doi et al., 2020; Zakrzewski, Lee & Lye, 2022; Gaugler et al, 2024; Yehya et al., 2024
Temperature	Zakrzewski, Lee & Lye, 2022
impeller tip speed	Xu et al., 2017b; Doi et al., 2020; Kreitmayer et al., 2023
Kolmogorov length	Kreitmayer et al., 2023
Shear stress	Xu et al., 2017b; Xing et al., 2024; Yehya et al., 2024
Reynolds	Kreitmayer et al., 2023
Geometrical aspects	Kreitmayer et al., 2023; Bernemann et al., 2024; Chaudhry, 2024
Sparger design, type, location	
Number of impellers	
Impeller design, type, location	
Vessel baffling configuration	
Vessel diameter	
Impeller diameter	
Aspect ratio	Van 't Riet & Van der Lans, 2011; Xu et al., 2017a; Xu et al., 2017b

Table 1: Overview of key bioprocess variables relevant to scale-up.

These included both directly measured variables and engineered features derived from vessel geometry and agitation characteristics, as well as factors for which mechanistic understanding remains limited. All influence process performance and product quality. Importantly, not all parameters were included in table 1, only the most relevant were retained. All selected parameters influence live biological systems, where even small changes can yield large effects.

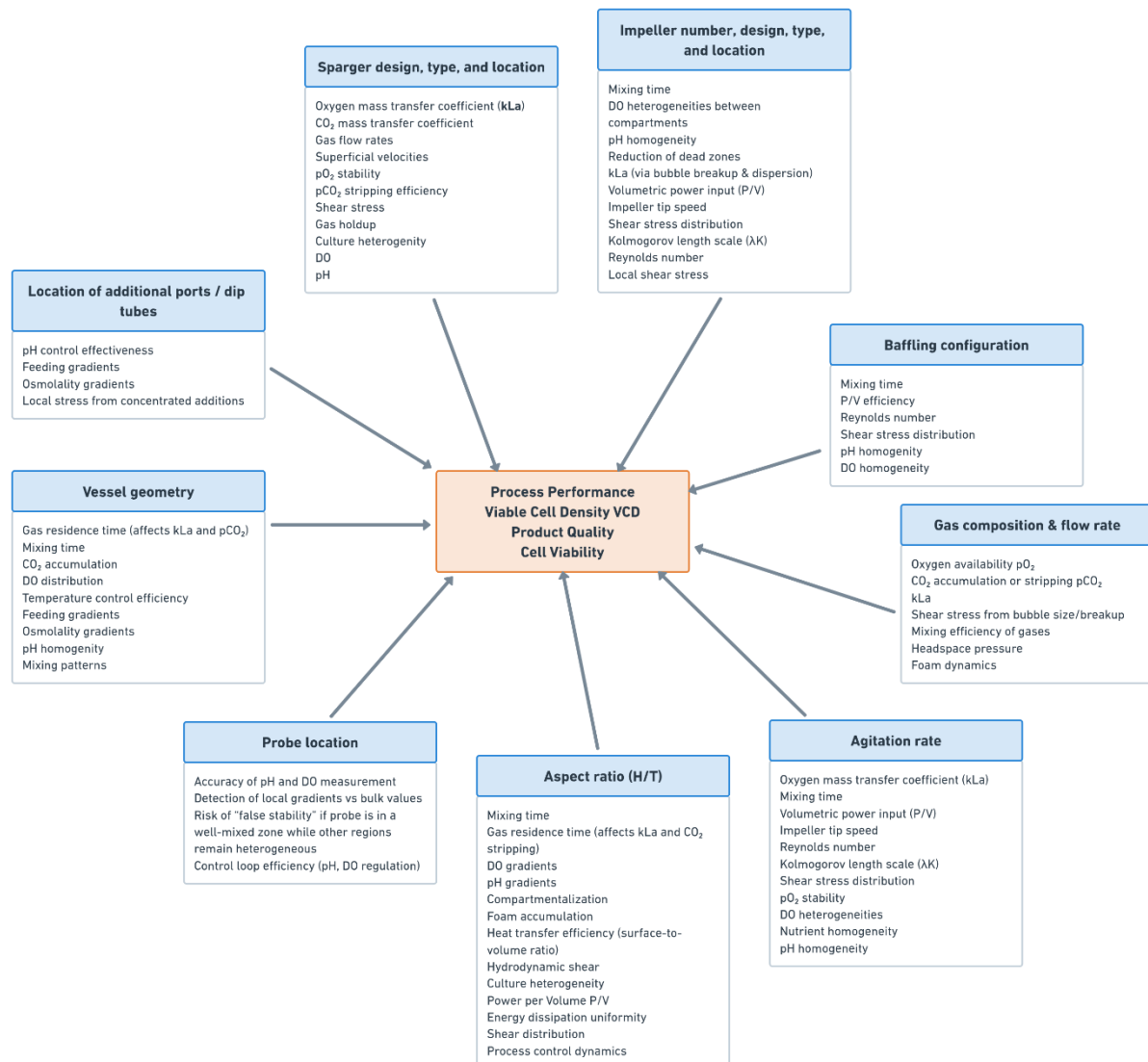


Figure 23: Key bioreactor design and operational factors influencing process performance.

Furthermore, while bioreactor design and geometry strongly affect performance as shown in figure 23, operational parameters such as seeding density, culture duration, and media composition also have a substantial impact on key performance indicators, including VCD, growth rate, and critical quality attributes (Karimi Alavijeh, Lee & Gras, 2024).

Scale-independent variables (e.g. pH, and media composition) were excluded, as they are typically held constant for a given clone. Similarly, cell-biology-specific parameters, such as lactate production or glucose uptake rate, were not included, as they vary between clones and cell types and are therefore not transferable across systems.

Given the dimensionality and complexity of bioprocess variables, it is not possible to collect and harmonise all relevant parameters from public sources for machine learning. Many are unmeasured or unreported, making them unsuitable for inclusion. Qualitative parameters, such as impeller type, were used only to extract measurable information (e.g., impeller diameter) or excluded altogether. Therefore, only variables that significantly influence cell behaviour, are scale dependent, and are either directly available or derivable by feature engineering, but not cell-specific, were chosen for preprocessing.

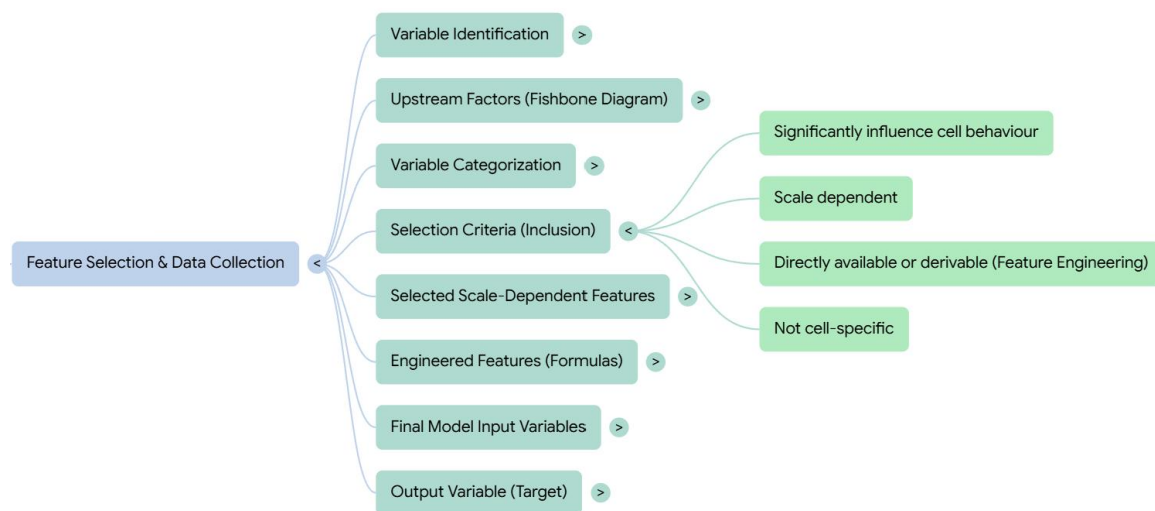


Figure 24: Project framework outlining the variable selection criteria.

The final variable set (Figure 25) included key scale-up parameters accessible from public sources, commonly used as process efficiency indicators and typically constant across scales. Additional scale-dependent variables influencing these features were engineered to enable fair cross-process comparisons. Mass transfer, hydrodynamics, and shear stress were thus represented quantitatively, ensuring relevance for ML-based scale-up.

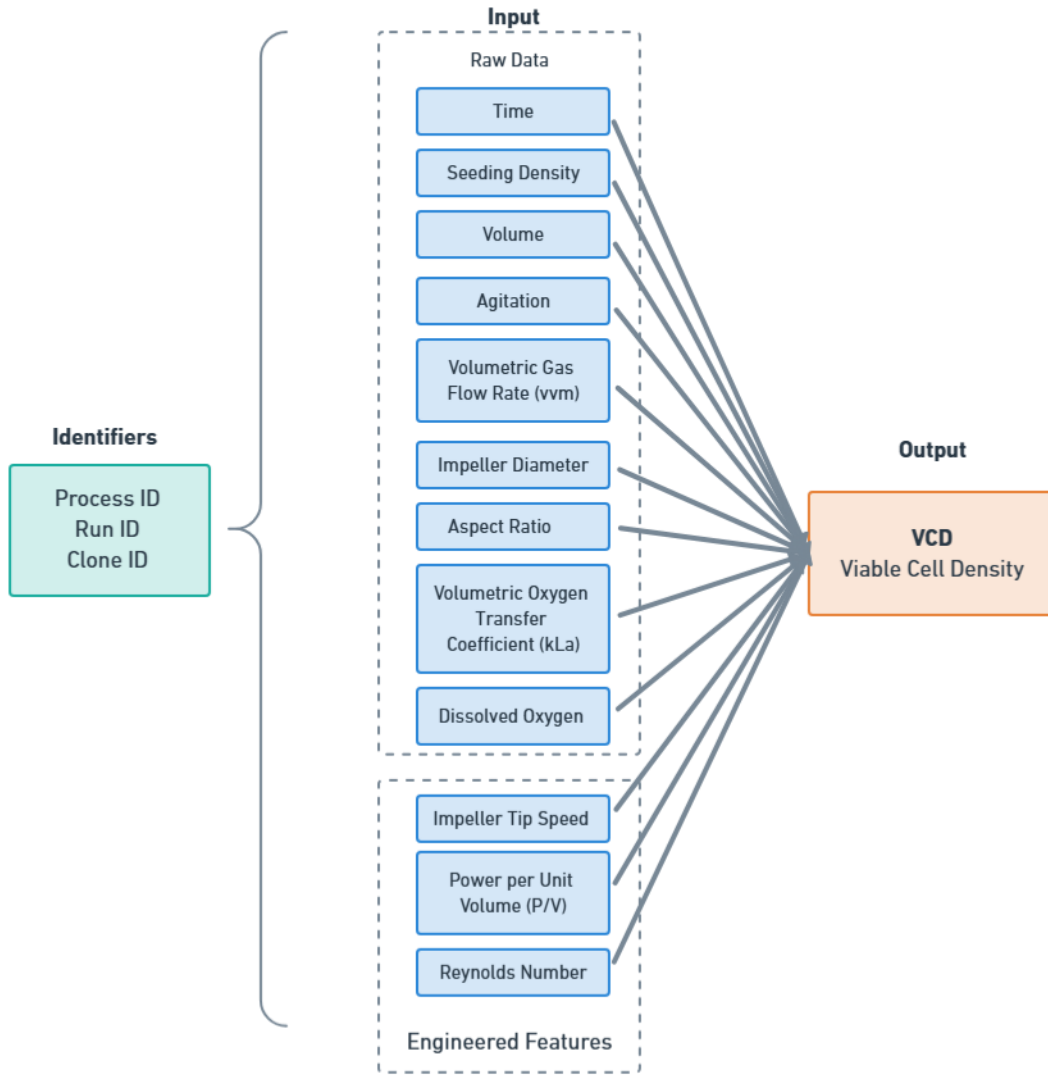


Figure 25: Features used for the ML model.

For unreported variables, feature engineering applied standard equations:

(1) impeller tip speed ($\frac{m}{s}$)

$$v_{impeller,tip} = \frac{\pi d_i n}{60}$$

where d_i = impeller diameter (m), n = agitation speed (rpm).

(2) Power per volume ($\frac{W}{m^3}$)

$$\frac{P}{V} = \frac{N_p \rho N^3 D^5}{V}$$

where N_p = impeller power number (dimensionless), ρ = fluid density ($\frac{kg}{m^3}$; $993 \frac{kg}{m^3}$ for water-like media at 37°C), N = agitation speed (rps or $\frac{rpm}{60}$), D = impeller diameter (m), V = working volume of liquid in the reactor (m^3).

(3) Reynolds number:

$$Re = \frac{\rho N D^2}{\mu}$$

where Re = Reynolds number (dimensionless), N = agitation speed (rps or $\frac{rpm}{60}$), D = impeller diameter (m), μ = fluid dynamic viscosity ($Pa \cdot s = \frac{kg}{ms}$; $0.0007 Pa \cdot s$ for water-like media at 37°C).

Most VCD data were digitised from published growth curves using WebPlotDigitizer to reconstruct time-series trajectories. Datasets used for the pilot and main studies are summarised in table 2.

Study	Dataset	Number of Processes	Number of Clones	Time Series	Features Number	Target Feature	Scales
Pilot study	source domain CHO	5	12	44	10	VCD	15mL-2000L
Pilot study	target domain iPSC	1	1	8	10	VCD	0.5L-6L
Main study	source domain CHO	26	33	118	11	VCD	15mL-5000L
Main study	target domain iPSC	1	1	16	11	VCD	0.5L-6L

Table 2: Overview of pilot and main study datasets for CHO (source domain) and iPSC (target domain).

4.3 EDA



Figure 26: Project framework outlining the EDA.

Both CHO and iPSC datasets were investigated for statistical distributions, missing values, and potential outliers. Data were examined across different processes, defined as sets of bioreactor runs at various scales for a specific clone under a defined protocol, including parameters such as feeding strategy, DO setpoint, etc. Multiple runs of the same process were also included.

A range of visualisations was employed. Histograms were used for continuous variables (volume, agitation, vvm, impeller diameter, aspect ratio, kLa, DO, tip speed, P/V, and Reynolds number), providing insight into overall distribution shapes and potential outliers. Boxplots were applied to VCD to investigate differences in mean, minimum, and maximum values across processes.

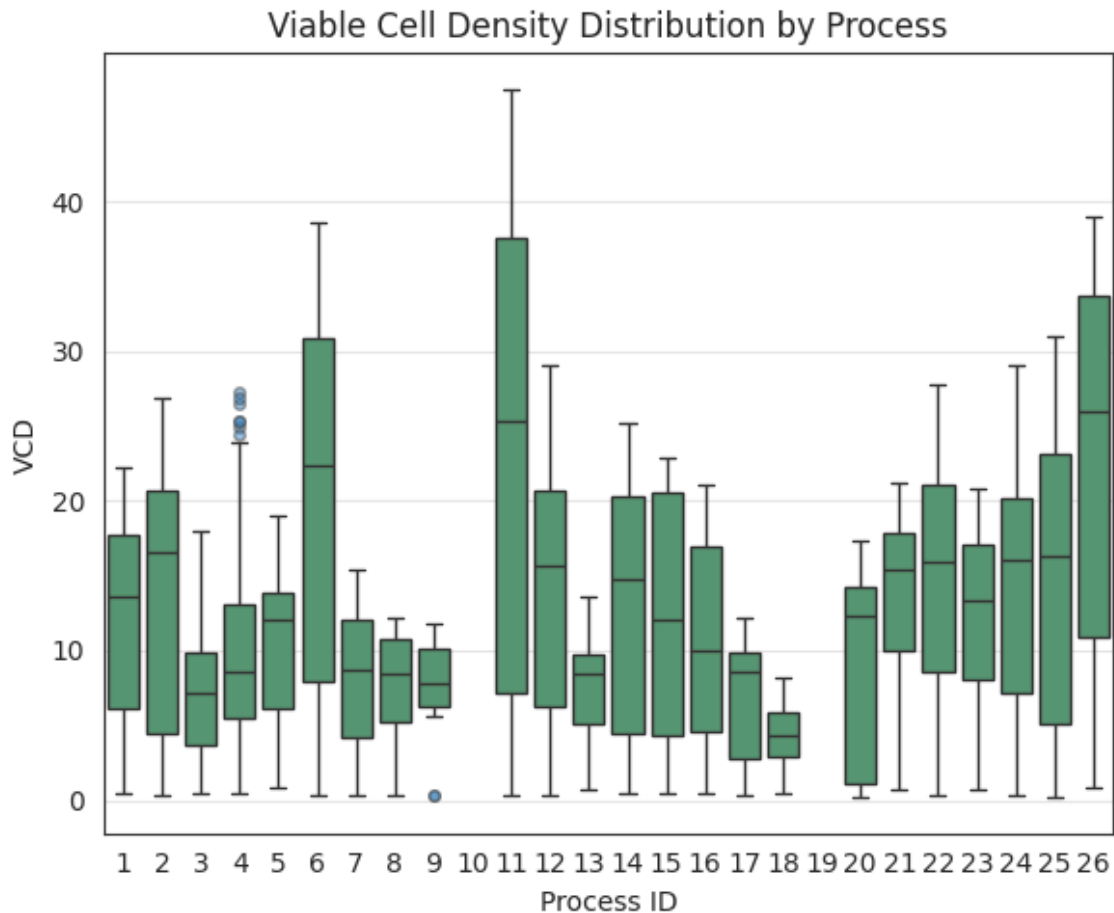


Figure 27: Example boxplot from EDA.

Although time is inherently continuous, it was analysed as discrete sampling intervals. Histograms were used to evaluate the distribution of sampling points and process durations. Importantly, outliers were not excluded, as they often represented meaningful biological phenomena, such as very high tip speed and P/V values in very small-scale reactors, or robust clones achieving unusually high VCD.

Given the time-series nature of bioprocesses, VCD trajectories were visualised with line plots to capture growth trends and sampling patterns. This approach enabled comparison of clones within the same process, variation between processes and scales, and identification of characteristic growth phases (lag, exponential, and stationary). Such plots also revealed differences in clone performance, highlighting more and less efficient growth profiles.

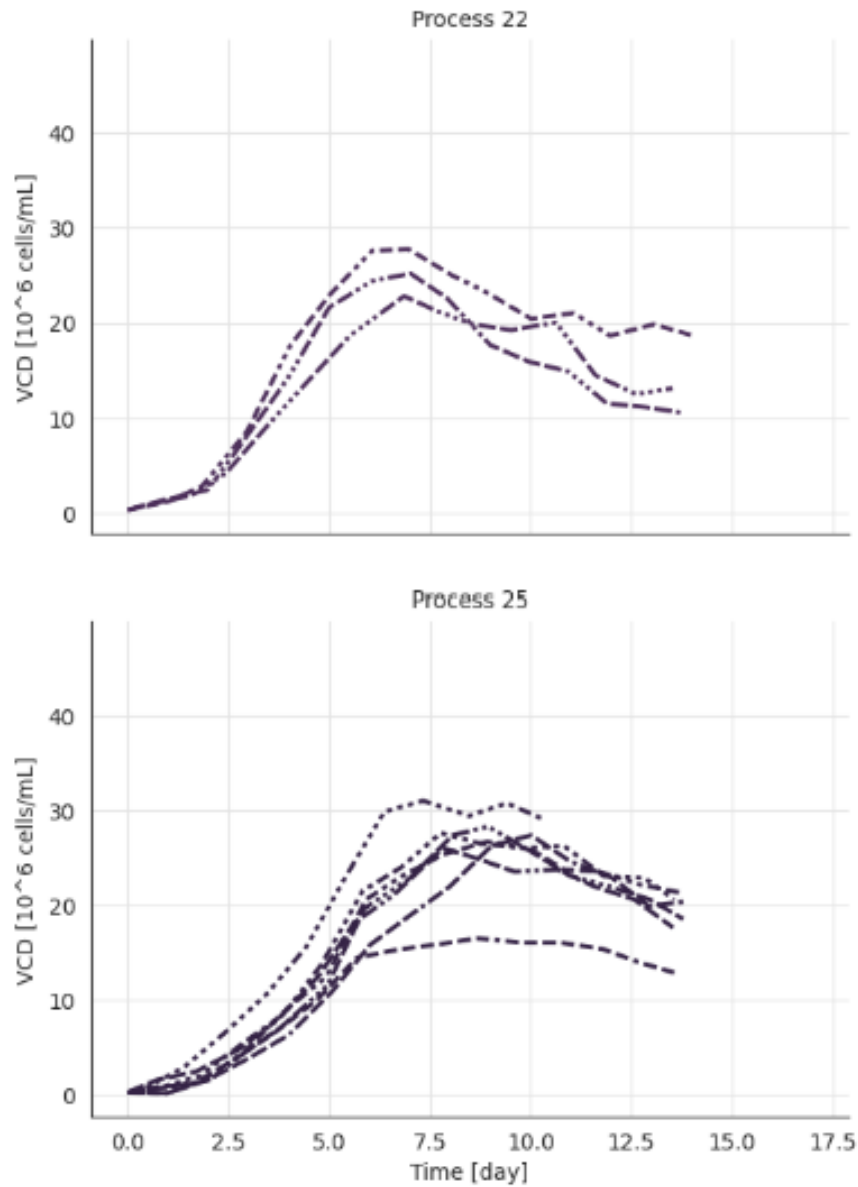


Figure 28: Example growth curves from EDA.

To detect multicollinearity, a correlation matrix was generated. This analysis confirmed expected strong correlations between physically linked variables, such as impeller diameter, agitation speed, and tip speed. The correlation matrix also served as a consistency check: strong correlations between physically related variables confirmed dataset reliability, while weaker than expected correlations indicated either process variability or possible data quality issues.

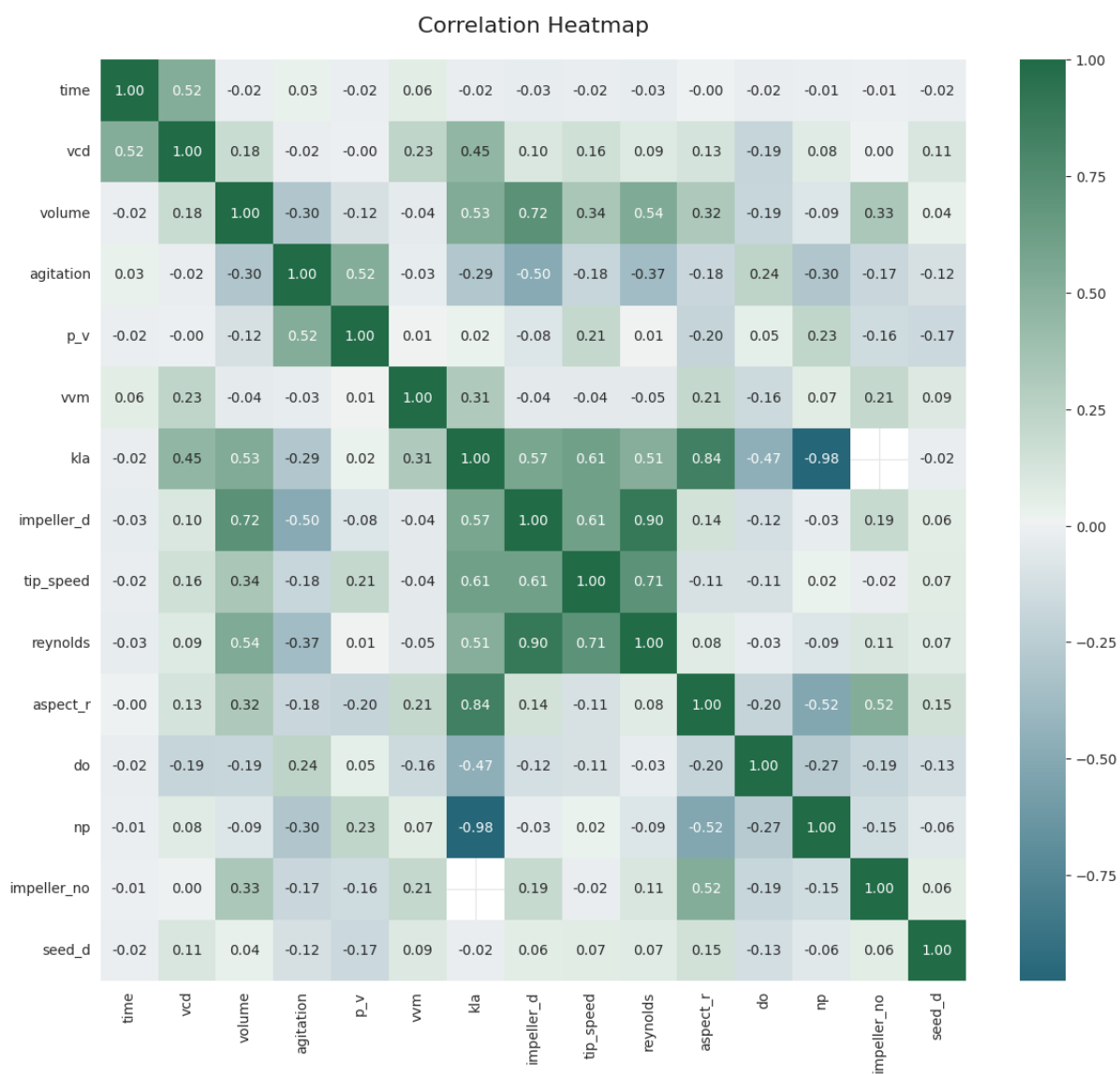


Figure 29: Example correlation heatmap from EDA.

4.4 Data Preprocessing

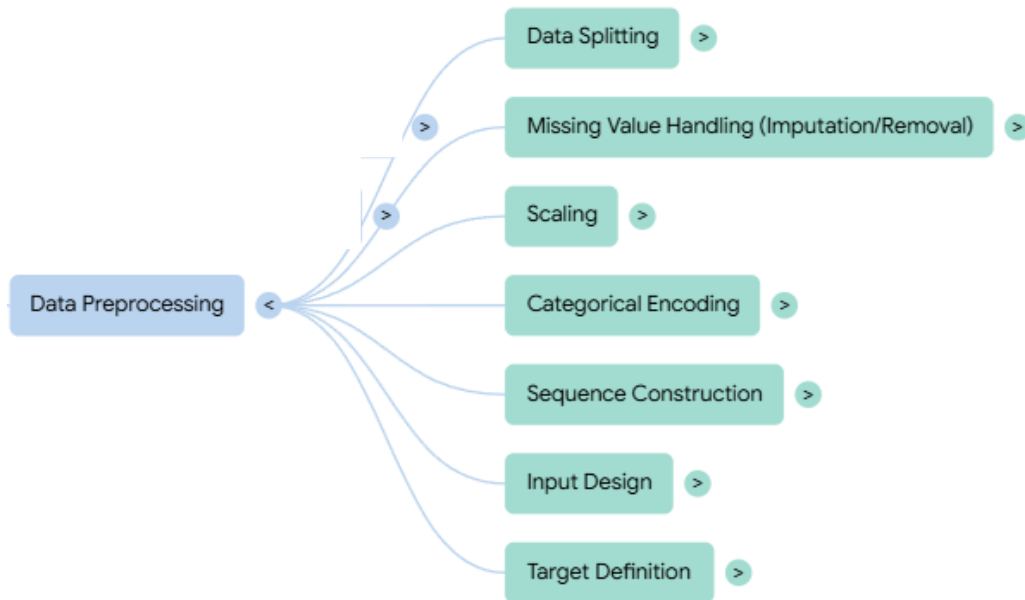


Figure 30: Project framework outlining data preprocessing.

After exploratory analysis, the next step of this study was data preprocessing for the RNN model based on LSTM layers. Since the chosen architecture required sequential inputs, preprocessing was directed towards preparing time-series sequences containing all relevant process variables as input data.

Data splitting

The CHO dataset was first shuffled and randomly divided into training (70%), validation (15%), and test (15%) subsets. This split provided sufficient data for model learning while reserving independent subsets for hyperparameter optimization and unbiased performance evaluation.

The CHO dataset was split so that the test set included the unseen 5000 L scale, ensuring that model generalisation could be evaluated. The iPSC dataset was partitioned to enable training on small-scale data and assessment of predictive performance on larger, previously unseen scales.

Handling missing values

The extent of missing values was systematically assessed. The variable k_La showed more than 50% missing values in pilot and main study and was therefore removed to preserve dataset quality. For other variables, imputation strategies were applied based on domain knowledge and typical bioprocess practices.

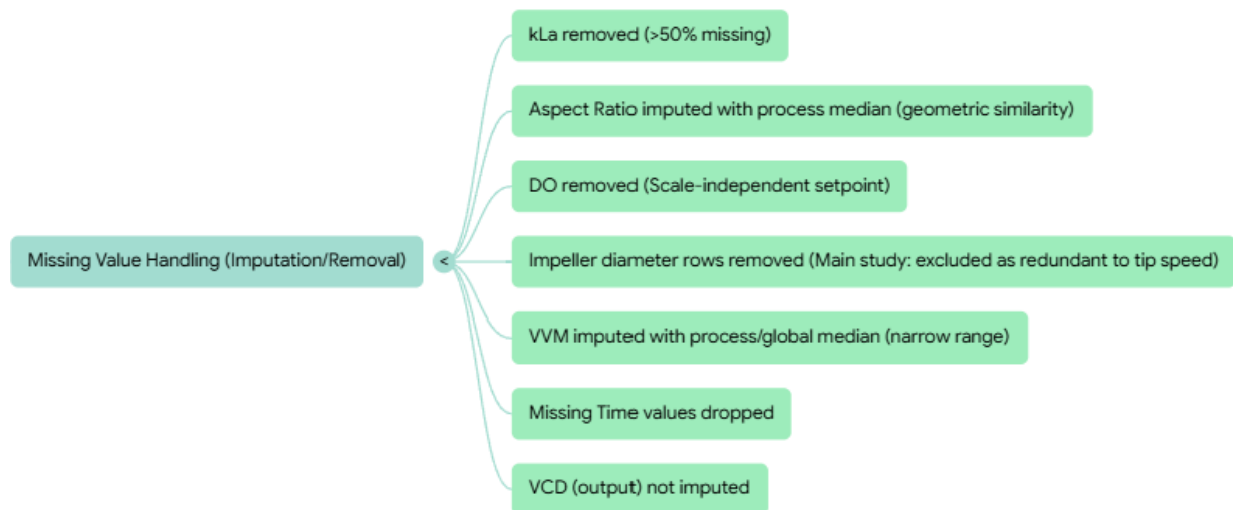


Figure 31: Project framework outlining the missing values handling.

- (1) Aspect ratio is usually kept constant within a scale-up process to maintain geometric similarity across scales, ensuring comparable hydrodynamic conditions such as mixing and mass transfer (Arias & Kotipalli, 2021). Missing values were imputed using the median aspect ratio within each process. For example, in the pilot dataset, aspect ratios ranged between 1.5 and 2.5, therefore imputing with the median per process provided a realistic and representative value aligned with bioreactor geometries.
- (2) DO represents a scale-independent process setpoint (e.g., 40% or 60% for a given runs in pilot study). Since it is a fixed input rather than an observed outcome, the DO column was removed to prevent the model from incorrectly relying on this variable.
- (3) Impeller diameter as a vessel-specific, scale-dependent variable could not be meaningfully imputed with averages or medians across scales. Given that only a small fraction of values was missing, the corresponding rows were removed to avoid introducing misleading information. In the main study, impeller diameter was excluded from the dataset as it represented redundant information. Impeller tip speed was retained, as it provides a more meaningful descriptor for modelling scale-up.

- (4) VVM values were consistently very low across processes, for instance 0.01–0.04 in pilot study. For processes with partial missing values, the process-specific median was used. For processes with entirely missing values, the global median was applied due to the narrow value range across the dataset.
- (5) Missing time values were dropped, as imputation could distort the sequence structure and introduce bias.
- (6) As the output variable, VCD was not imputed. Missing values were excluded from the dataset.

This preprocessing strategy ensured that input variables were consistent with bioprocess engineering principles and avoided unrealistic imputations.

Data Augmentation

To overcome the limited number of experimental runs, data augmentation was applied to the training datasets. The augmentation was designed to simulate natural experimental variability (noise = 0.05) in scale-dependent process parameters, while preserving temporal structure and physical relationships between features.

```
np.random.seed(42)

def augment_sequence(df_group, noise = 0.05):
    aug_df = df_group.copy()
    for col in ['p_v', 'vvm', 'tip_speed', 'reynolds']:
        aug_df[col] += aug_df[col] * np.random.uniform(-noise, noise)
    return aug_df

train_df['group_id'] = train_df.groupby(['p_id', 'r_id', 'clone_id', 'group_volume']).ngroup()

train_aug = []

for group_id, df_group in train_df.groupby('group_id'):
    for i in range(2):
        augmented = augment_sequence(df_group)
        augmented['group_id'] = f'{group_id}_aug{i}'
        train_aug.append(augmented)
train_df['group_id'] = train_df['group_id'].astype(str)
train_aug = pd.concat([train_df] + train_aug, ignore_index=True)
```

Figure 32: Code snippet showing data augmentation.

Scaling

Because RNNs are sensitive to differences in feature magnitude, all variables were standardised. To avoid vanishing gradients and ensure comparability across features, values were scaled to the range $[-1, 1]$, corresponding to the tanh activation function.

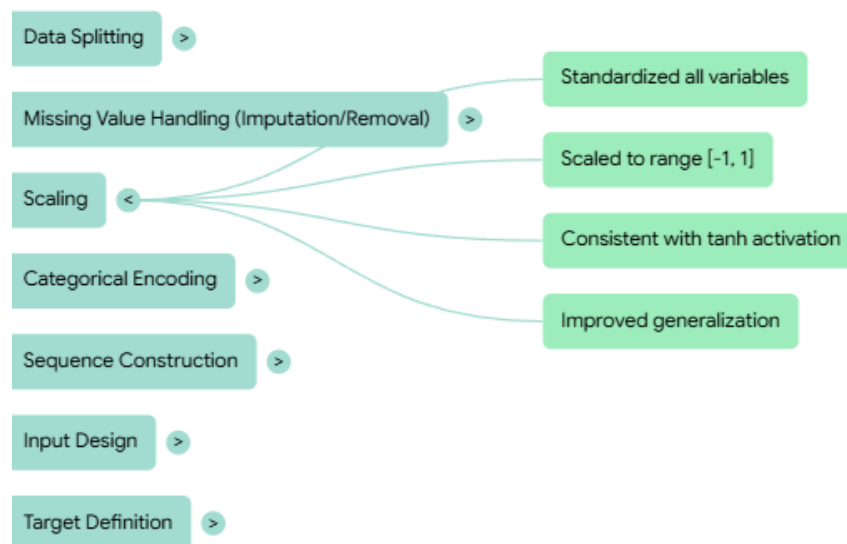


Figure 33: Project framework outlining scaling approach.

This normalisation removed the influence of measurement units, enabling the model to focus on relationships rather than absolute magnitudes and thereby improving generalisation. Scaling parameters were fitted for both CHO and iPSC training datasets.

```
num_cols = ['time', 'vcd', 'volume', 'p_v', 'vvm', 'tip_speed', 'aspect_r', 'reynolds', 'seed_d']

scaling_df = pd.concat([train_aug[num_cols], train_aug_ipsc[num_cols]], ignore_index=True)

scaler = MinMaxScaler(feature_range=(-1, 1))
scaler.fit(scaling_df)

train_aug[num_cols] = scaler.transform(train_aug[num_cols])

joblib.dump(scaler, 'scaler_combined_cho_ipsc_main_study.pkl')
```

Figure 34: Code snippet showing feature scaling.

Categorical encoding

In the pilot study, clone identity was the only categorical variable included in the model. Due to the large number of unique clones, an ordinal encoder was applied. Unseen clones in the validation and test sets were assigned to a dedicated unknown category to maintain model robustness and prevent evaluation errors.

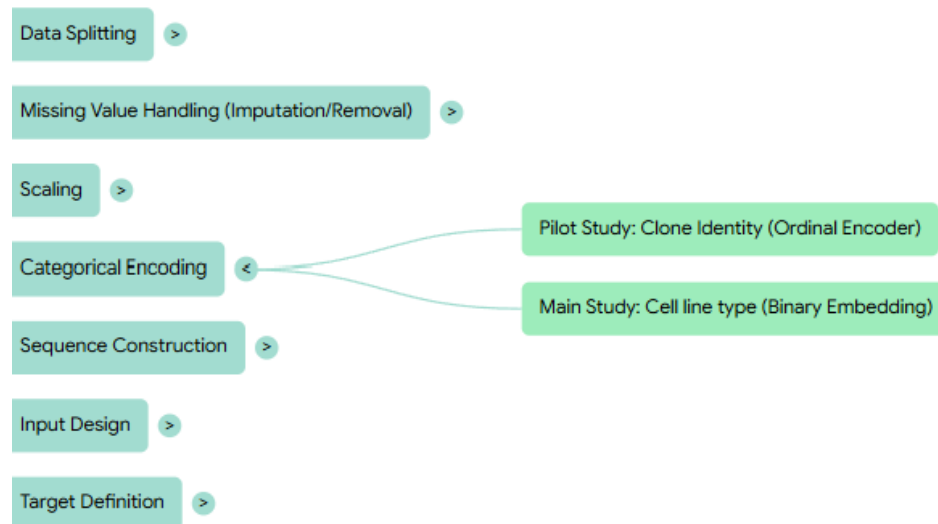


Figure 35: Project framework outlining the categorical encoding.

In the main study, clone identity was removed from the input features, as its inclusion did not enhance predictive accuracy. Instead, the cell line type (CHO or iPSC) was encoded as a categorical variable using binary embedding.

```

category_df = pd.concat([train_aug[['cell_type']], train_aug_ipsc[['cell_type']], ignore_index=True)

enc = OneHotEncoder(
    categories=[['CHO', 'iPSC']],
    drop=None,
    handle_unknown='ignore',
    sparse_output=False
)
enc.fit(category_df[['cell_type']])
Xohe_cho = enc.transform(train_aug[['cell_type']])
Xohe_ipsc = enc.transform(train_aug_ipsc[['cell_type']])
ohe_cols = [f'cell_type_{c}' for c in enc.categories_[0]]
train_aug[ohe_cols] = Xohe_cho
train_aug_ipsc[ohe_cols] = Xohe_ipsc
joblib.dump(enc, 'cell_type_enc.joblib')
files.download('cell_type_enc.joblib')

```

Figure 36: Code snippet showing one-hot encoding.

Sequence Construction and Autoregressive Input Design

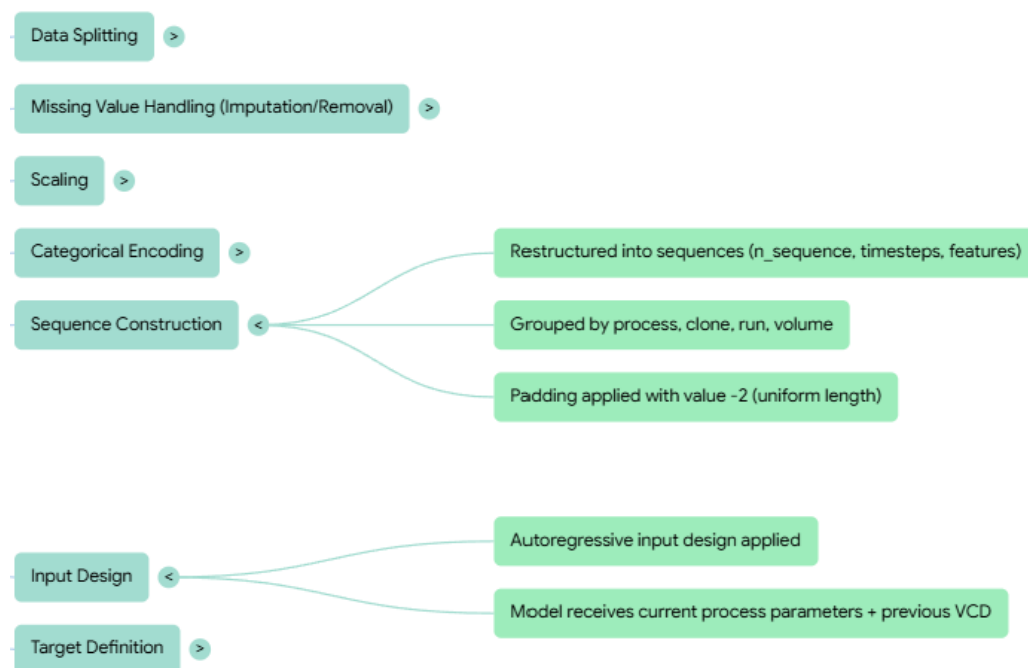


Figure 37: Project framework outlining the sequence construction and input design.

Because the model output was defined as a sequence of VCD values over time, the dataset was restructured into sequences grouped by process number, clone number, run number, and volume. This ensured that each sequence corresponded to a single run. Sequences were represented in the format (n_sequence, timesteps, features). To allow all sequences to have

uniform length, padding was applied with the value -2 , consistent with the normalised input range.

```
def build_ar_sequences(df, group_cols, feature_cols, seed_col='seed_d', vcd_col='vcd'):
    X_list, y_list = [], []
    df_sorted = df.sort_values(group_cols + ['time'])
    other_cols = [c for c in feature_cols if c != vcd_col]

    for _, g in df_sorted.groupby(group_cols, sort=False):
        g = g.reset_index(drop=True)
        v = g[vcd_col].to_numpy().astype('float32')
        seed = g[seed_col].iloc[0]
        v_prev = np.concatenate([[seed], v[:-1]]).astype('float32')
        other = g[other_cols].to_numpy().astype('float32')
        X = np.concatenate([other, v_prev[:, None]], axis=1)
        y = v[:, None]
        X_list.append(X)
        y_list.append(y)
    return X_list, y_list

ar_X_train, ar_y_train = build_ar_sequences(train_aug, group_cols, feature_cols)
ar_X_val, ar_y_val = build_ar_sequences(val_df, group_cols, feature_cols)
ar_X_test, ar_y_test = build_ar_sequences(test_df, group_cols, feature_cols)

for i, (x_seq, y_seq) in enumerate(zip(ar_X_train, ar_y_train)):
    print(f"X shape = {x_seq.shape}, y shape = {y_seq.shape}")

X shape = (11, 11), y shape = (11, 1)
X shape = (11, 11), y shape = (11, 1)
X shape = (11, 11), y shape = (11, 1)
X shape = (9, 11), y shape = (9, 1)
X shape = (9, 11), y shape = (9, 1)
X shape = (9, 11), y shape = (9, 1)
X shape = (9, 11), y shape = (9, 1)
X shape = (9, 11), y shape = (9, 1)
X shape = (9, 11), y shape = (9, 1)
X shape = (9, 11), y shape = (9, 1)
X shape = (10, 11), y shape = (10, 1)
X shape = (10, 11), y shape = (10, 1)
```

Figure 38: Code snippet for building autoregressive time-series sequences

```

def trim_pad(X_list, y_list, max_len, pad):
    X_list = [x[:max_len, :] for x in X_list]
    y_list = [np.asarray(y, dtype='float32')[:max_len].reshape(-1, 1) for y in y_list]

    X = tf.ragged.constant(X_list).to_tensor(default_value=pad, shape=[None, max_len, X_list[0].shape[1]])
    y = tf.ragged.constant(y_list).to_tensor(default_value=pad, shape=[None, max_len, 1])
    return X, y

max_len = 14
pad = -2.0

X_train, y_train = trim_pad(ar_X_train, ar_y_train, max_len, pad)
X_val, y_val = trim_pad(ar_X_val, ar_y_val, max_len, pad)
X_test, y_test = trim_pad(ar_X_test, ar_y_test, max_len, pad)

X_train, y_train = X_train.numpy(), y_train.numpy()
X_val, y_val = X_val.numpy(), y_val.numpy()
X_test, y_test = X_test.numpy(), y_test.numpy()

print(y_train.shape)

(252, 14, 1)

print(X_train.shape)
print(X_test.shape)
print(y_test.shape)

(252, 14, 11)
(17, 14, 11)
(17, 14, 1)

```

Figure 39: Code snippet for trimming and padding time-series sequences.

To enable temporal prediction of VCD, an autoregressive input design was applied. The model receives process parameters from the current time step together with the VCD from the previous step, allowing it to learn temporal dependencies directly from the data and predict the next VCD based on recent system states and environmental conditions.

Target definition

VCD, as the output variable, was removed from the input feature set and defined as the target for prediction.

4.5 CHO Source Model Development

In this study, an RNN was implemented to capture the multidimensional interactions underlying cell growth in bioreactors across scales. Due to the time-series nature of bioprocess data, RNNs and particularly LSTMs, are well suited for forecasting tasks, as they can learn temporal dependencies and handle noisy sequential inputs. Since later outcomes in cell culture strongly depend on early-stage conditions, the use of RNNs was appropriate, and LSTM layers were incorporated to provide robustness and the ability to simulate entire VCD trajectories under varying conditions (Hewamalage, Bergmeir & Bandara, 2021; Ma et al., 2022; Mienye, Swart & Obaido, 2024).

Pilot study (CHO Source Model)

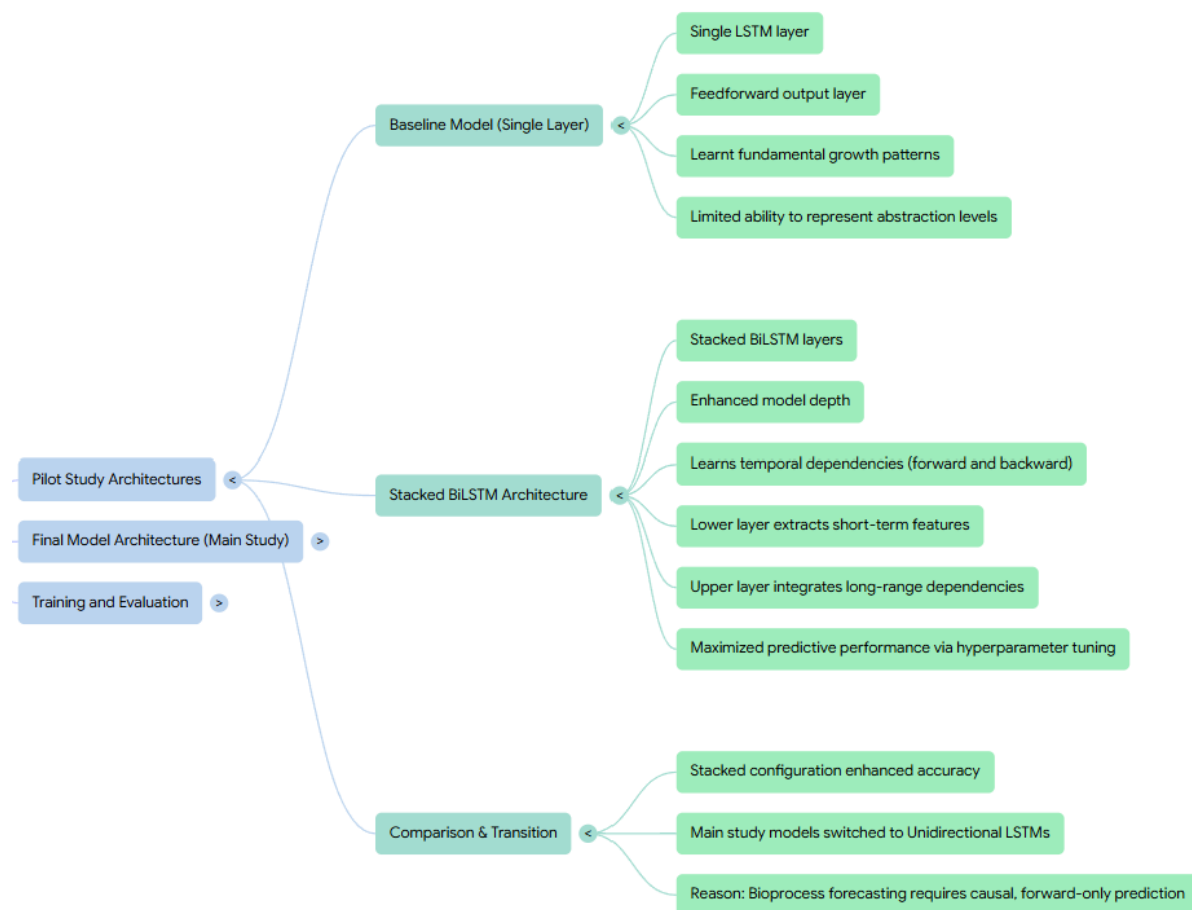


Figure 40: Project framework outlining the pilot study models architecture.

In the pilot study, a baseline model was implemented consisting of a single LSTM layer followed by a feedforward output layer. This architecture represents the standard application of LSTMs in sequence modelling tasks, where the objective is to predict a single continuous variable (Ma et al., 2022).

```
model = tf.keras.Sequential([
    tf.keras.layers.Masking(mask_value=-2.0, input_shape=(X_train.shape[1], X_train.shape[2])),
    tf.keras.layers.LSTM(64, return_sequences=True),
    tf.keras.layers.Dropout(0.2),
    tf.keras.layers.Dense(32, activation='tanh'),
    tf.keras.layers.Dense(1)
])
```

Figure 41: LSTM model architecture in 1st pilot study stage.

As a next step, a stacked BiLSTM architecture to enhance model depth and evaluate whether learning temporal dependencies in both forward and backward directions would improve the accuracy of CHO cell growth predictions.

```
model = models.Sequential([
    layers.Masking(mask_value=-2.0, input_shape=(timesteps, n_features)),

    layers.Bidirectional(layers.LSTM(64, return_sequences=True, dropout=0.5, recurrent_dropout=0.5)),
    layers.LayerNormalization(),
    layers.Dropout(0.5),

    layers.Bidirectional(layers.LSTM(32, return_sequences=True, dropout=0.5, recurrent_dropout=0.5)),
    layers.LayerNormalization(),
    layers.Dropout(0.4),

    layers.Dense(64, activation='tanh', kernel_regularizer=regularizers.l2(0.01)),
    layers.Dense(1, activation='tanh')
])
```

Figure 42: LSTM model architecture in 2nd pilot study stage.

A stacked architecture is considered more advanced, as it is designed to handle complex datasets by learning hierarchical representations at different levels of abstraction (Hewamalage, Bergmeir & Bandara, 2021; Ma et al., 2022). A single LSTM layer is capable of capturing temporal patterns but remains limited in its ability to represent multiple levels of abstraction. By stacking two LSTM layers, the lower layer extracts short-term temporal features directly from the input signals, while the upper layer integrates these features into higher-level, long-range dependencies. The stacked LSTM network was therefore evaluated as a more

advanced architecture, specifically designed to capture deeper temporal dependencies and hierarchical representations.

For Pilot study BiLSTM different hyperparameter configurations were tested to identify the optimal model architecture that maximised predictive performance.

```

model = models.Sequential([
    layers.Masking(mask_value=-2.0, input_shape=(timesteps, n_features)),

    layers.Bidirectional(layers.LSTM(64, return_sequences=True, dropout=0.5, recurrent_dropout=0.5)),
    layers.LayerNormalization(),
    layers.Dropout(0.5),

    layers.Bidirectional(layers.LSTM(32, return_sequences=True, dropout=0.5, recurrent_dropout=0.5)),
    layers.LayerNormalization(),
    layers.Dropout(0.5),

    layers.Dense(64, activation='tanh', kernel_regularizer=regularizers.l2(0.01)),
    layers.Dense(1, activation='tanh')
])

```

Ausführungsausgabe vom 2. Aug. 2025 14:18

```

0KB
Stream
- RMSE = 9.7% of the VCD max in test set.
- MAE = 7.5% of the VCD max in test set.

```

```

model = models.Sequential([
    layers.Masking(mask_value=-2.0, input_shape=(timesteps, n_features)),

    layers.Bidirectional(layers.LSTM(64, return_sequences=True, dropout=0.5, recurrent_dropout=0.5)),
    layers.LayerNormalization(),
    layers.Dropout(0.5),

    layers.Bidirectional(layers.LSTM(32, return_sequences=True, dropout=0.4, recurrent_dropout=0.4)),
    layers.LayerNormalization(),
    layers.Dropout(0.4),

    layers.Dense(64, activation='tanh', kernel_regularizer=regularizers.l2(0.01)),
    layers.Dense(1, activation='tanh')
])

```

Ausführungsausgabe vom 2. Aug. 2025 14:18

```

0KB
Stream
- RMSE = 9.7% of the VCD max in test set.
- MAE = 7.5% of the VCD max in test set.

```

```

model = models.Sequential([
    layers.Masking(mask_value=-2.0, input_shape=(timesteps, n_features)),

    layers.Bidirectional(layers.LSTM(64, return_sequences=True, dropout=0.4, recurrent_dropout=0.4)),
    layers.LayerNormalization(),
    layers.Dropout(0.4),

    layers.Bidirectional(layers.LSTM(32, return_sequences=True, dropout=0.3, recurrent_dropout=0.3)),
    layers.LayerNormalization(),
    layers.Dropout(0.3),

    layers.Dense(64, activation='tanh', kernel_regularizer=regularizers.l2(0.01)),
    layers.Dense(1, activation='tanh')
])

```

Ausführungsausgabe vom 25. Juli 2025 09:28

```

0KB
Stream
- RMSE = 11.0% of the VCD max in test set.
- MAE = 8.3% of the VCD max in test set.

```

Figure 43: Code showing different BiLSTM hyperparameter configurations tested during the pilot study.

```

model = models.Sequential([
    layers.Masking(mask_value=-2.0, input_shape=(timesteps, n_features)),

    layers.Bidirectional(layers.LSTM(64, return_sequences=True, dropout=0.5, recurrent_dropout=0.5)),
    layers.LayerNormalization(),
    layers.Dropout(0.5),

    layers.Bidirectional(layers.LSTM(32, return_sequences=True, dropout=0.5, recurrent_dropout=0.5)),
    layers.LayerNormalization(),
    layers.Dropout(0.4),

    layers.Dense(64, activation='tanh', kernel_regularizer=regularizers.l2(0.01)),
    layers.Dense(1, activation='tanh')
])

model.compile(optimizer='adam', loss='mse', metrics=['mae'])

early_stop = EarlyStopping(patience=30, restore_best_weights=True)

history = model.fit(
    X_train, y_train,
    validation_data=(X_val, y_val),
    epochs=100,
    batch_size=8,
    callbacks=[early_stop]
)

```

Ausführungsausgabe vom 2. Aug. 2025 22:21

Ausführungsausgabe vom 2. Aug. 2025 22:22

0KB

```

Stream
    RMSE = 12.1% of the VCD max in test set.
    MAE = 8.2% of the VCD max in test set.

```

Figure 44: Code showing final pilot study BiLSTM.

Model performance was evaluated not only based on loss curves and minimal error values, but also on the shape of the predicted growth trajectories. A configuration with a slightly higher error was ultimately preferred, as its predicted curve most accurately reflected the realistic pattern of cell growth.

Both, basic Pilot study LSTM and stacked BiLSTM architectures are illustrated in figure 45.

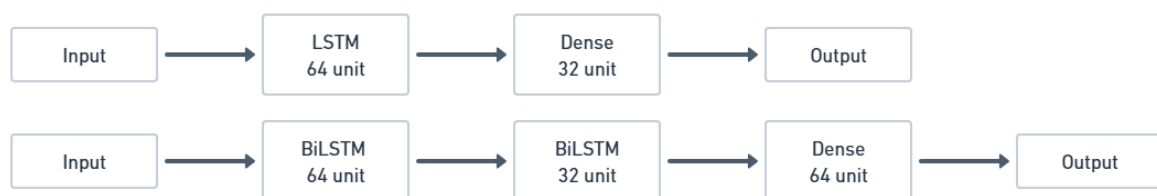


Figure 45: A schematic comparison for source models in pilot study.

Pilot experiments with a single LSTM layer demonstrated that while fundamental growth patterns could be learnt, predictive accuracy was enhanced when a stacked configuration was

applied. However, since bioprocess forecasting requires causal, forward-only prediction, later models in the main study were implemented as unidirectional, autoregressive LSTMs to ensure alignment with real-time applications.

Main study (CHO Source Model)

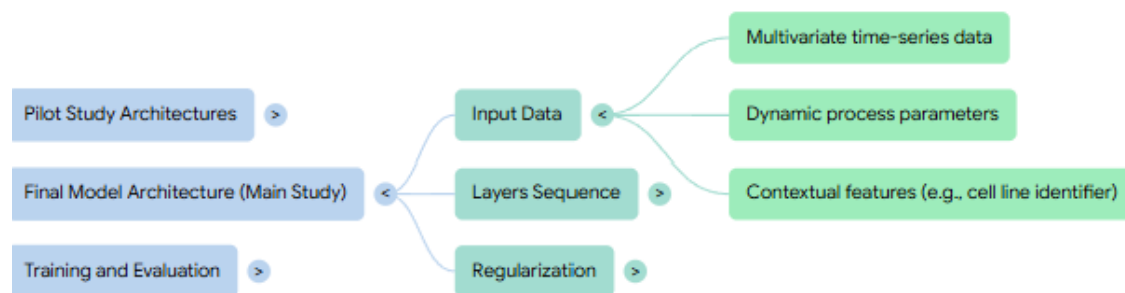


Figure 46: Project framework outlining the main study input data.

The final model processed multivariate time-series data, with each timestep including dynamic process parameters and contextual features such as the cell line identifier. The architecture comprised a masking layer to exclude padded values, followed by two stacked LSTM layers and

dense layers that mapped temporal features into growth trajectory predictions. Dropout regularisation and early stopping were applied to prevent overfitting.

```
model = models.Sequential([
    layers.Masking(mask_value=-2.0, input_shape=(timesteps, n_features)),

    layers.LSTM(64, return_sequences=True, dropout=0.5, recurrent_dropout=0.5),
    layers.LayerNormalization(),
    layers.Dropout(0.5),

    layers.LSTM(32, return_sequences=True, dropout=0.5, recurrent_dropout=0.5),
    layers.LayerNormalization(),
    layers.Dropout(0.4),

    layers.Dense(64, activation='tanh', kernel_regularizer=regularizers.l2(0.01)),
    layers.Dense(1, activation='tanh')
])

model.compile(optimizer='adam', loss='mse', metrics=['mae'])

early_stop = EarlyStopping(patience=30, restore_best_weights=True)

history = model.fit(
    X_train, y_train,
    sample_weight=sw_train,
    validation_data=(X_val, y_val, sw_val),
    epochs=100,
    batch_size=8,
    callbacks=[early_stop]
)
```

Figure 47: Stacked LSTM model architecture used in main study.

Since the sequences in the dataset were partially padded, the model architecture began with a masking layer to ignore padded values, ensuring that only valid observations contributed to learning. The core of the network consisted of two stacked LSTM layers, where the output sequence of the first layer was passed as input to the second. At each timestep, the input vector x_t entered the first LSTM layer with 64 units, which produced a sequence of hidden states h_{t1} .

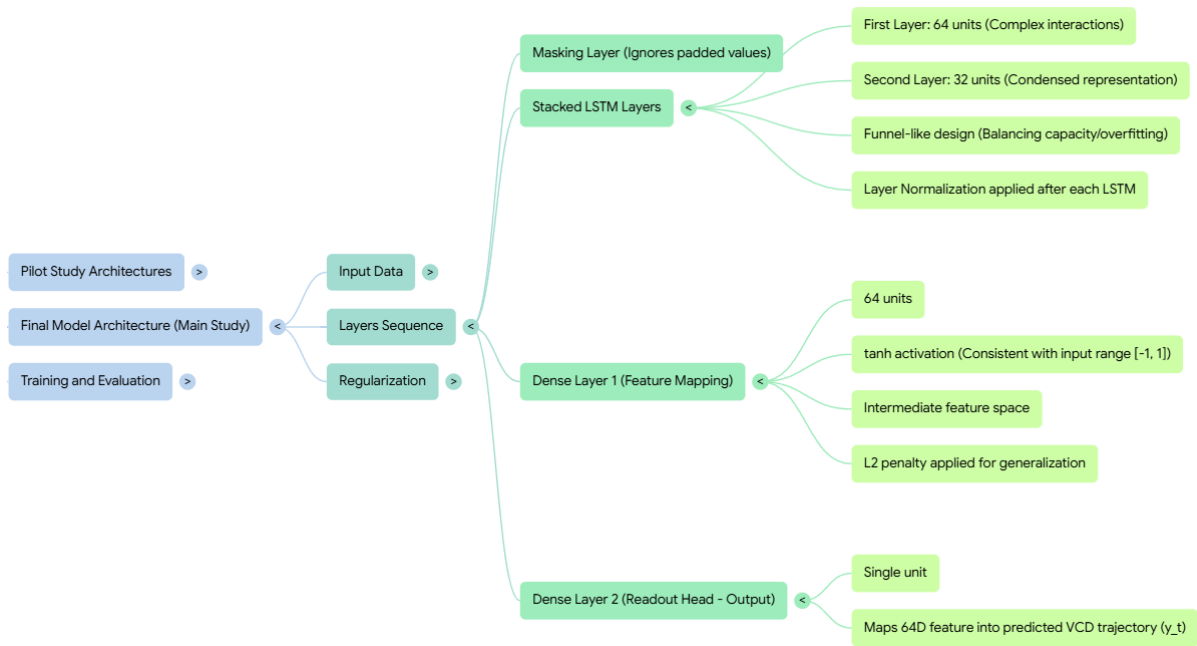


Figure 48: Project framework outlining the main study model architecture.

Dropout and recurrent dropout were applied at this stage to reduce the risk of overfitting (Mishra et al., 2020). The second LSTM layer, with 32 units, further transformed the temporal representation into hidden states h_{t2} , which were then passed to a dense layer with 64 units for feature mapping.

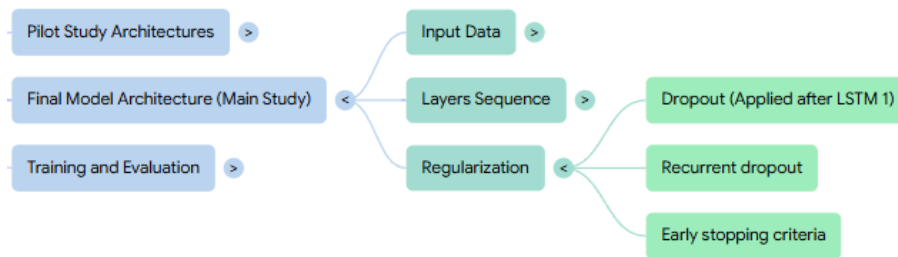


Figure 49: Project framework outlining the main study model regularisation mechanism.

The decision to use 64 units in the first LSTM layer and 32 units in the second was guided by the need to balance model capacity with the risk of overfitting (Dobrzycki, Bernardos & Casar, 2025). The first layer was assigned higher dimensionality to capture the complex temporal interactions among multiple process parameters. The second layer condensed this representation into a more compact form, ensuring that only the most relevant dependencies

were propagated forward. This funnel-like design reduced computational complexity while supporting model generalisation. Pilot experiments with alternative configurations (e.g. 32–16 or 128–64) resulted in either underfitting or overfitting, confirming that the 64 - 32 configuration offered the most effective trade-off.

Layer normalisation was applied after each LSTM layer to stabilise hidden state dynamics and enhance training efficiency. Unlike batch normalisation, which depends on batch-level statistics, layer normalisation operates across features within each timestep, making it more suitable for recurrent models. In time-series modelling, it mitigates vanishing or exploding gradients, reduces sensitivity to initialisation, and improves generalisation by providing stable temporal representations before dense layer processing (Ba et al., 2016; Tolic et al., 2024).

Following the LSTM layers, a dense layer with 64 units and tanh activation was included to map the temporal representations into a compact feature space. An explicit tanh activation was chosen here to remain consistent with the normalised input range $[-1, 1]$, ensuring stable gradient flow and preventing unbounded outputs. Introducing a dense layer with 64 units before the final output layer provided an intermediate feature space where regularization (via L2 penalty) could be applied to improve generalisation. Pilot experiments confirmed that inclusion of the dense layer improved performance metrics and reduced variance between training and validation losses.

Finally, a single-unit dense layer served as the readout head, mapping the 64-dimensional feature representation into outputs y_t at each timestep. This output layer translated the abstract temporal patterns learnt by the network into the predicted VCD trajectory over time. A schematic representation of the model architecture is provided in figure 50.

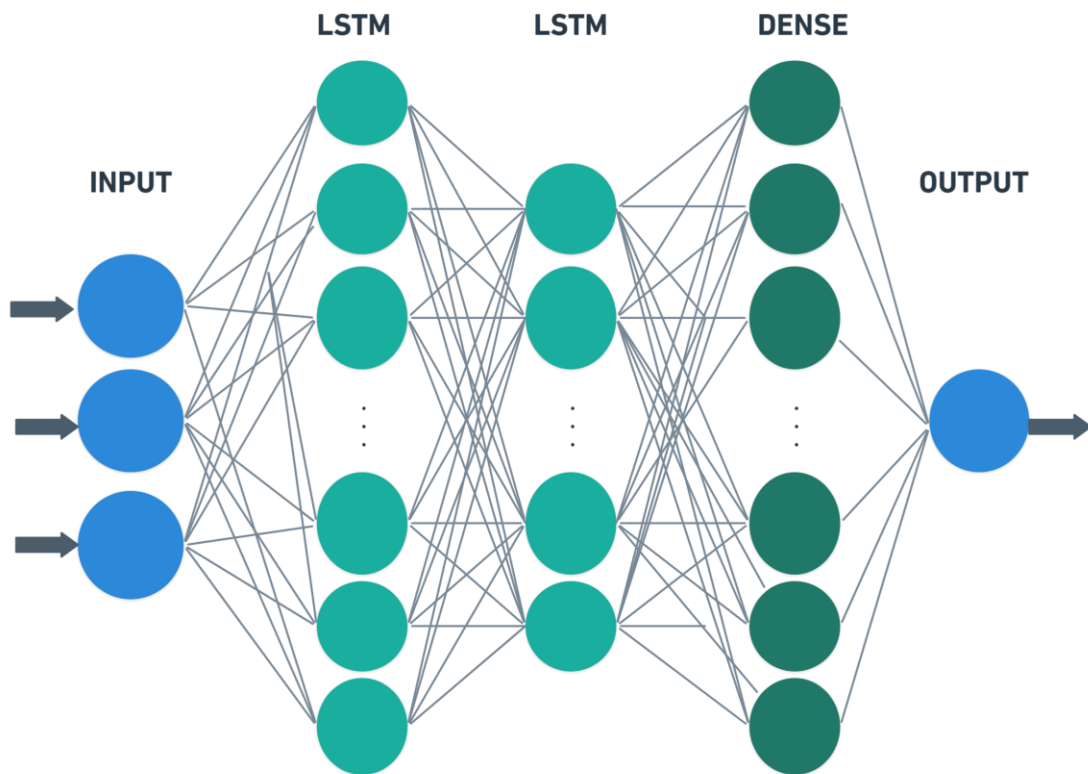


Figure 50: The structure of the final stacked LSTM model.

Hyperparameters were selected through iterative experimentation with the objective of balancing predictive accuracy and generalisation capacity. The parameters tuned included the number of LSTM units, batch size, dropout rate, and learning rate.

Moreover, the additional function that generated the sample-weight masks further improved model learning in the main study.

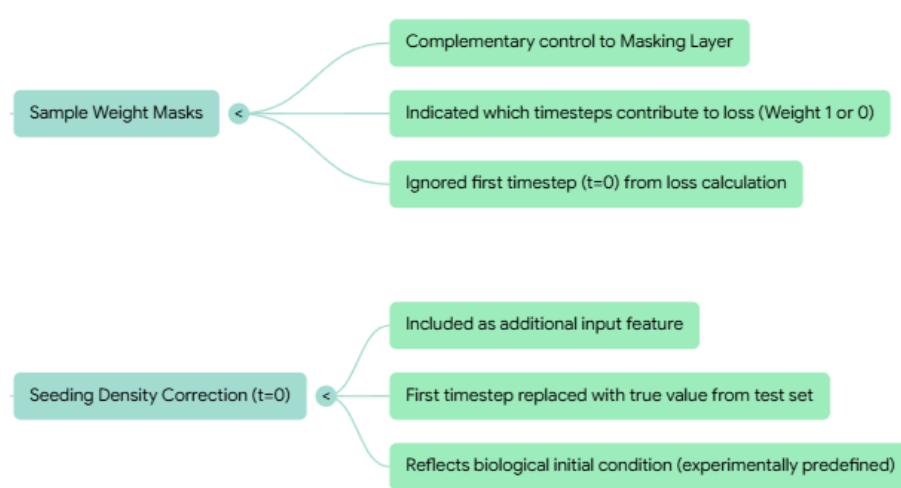


Figure 51: Project framework outlining the additional sample-weight masks function.

This function indicated which time steps in each sequence should contribute to the loss calculation during training, validation, and testing. It served as a complementary control to the Masking layer, which ensured that padded regions did not influence how temporal dependencies were learned.

```

def build_sample_weights(y, pad_value=-2.0, k=1):
    y2 = np.squeeze(y, axis=-1)
    valid = (y2 != pad_value)
    T = y2.shape[1]
    after_k = np.arange(T)[None, :] >= k
    return (valid & after_k).astype('float32')

sw_train = build_sample_weights(y_train, pad, k=1)
sw_val = build_sample_weights(y_val, pad, k=1)
sw_test = build_sample_weights(y_test, pad, k=1)
  
```

Figure 52: Code snippet showing sample-weight masks function.

The sample weights were applied during the loss computation stage, assigning a weight of 1 to valid time steps and 0 to padded or excluded ones. The function not only excluded padded values but also ignored the first time step ($t = 0$) from the loss calculation, as the model did not capture the initial point accurately. To retain this information, the seeding density (VCD at $t = 0$) was therefore included as an additional input feature and the first time step was replaced with the corresponding true value from the test dataset to ensure a smooth and continuous growth trajectory.

```

valid = (y_test[..., 0] != pad)

for i in range(predicted_vcd.shape[0]):
    idx = np.where(valid[i])[0]
    if idx.size:
        j = idx[0]
        predicted_vcd[i, j, 0] = y_test[i, j, 0]

predicted_vcd = predicted_vcd.squeeze(-1)
y_test = y_test.squeeze(-1)

```

Figure 53: Code snippet showing replacement of the VCD value at $t = 0$.

This correction reflects the biological meaning of the seeding density (VCD at $t = 0$), which is experimentally predefined and serves as an initial condition rather than a model-predicted variable.

Summary of model architectures are shown in table 3.

CHO Source Model	Model architecture	Dataset split	Optimizer	Batch Size	Epochs (max)	Loss function	Evaluation metrics	Dropout	Sample weighting
Pilot study 1st stage	LSTM, Masking Layer, Dense Layers	training (70%), validation (15%), and testing (15%)	Adam (default learning rate = 0.001)	8	100	MSE	MAE, RMSE	0.2	-
Pilot study 2nd stage	Stacked BiLSTM network, Masking Layer, Layer Normalization and Dense Layers	training (70%), validation (15%), and testing (15%)	Adam (default learning rate = 0.001)	8	100	MSE	MAE, RMSE	0.4-0.5	-
Main study	Stacked LSTM network, Masking Layer, Layer Normalization and Dense Layers	training (70%), validation (15%), and testing (15%, containing one unseen bioreactor scale)	Adam (default learning rate = 0.001)	8	100	MSE	MAE, RMSE	0.4-0.5	Exclude padded and initial ($t=0$) time steps from loss computation

Table 3: Summary of each model architecture

Training and Evaluation (CHO Source Model)

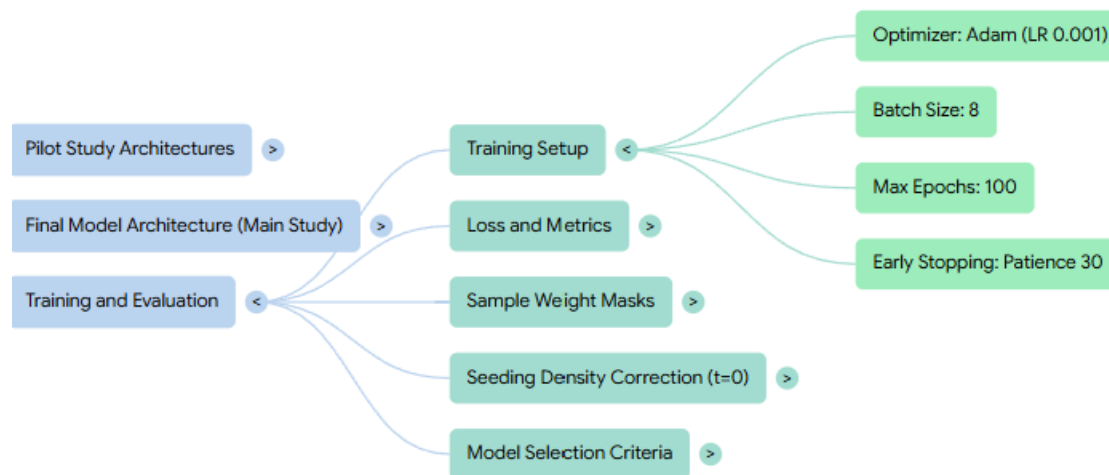


Figure 54: Project framework outlining the training setup

Training was conducted using the Adam optimizer with default parameters, including a learning rate of 0.001. A batch size of 8 was applied consistently across experiments. The maximum number of epochs was set to 100, with early stopping employed using a patience of 30 epochs and automatic restoration of the best weights. These settings were used for the pilot study and were reevaluated for the main dataset confirming their suitability.

```

model.compile(optimizer='adam', loss='mse', metrics=['mae'])

early_stop = EarlyStopping(patience=30, restore_best_weights=True)

history = model.fit(
    X_train, y_train,
    sample_weight=sw_train,
    validation_data=(X_val, y_val, sw_val),
    epochs=100,
    batch_size=8,
    callbacks=[early_stop]
)
  
```

Figure 55: Code snippet showing the model training.

The models were trained using MSE as the loss function, with MAE monitored as complementary performance metrics. To ensure consistent evaluation, padded timesteps corresponding to the masking value of -2.0 were excluded from metric calculations.

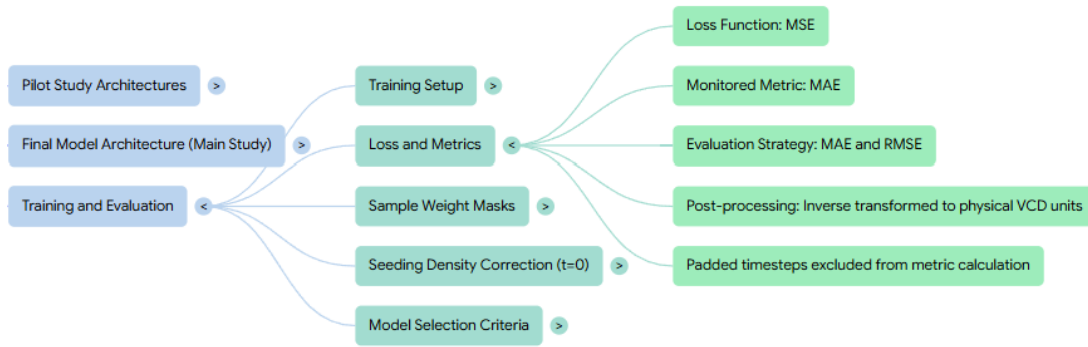


Figure 56: Project framework outlining the training metrics.

Model predictions were inverse transformed from the normalised range back into physical units of viable cell density prior to error computation, ensuring that reported metrics reflected biologically interpretable values.

```

n_samples, time_steps = predicted_vcd.shape
vcd_idx = list(scaler.feature_names_in_).index('vcd')

flat_pred = predicted_vcd.flatten()
flat_true = y_test.flatten()

dummy_pred = np.zeros((n_samples * time_steps, len(scaler.feature_names_in_)))
dummy_true = np.zeros_like(dummy_pred)

dummy_pred[:, vcd_idx] = flat_pred
dummy_true[:, vcd_idx] = flat_true

inv_pred = scaler.inverse_transform(dummy_pred[:, vcd_idx])
inv_true = scaler.inverse_transform(dummy_true[:, vcd_idx])

vcd_pred_original = inv_pred.reshape(n_samples, time_steps)
vcd_true_original = inv_true.reshape(n_samples, time_steps)

```

Figure 57: Code snippet showing inverse transformation.

The evaluation strategy was based on MAE and RMSE, both computed after inverse transformation of the predictions into physical units of viable cell density.

Global average error:

```
mask = y_test != -2.0

masked_true = vcd_true_original[mask]
masked_pred = vcd_pred_original[mask]

mse = mean_squared_error(masked_true, masked_pred)
rmse = np.sqrt(mse)
mae = mean_absolute_error(masked_true, masked_pred)

print(f"Test RMSE= {rmse:.2f} 10^6 cells/mL")
print(f"Test MAE= {mae:.2f} 10^6 cells/mL")
```

```
Test RMSE= 3.55 10^6 cells/mL
Test MAE= 2.66 10^6 cells/mL
```

```
rmse_p = (rmse / vcd_true_original.max()) * 100
mae_p = (mae / vcd_true_original.max()) * 100

print(f"RMSE = {rmse_p:.1f}% of the VCD max in test set.")
print(f"MAE = {mae_p:.1f}% of the VCD max in test set.")
```

```
RMSE = 9.6% of the VCD max in test set.
MAE = 7.2% of the VCD max in test set.
```

Figure 58: Project framework outlining the RMSE and MAE computation.

4.6 Transfer Learning to iPSC target domain

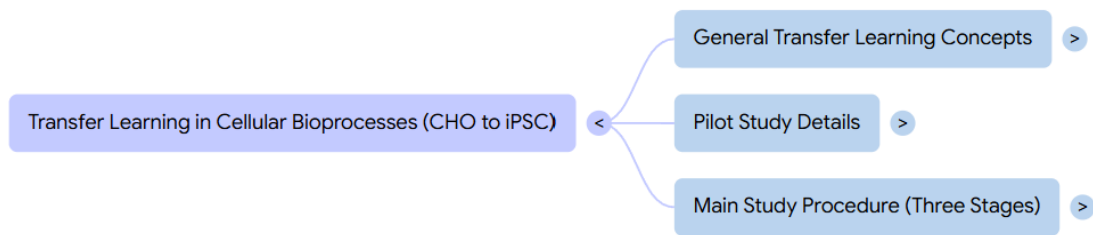


Figure 59: Project framework outlining the transfer learning.

The model developed on the CHO dataset was transferred to the iPSC dataset to assess its performance across different cell lines. The same preprocessing pipeline was applied to the iPSC data, including sequence padding, masking, and feature normalisation.

Transfer was implemented using a layer-freezing strategy. Freezing the initial layers reduced the number of trainable parameters, thereby lowering the risk of overfitting on the smaller iPSC dataset (Dobrzycki, Bernardos & Casar, 2025), while allowing efficient adaptation to the new task.

Pilot Study (iPSC target domain)

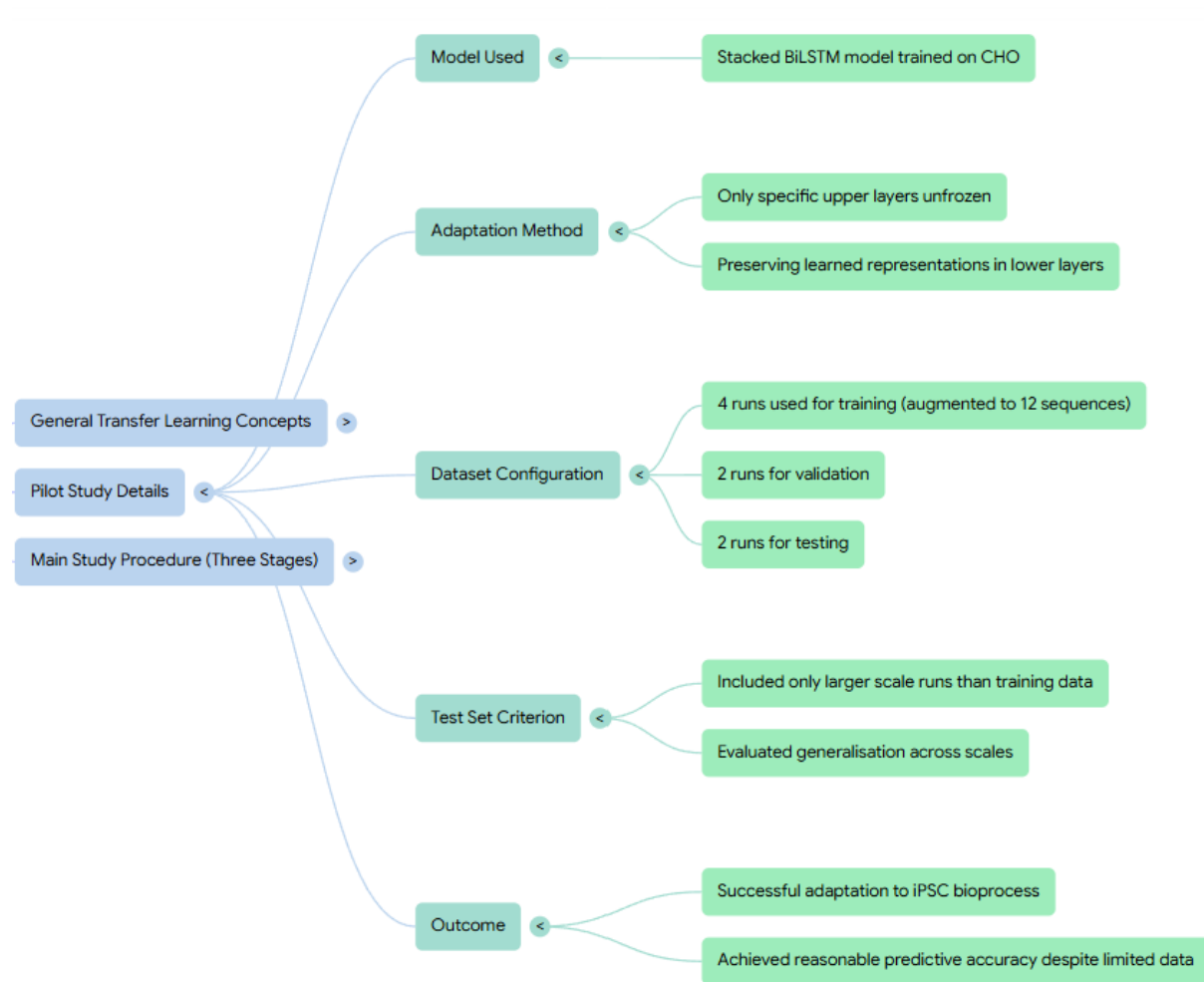


Figure 60: Project framework outlining the transfer learning in pilot study.

In the final stage of the pilot study, the trained CHO model was transferred to an iPSC dataset through a transfer learning approach to evaluate the reusability of learned temporal representations across mammalian systems. The stacked BiLSTM model trained on the CHO dataset was loaded and only specific upper layers were unfrozen to adapt pretrained model to new data while preserving the learned representations in lower layers.

Model: "sequential"

Layer (type)	Output Shape	Param #
masking (Masking)	(None, 12, 8)	0
bidirectional (Bidirectional)	(None, 12, 128)	37,376
layer_normalization (LayerNormalization)	(None, 12, 128)	256
dropout (Dropout)	(None, 12, 128)	0
bidirectional_1 (Bidirectional)	(None, 12, 64)	41,216
layer_normalization_1 (LayerNormalization)	(None, 12, 64)	128
dropout_1 (Dropout)	(None, 12, 64)	0
dense (Dense)	(None, 12, 64)	4,160
dense_1 (Dense)	(None, 12, 1)	65

Total params: 249,605 (975.02 KB)
 Trainable params: 83,201 (325.00 KB)
 Non-trainable params: 0 (0.00 B)
 Optimizer params: 166,404 (650.02 KB)

```

for layer in model.layers:
    if layer.name in ['bidirectional_1', 'layer_normalization_1', 'dense', 'dense_1']:
        layer.trainable = True
    else:
        layer.trainable = False

model.compile(optimizer='adam', loss='mse', metrics=['mae'])

from tensorflow.keras.callbacks import EarlyStopping

early_stop = EarlyStopping(patience=30, restore_best_weights=True)

history = model.fit(
    X_finetune, y_finetune,
    validation_data=(X_val, y_val),
    epochs=100,
    batch_size=8,
    callbacks=[early_stop]
)

```

Figure 61: Code snippet showing TL model architecture

The model was trained on data from 4 experimental runs (augmented to 12 sequences), with 2 runs for validation and 2 for testing. The test set included only larger-scale runs to assess cross-scale generalisation. The trained model was then adapted to the iPSC dataset, confirming the feasibility of knowledge transfer for the main study.

Main study (iPSC target domain)

In the main study the transfer procedure comprised three stages:

(1) Zero-shot evaluation:

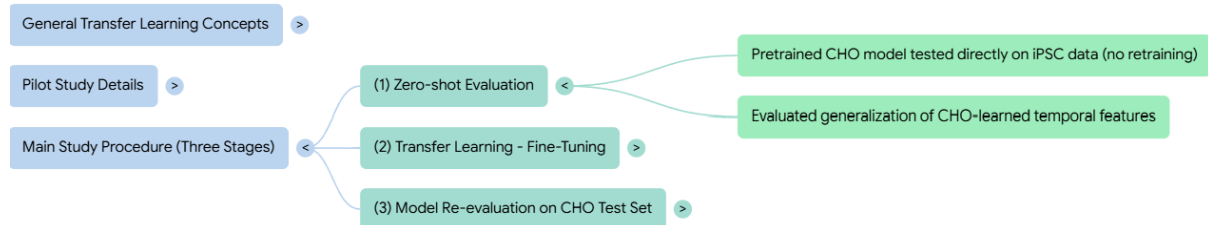


Figure 62: Project framework outlining the TL zero-shot evaluation.

The pretrained CHO model was directly tested on iPSC data without retraining to evaluate how well CHO-learned temporal features generalised across cell types and process scales.

```
model = load_model('model_main_study.keras')
model.summary()
```

Model: "sequential"

Layer (type)	Output Shape	Param #
masking (Masking)	(None, 14, 11)	0
lstm (LSTM)	(None, 14, 64)	19,456
layer_normalization (LayerNormalization)	(None, 14, 64)	128
dropout (Dropout)	(None, 14, 64)	0
lstm_1 (LSTM)	(None, 14, 32)	12,416
layer_normalization_1 (LayerNormalization)	(None, 14, 32)	64
dropout_1 (Dropout)	(None, 14, 32)	0
dense (Dense)	(None, 14, 64)	2,112
dense_1 (Dense)	(None, 14, 1)	65

Total params: 102,725 (401.27 KB)
Trainable params: 34,241 (133.75 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 68,484 (267.52 KB)

zero-shot evaluation

```
model.compile(optimizer='adam', loss='mse', metrics=['mae'])
```

```
model.evaluate(X_test, y_test, sample_weight=sw_test)
predicted_vcd = model.predict(X_test)
```

Figure 63: Code snippet showing the TL zero-shot evaluation.

(2) Transfer Learning - fine-tuning on small scale iPSC dataset:

The masking layer, the first LSTM layer (64 units), and the subsequent layer normalisation were frozen, preserving the representations learnt from the CHO dataset that capture general temporal dependencies (figure 64).

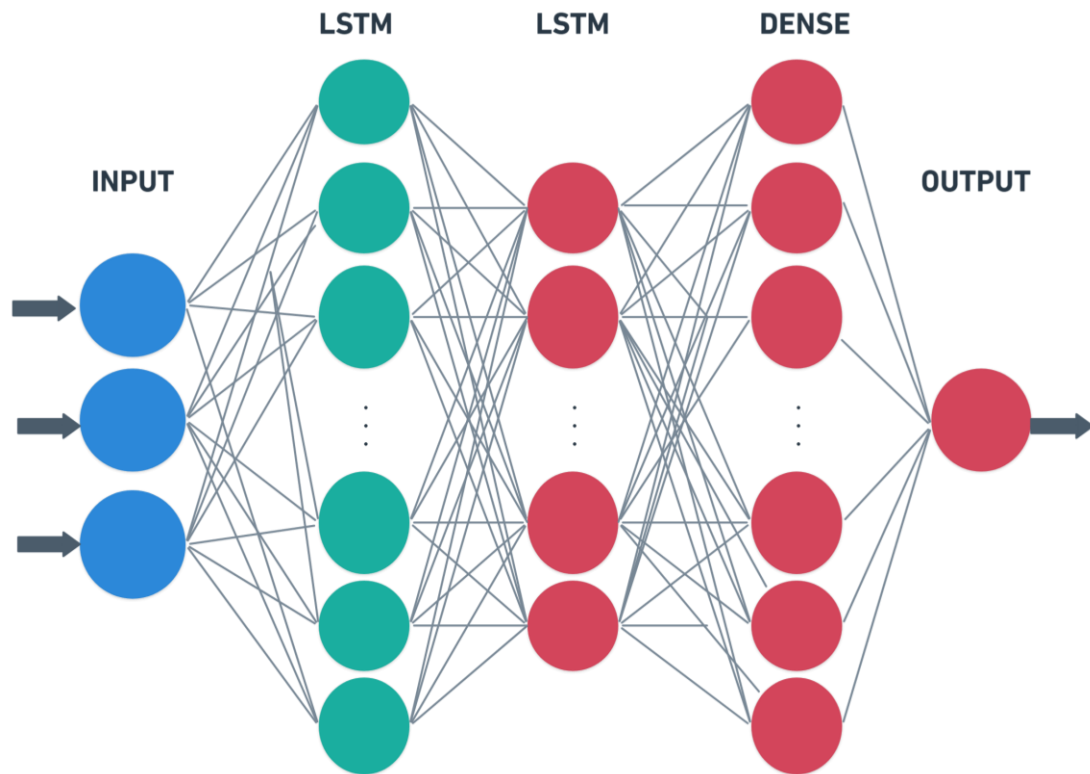


Figure 64: Topology of TL model.

The later layers, including the second LSTM with 32 units, the dense layer with 64 units, and the final output layer, were retrained.

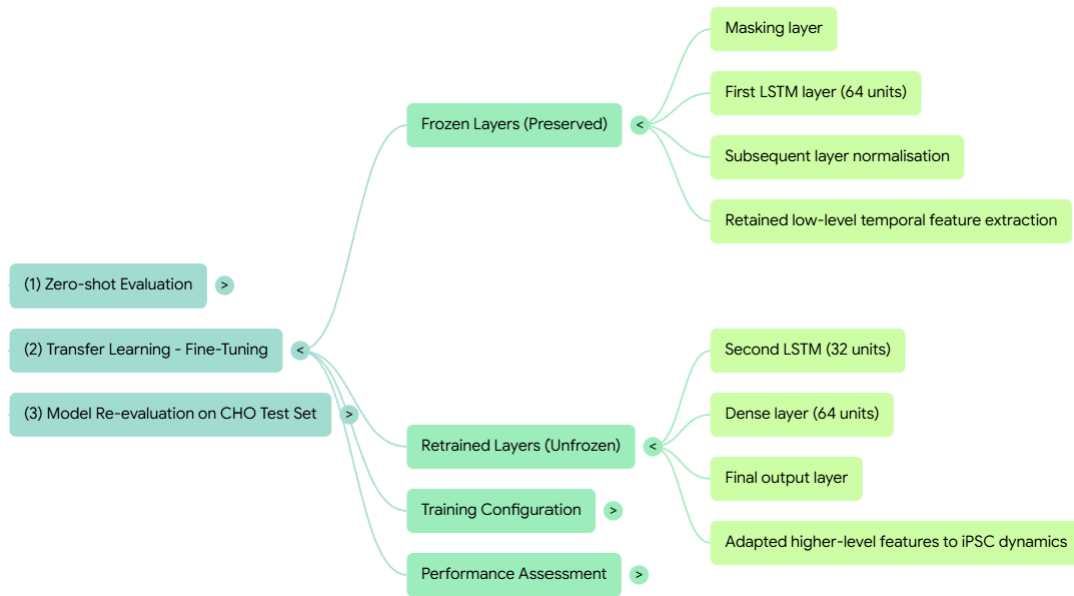


Figure 65: Project framework outlining the TL model architecture.

This allowed the model to adapt higher-level features to iPSC-specific dynamics while retaining low-level temporal feature extraction and fine-tuning higher-level abstractions to capture the specific growth dynamics of iPSC cultures. Comparable approaches, where only the final hidden and output layers were retrained, have demonstrated significant reductions in prediction error and uncertainty in bioprocess datasets (Rogers et al., 2022).

Model: "sequential"

Layer (type)	Output Shape	Param #
masking (Masking)	(None, 14, 11)	0
lstm (LSTM)	(None, 14, 64)	19,456
layer_normalization (LayerNormalization)	(None, 14, 64)	128
dropout (Dropout)	(None, 14, 64)	0
lstm_1 (LSTM)	(None, 14, 32)	12,416
layer_normalization_1 (LayerNormalization)	(None, 14, 32)	64
dropout_1 (Dropout)	(None, 14, 32)	0
dense (Dense)	(None, 14, 64)	2,112
dense_1 (Dense)	(None, 14, 1)	65

Total params: 102,725 (401.27 KB)
Trainable params: 34,241 (133.75 KB)
Non-trainable params: 0 (0.00 B)
Optimizer params: 68,484 (267.52 KB)

```
for layer in model.layers:
    if layer.name in ['lstm_1', 'layer_normalization_1', 'dense', 'dense_1']:
        layer.trainable = True
    else:
        layer.trainable = False
```

```
model.compile(optimizer='adam', loss='mse', metrics=['mae'])
```

```
early_stop = EarlyStopping(patience=30, restore_best_weights=True)
```

```
history_t11 = model.fit(
    X_train, y_train,
    sample_weight=sw_train,
    validation_data=(X_val, y_val, sw_val),
    epochs=60,
    batch_size=8,
    callbacks=[early_stop]
)
```

Figure 66: Code snippet showing the TL model.

Training followed the same procedure as for the CHO dataset, using the Adam optimizer, a batch size of 8, early stopping with a patience of 30 epochs, and MSE as the loss function with MAE as a complementary evaluation metric.

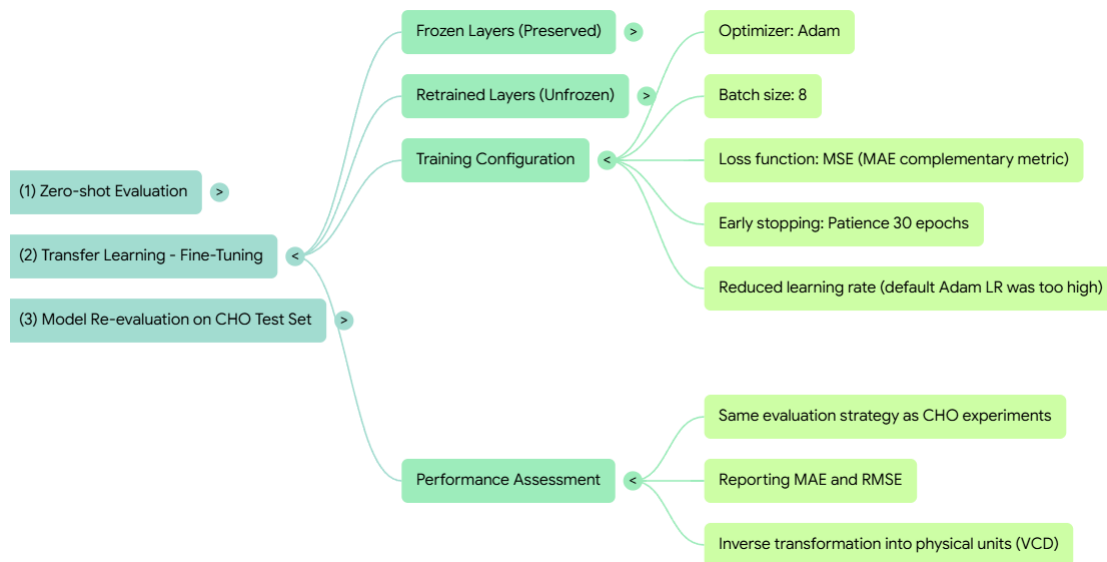


Figure 67: Project framework outlining the TL training configuration.

Importantly, the learning rate was reduced to 0.00001 after the initial trial-and-error run, as the model restored the best weights at epoch 0, indicating that the default Adam optimizer learning rate was too high.

Model performance was assessed with the same evaluation strategy as in the CHO experiments, reporting both MAE and RMSE after inverse transformation into physical units of VCD.

(3) Model re-evaluation on the initial CHO test set

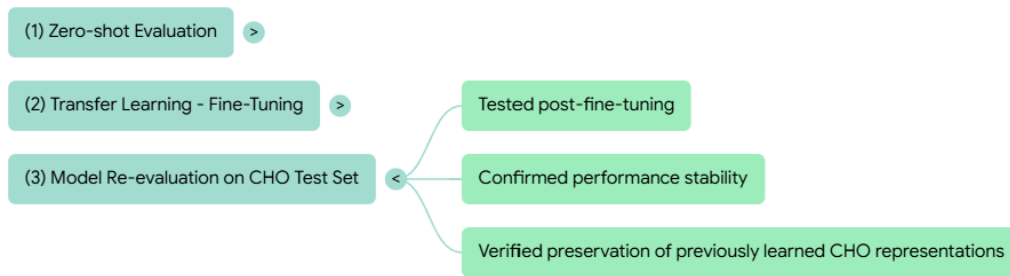


Figure 68: Project framework outlining the TL re-evaluation.

Following fine-tuning, the model was re-evaluated on the original source domain test set to confirm that performance remained stable and that previously learned representations were preserved.

```

from tensorflow.keras.models import load_model
model = load_model('model_main_study_tl1_low_LR.keras')
model.summary()

```

Model: "sequential"

Layer (type)	Output Shape	Param #
masking (Masking)	(None, 14, 11)	0
lstm (LSTM)	(None, 14, 64)	19,456
layer_normalization (LayerNormalization)	(None, 14, 64)	128
dropout (Dropout)	(None, 14, 64)	0
lstm_1 (LSTM)	(None, 14, 32)	12,416
layer_normalization_1 (LayerNormalization)	(None, 14, 32)	64
dropout_1 (Dropout)	(None, 14, 32)	0
dense (Dense)	(None, 14, 64)	2,112
dense_1 (Dense)	(None, 14, 1)	65

```

Total params: 63,557 (248.27 KB)
Trainable params: 14,657 (57.25 KB)
Non-trainable params: 19,584 (76.50 KB)
Optimizer params: 29,316 (114.52 KB)

```

```

model.compile(optimizer='adam', loss='mse', metrics=['mae'])

```

```

model.evaluate(X_test, y_test, sample_weight=sw_test)
predicted_vcd = model.predict(X_test)

```

Figure 69: Code snippet showing the TL re-evaluation.

5. Results

This chapter provides a critical analysis of the empirical results derived from the CHO and iPSC modelling stages. It examines the influence of the pilot study on the experimental design of the main analyses, evaluates predictive performance across model configurations, and assesses the effectiveness of transferring temporal representations between mammalian cell lines with distinct physiological and process characteristics. The discussion outlines how the pilot phase established the methodological feasibility and baseline performance of the framework, while the main study demonstrated successful knowledge transfer and enhanced cross-scale generalisation.

5.1 Role and Insights of the Pilot Study

To address the uncertainties surrounding the application of machine learning to bioprocess scale-up, a pilot study was conducted to evaluate feasibility, test the data workflow, and assess the model's capacity to represent cell growth dynamics.

The pilot study comprised three sequential stages of model development:

- (1) an initial LSTM model trained on CHO data,
- (2) an enhanced configuration integrating two BiLSTM layers and data augmentation, and
- (3) a transfer learning model applied to iPSC data.

The first model (stage 1) exhibited evident overfitting, effectively memorising training patterns, while failing to generalise to unseen data (figure 70). The training loss converged towards zero, whereas the validation loss remained higher and showed noticeable fluctuations. In addition, substantial run-to-run variability was observed despite identical hyperparameter settings and dataset composition, likely reflecting stochastic effects inherent to neural network training and the limited representational capacity of the baseline model.

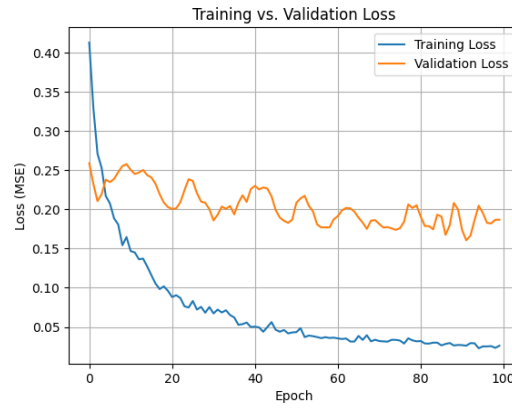


Figure 70: Loss function in 1st stage pilot.

To address overfitting and high variance observed in the baseline model, the second stage incorporated data augmentation and an optimised architecture comprising BiLSTM layers, layer normalisation, and additional dropout. These modifications enhanced convergence stability and predictive accuracy, as evidenced by smoother and more consistent validation loss trajectories (figure 71). The refined model demonstrated improved generalisation and revealed stronger temporal interdependence within the CHO growth data. Although the bidirectional design utilised non-causal information, the overall performance gains highlighted the presence of robust temporal correlations, supporting the suitability of the sequence-to-sequence LSTM framework for the main study.

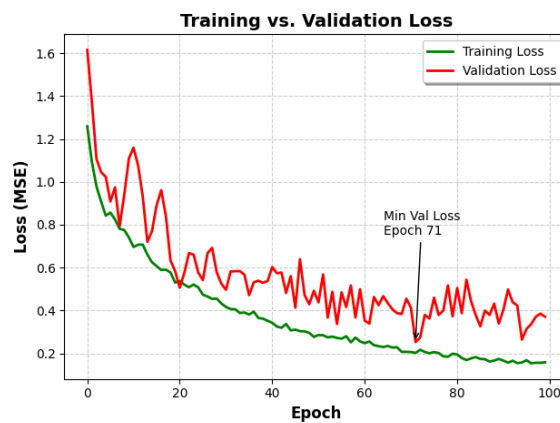


Figure 71: Loss function in 2nd stage pilot.

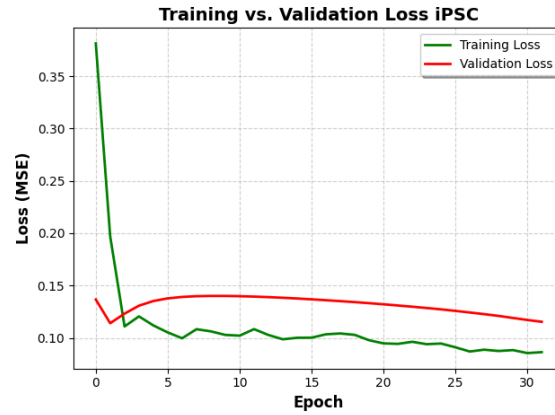


Figure 72: Loss function in 3rd stage pilot.

The final stage implemented transfer learning using a limited iPSC dataset. The model was fine-tuned on small-scale bioreactor runs and subsequently evaluated at a larger scale. Figure 72 shows, the loss curves remained relatively stable, indicating minimal overfitting despite the restricted data volume. Although the validation loss was consistently higher than the training loss, its gradual decline suggested satisfactory generalisation under data-constrained conditions.

To compare stages (1) and (2), figure 73 presents the model outputs excluding padded values from evaluation. Both configurations accurately reproduced the exponential growth phase (runs 1 and 8), showing strong agreement with experimental trends. However, neither model successfully predicted the initial seeding density at $t=0$, indicating a limitation in representing early-stage conditions within the input features.

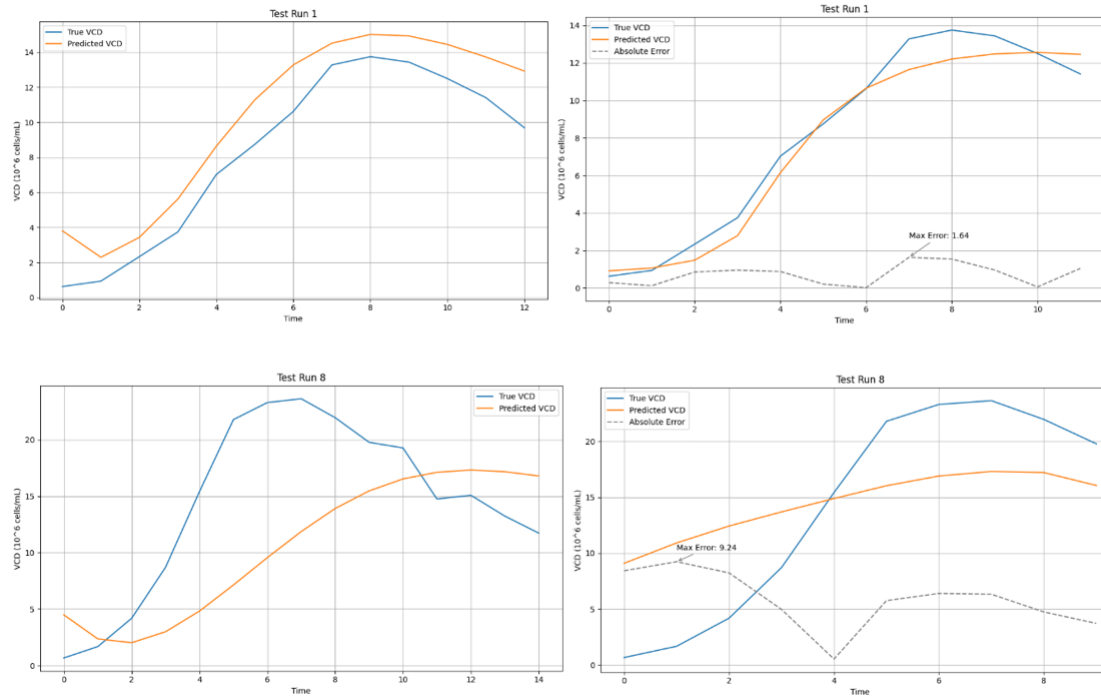


Figure 73: 1st (left) and 2nd (right) stage the model output.

The transfer learning stage successfully reproduced the iPSC training dynamics but demonstrated reduced predictive accuracy when extrapolated to unseen large-scale bioreactor conditions, as shown in figure 74.

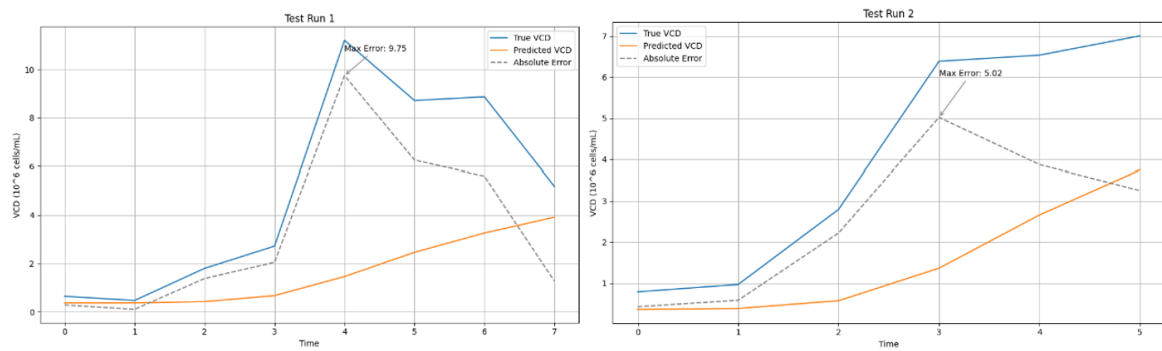


Figure 74: 3rd stage model output.

The pilot study established a validated foundation for the main investigation. The proposed framework effectively integrated exploratory analysis, data preprocessing, baseline CHO model training, and initial transfer to iPSC data. Quantitative evaluation revealed improved predictive accuracy in stage (2) compared with stage (1), while the transfer model stage (3) achieved reasonable yet less precise results. These outcomes confirm the feasibility of knowledge transfer between CHO and iPSC systems, though further optimisation and larger datasets are needed to enhance generalisation. The study also identified key architectural refinements, notably the advantage of bidirectional layers for learning temporal dependencies. However, as the bidirectional design relies on future time steps, it remains non-causal and unsuitable for real-time prediction or process control. This limitation motivated a shift to a causal, autoregressive LSTM in the main experiments, which further examined the transfer learning hypotheses derived from the pilot phase.

The corresponding results are summarised in table 4 and visualised in figure 75.

Pilot study model stage	Training Dataset	MAE [10^6 cells/mL]	RMSE [10^6 cells/mL]	MAPE	%RMSE
Initial LSTM	CHO	3.32	4.32	13.3%	17.3%
BiLSTM	CHO	2.04	3.01	8.2%	12.1%
Transfer Learning	iPSC	3.01	4.05	26.8%	36.1%

Table 4: Pilot study performance of each model architecture.

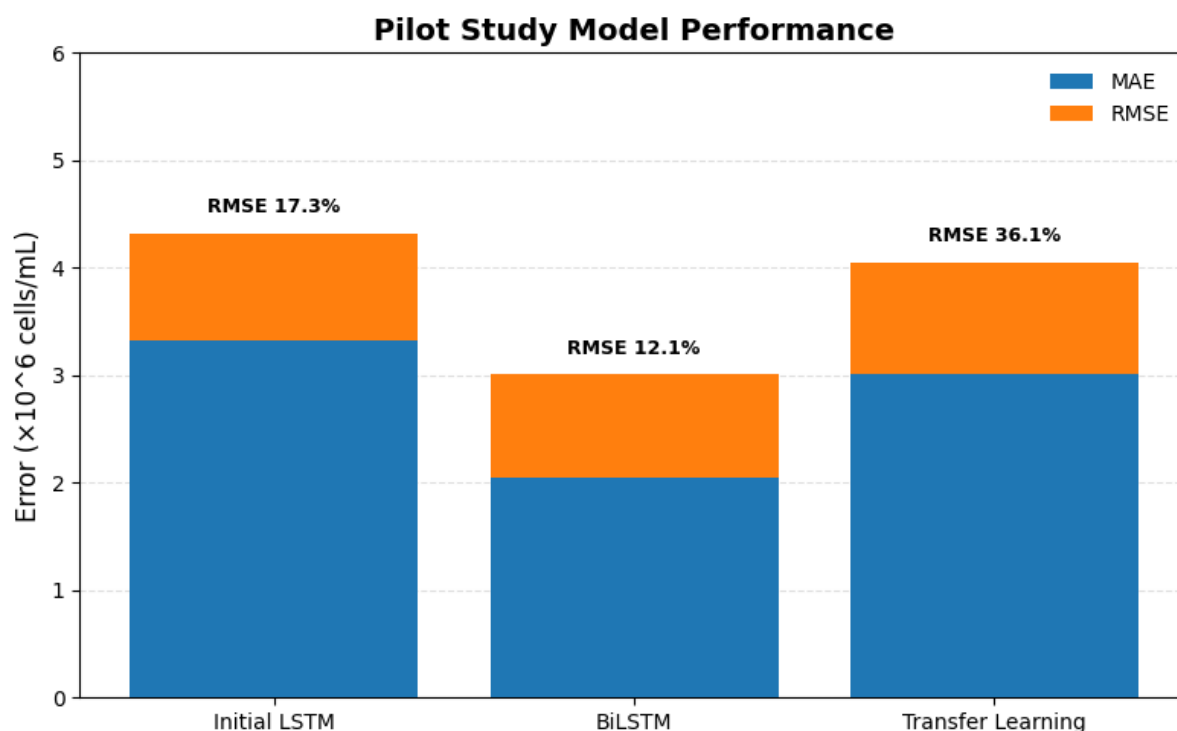


Figure 75: Comparison of pilot study models performance.

5.2 Quality and Validity of the Source Model (CHO) in the main study

Following the pilot phase, the quality and validity of the CHO source model used in the main study were evaluated to confirm its suitability for transfer learning. Model performance was assessed using training and validation loss trajectories alongside error metrics for VCD. Figure 77 shows, both loss curves exhibit a smooth and consistent decline throughout the epochs, with no indication of overfitting. The validation loss closely mirrors the training loss, demonstrating stable convergence and strong generalisation. The minimum validation loss occurred at epoch 98. No late-epoch drift was observed. During the initial epochs, a sharp decline in both losses reflects the model's rapid acquisition of key data relationships, while in the later phase the losses stabilised at a low MSE, indicating robust and well-regularised learning. The CHO model displayed stability, efficient learning dynamics, and high predictive accuracy, establishing a reliable foundation for subsequent transfer to iPSC systems.

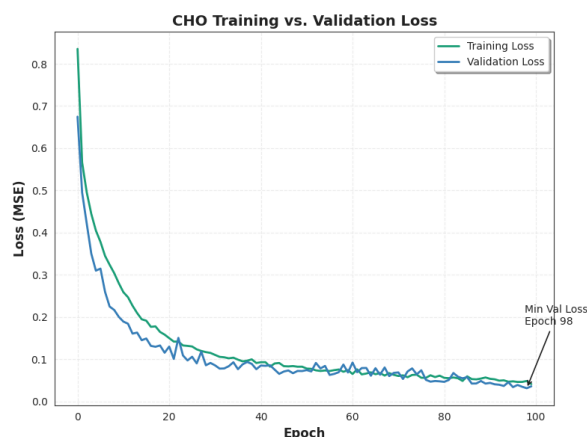


Figure 76: Main study loss curves for source domain.

Building on the loss analysis, the CHO model accurately captured the characteristic sigmoidal growth trajectory, reproducing the exponential rise and subsequent plateau. The predicted and observed VCD profiles showed strong alignment across most test runs, though minor overestimations appeared at higher cell densities near the saturation phase (figure 78). The absolute error curve indicates that deviations were most pronounced during the mid-to-late growth phases, whereas the overall trend indicates stable and generalisable predictive performance.

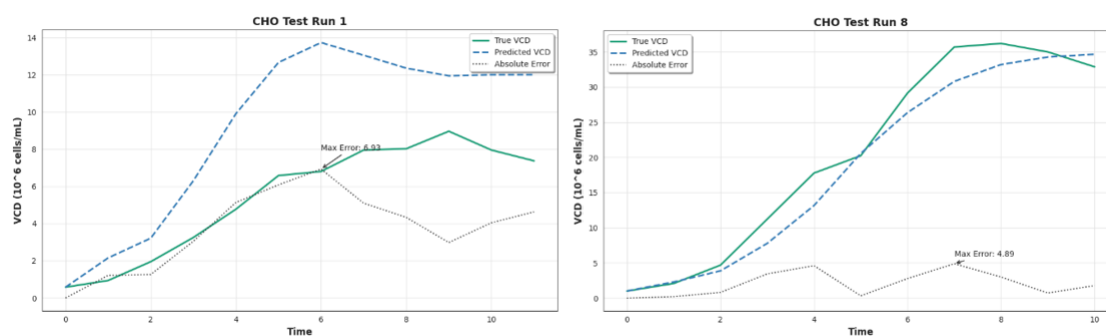


Figure 77: Main study model output for source domain.

The model achieved a global test RMSE 9.6% and MAE of 7.2% of the maximum VCD, indicating high predictive accuracy. While performance remained robust across runs, moderate variability was observed, likely reflecting bioprocess heterogeneity and clone-specific growth behaviour (figure 79).

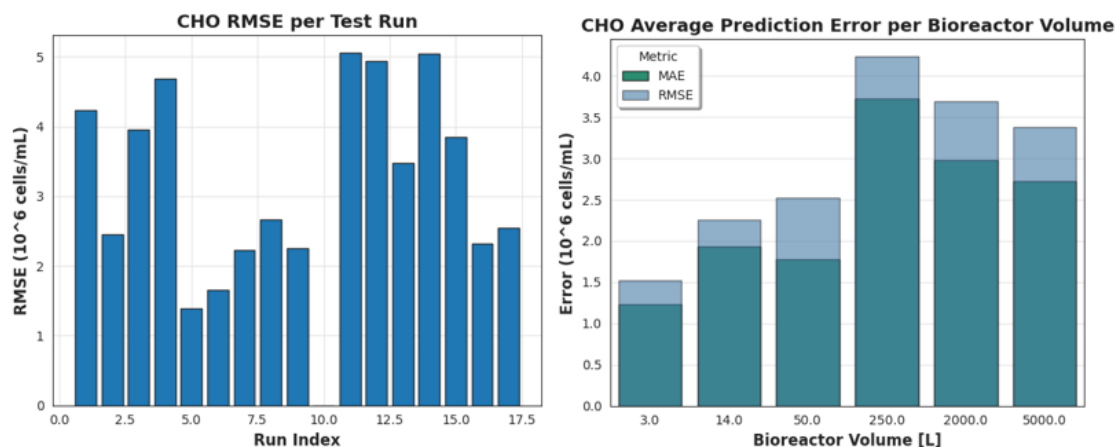


Figure 78: Main study source model error comparison.

When assessed across multiple working volumes (figure 79), the model maintained consistent predictive accuracy, with MAE values ranging from 1.2×10^6 to 3.7×10^6 cells/mL. The best performance was observed at smaller scales (3 L–50 L), with a slight decline at intermediate volumes, likely reflecting scale-specific process variability. The weakest accuracy occurred at the 250 L scale. Remarkably, the model accurately reproduced VCD trajectories for the 5000 L bioreactor, a scale absent from the training data with performance comparable to known volumes. This outcome suggests that the model captured transferable temporal and process relationships rather than memorising scale-dependent features. The inclusion of physically meaningful scale descriptors likely supported this capability, enabling effective extrapolation to unseen operating conditions. Such cross-scale generalisation is essential for predictive modelling in process development and scale-up, indicating that the trained network successfully encoded scalable bioprocess behaviour.

The CHO model demonstrated high predictive accuracy and effectively captured key biological growth features, including the exponential rise and transition to the stationary phase. The consistent overestimation near the plateau likely reflects unmodelled process heterogeneity or metabolic constraints. Its strong generalisation across scales and runs indicates that the network learned core bioprocess principles rather than dataset-specific patterns. Collectively, these results validate the CHO LSTM as a robust foundation for transfer learning to iPSC data, enabling efficient reuse and refinement of temporal and process representations.

5.2 Cross-Domain Evaluation

Building on the validated CHO model, a zero-shot evaluation was performed to quantify the domain gap between CHO and iPSC systems. The pretrained CHO model was applied directly to the iPSC test dataset without any architectural modifications or additional training. This assessment aimed to examine the model's ability to generalise learned temporal dependencies and process relationships to a biologically distinct system.

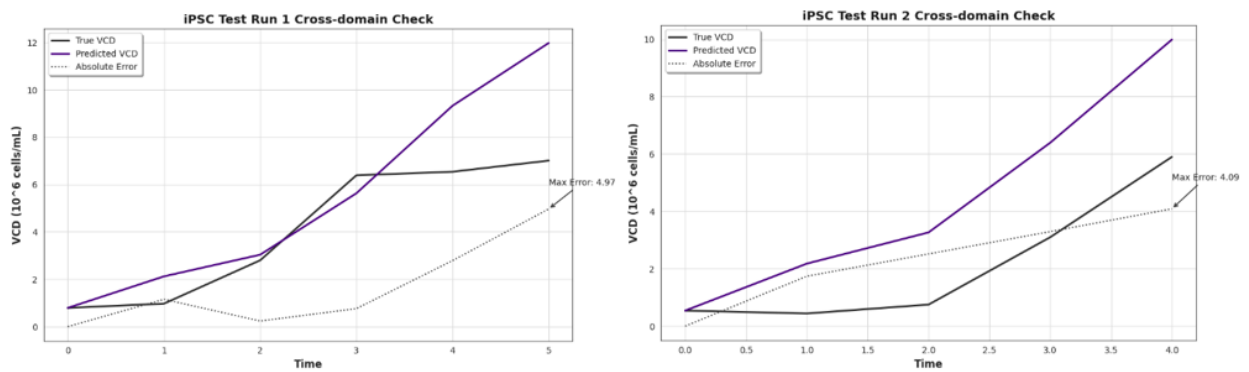


Figure 79: Main study model output for zero-shot evaluation.

Figure 80 shows, the model reproduced the overall iPSC growth trajectory but consistently overestimated cell densities, particularly toward the end of culture. Prediction errors reached 27.9% MAE and 36.3% RMSE relative to the maximum VCD, with slightly smaller deviations at the 6 L scale (figure 81).

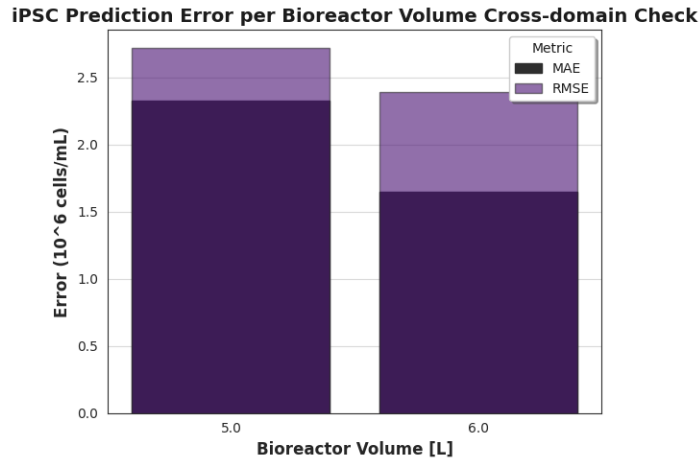


Figure 80: Zero-shot evaluation error.

Despite being trained on a different cell line, the model reproduced key temporal dynamics, demonstrating partial cross-domain generalisation. This indicates that the recurrent layers learned fundamental temporal relationships, such as the transition from exponential to stationary growth, shared across mammalian systems. However, the consistent overestimation of cell density reflects intrinsic biological and kinetic differences between CHO and iPSC cultures, likely arising from variations in growth rates and process control strategies. These discrepancies reveal a measurable domain gap that constrains zero-shot performance and highlight the need for domain adaptation to refine the learned feature space and improve predictive accuracy for iPSC systems.

5.3 Fine-Tuned Model Performance on iPSC

To extend the model from CHO to iPSC growth prediction, transfer learning was applied using a small-scale iPSC dataset. The pretrained CHO model was fine-tuned with frozen early layers and a reduced learning rate to preserve previously learned temporal representations and prevent overfitting.

Interestingly, the initial fine-tuning attempt performed on the small-scale dataset without learning-rate adjustment effectively reproduced the zero-shot evaluation. When the same pretrained weights were re-evaluated within the full transfer-learning pipeline, using identical sequence construction, masking, normalisation, and autoregressive rollout, early stopping restored the best epoch at 0 (figure 82), indicating the absence of beneficial parameter updates.

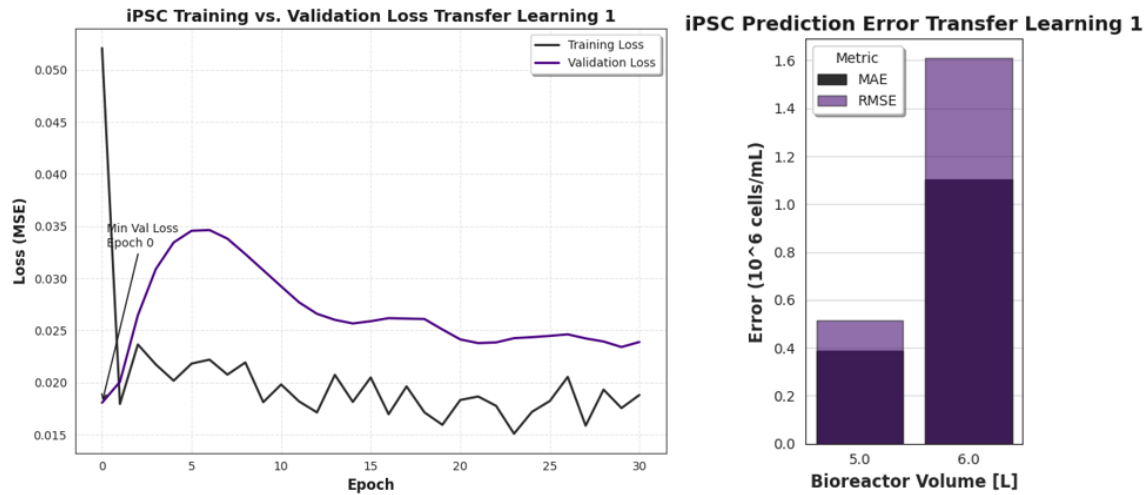


Figure 81: Loss curves and model error of main study transfer learning (1).

Despite this, performance improved markedly, achieving 17.4% RMSE and 11.3% MAE of the maximum VCD (figure 86). The predicted growth trajectories closely matched the true VCD profiles over time, particularly across the exponential phase (figure 83).

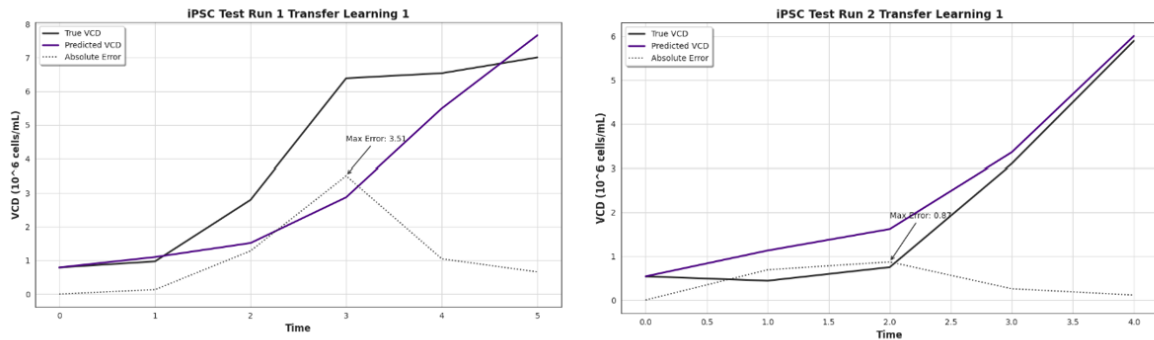


Figure 82: Main study model output for transfer learning (1).

Although both the true zero-shot and the frozen-layer evaluations used identical pretrained weights and preprocessing, the latter was recompiled for inference with the lower LSTM layers frozen. This configuration yielded a substantially more accurate result. The improvement reflects a stabilisation of inference behaviour rather than an update of learned parameters,

indicating that the pretrained CHO LSTM had already captured transferable temporal representations suitable for forecasting iPSC growth dynamics.

In the next stage of transfer learning, the learning rate was substantially reduced to preserve the pretrained temporal representations derived from CHO data. Using a default learning rate led to catastrophic forgetting, where previously learned features were rapidly overwritten, resulting in poorer validation performance (figure 82). Conversely, the lower learning rate allowed gradual and stable adaptation to iPSC-specific dynamics while maintaining the underlying growth behaviour, an essential consideration when fine-tuning on a limited dataset (figure 84).

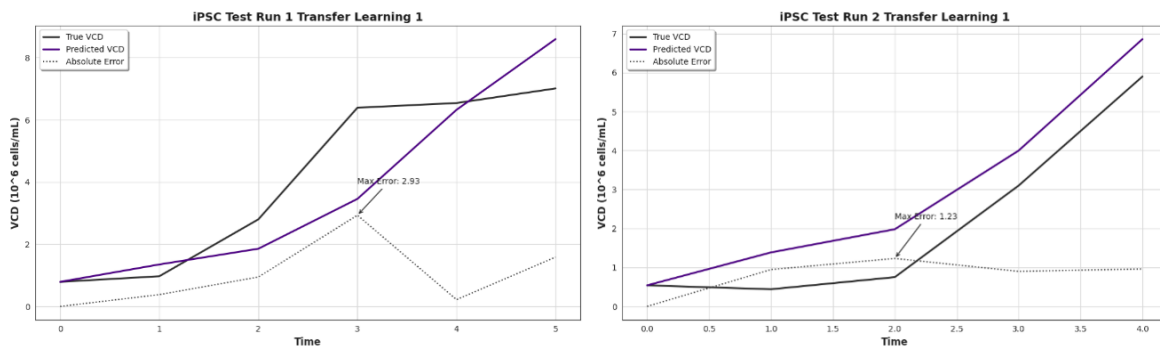


Figure 83: Main study model output for transfer learning (2).

When the learning rate was reduced by two orders of magnitude, the validation loss decreased steadily across training, while the training loss remained variable due to the limited size of the fine-tuning dataset (figure 85). This behaviour reflects stable and progressive adaptation to iPSC-specific growth dynamics without catastrophic forgetting of the CHO-learned temporal representations. In contrast to the high-learning-rate configuration, which rapidly degraded validation performance, the reduced-learning-rate setup enabled controlled refinement of the model while preserving its capacity to predict long-horizon growth trajectories. The minimum validation loss was achieved only after extended fine-tuning, around epoch 99.

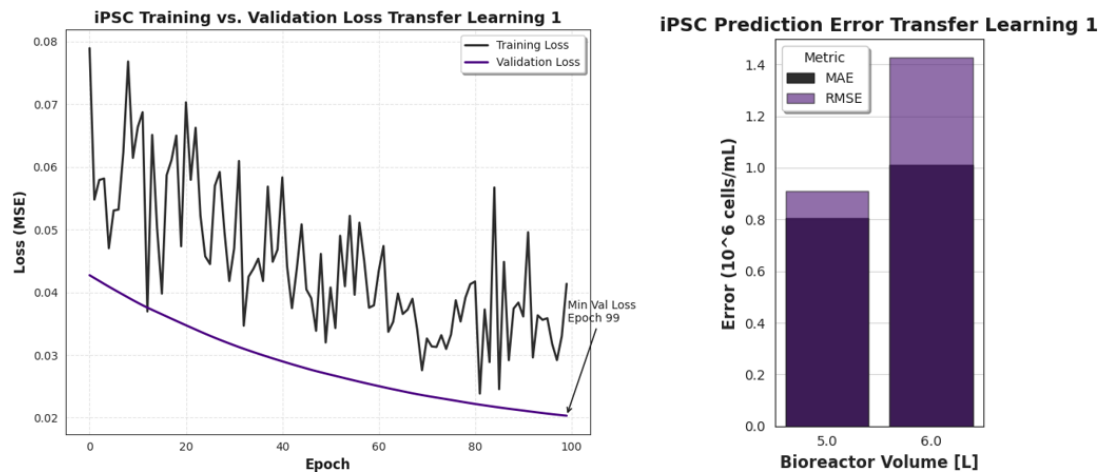


Figure 84: Loss curves and model error of main study transfer learning (2).

With the reduced learning rate applied during fine-tuning, the pretrained CHO model achieved a test RMSE of 17.4% and MAE of 13.1% relative to the maximum VCD in the iPSC test set (figure 86). These results represent a substantial improvement over the initial zero-shot evaluation and confirm that the pretrained temporal representations were effectively adapted to the specific growth dynamics of iPSC cultures.

5.4 Comparison of Predictive Performance on the Target Cell Line

Building on the fine-tuning results, figure 86 compares the prediction accuracy of the CHO-pretrained model on iPSC data across the three stages of transfer learning. The lower error observed after fine-tuning demonstrates that conservative adaptation allowed the model to retain stable long-horizon autoregressive behaviour while partially adjusting to iPSC-specific growth dynamics.

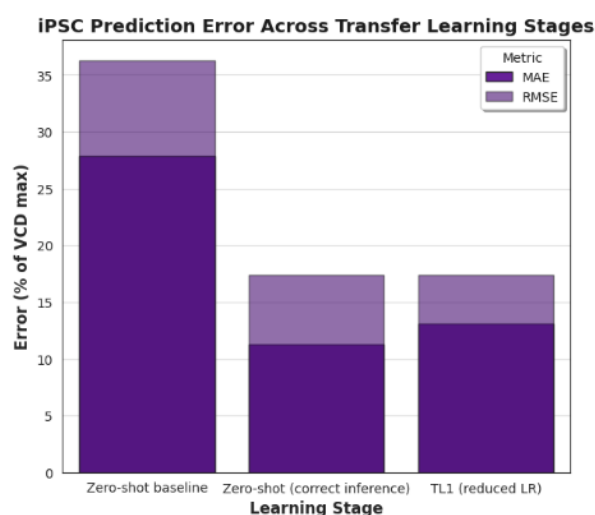


Figure 85: Comparison of predictive performance of the main study model on the target domain.

Furthermore, after fine-tuning on iPSC data, the CHO model was re-evaluated on its original test set and achieved an RMSE of 9.0% and MAE of 6.3% of the maximum VCD (table 5), values comparable to the original CHO performance. This supports that the model preserved its learned representations without exhibiting catastrophic forgetting. The consistent predictive accuracy across scales suggests that core growth kinetics and metabolic transitions are governed by conserved temporal dependencies shared among mammalian cell systems.

Main study model stage	Test Dataset	MAE [10^6 cells/mL]	RMSE [10^6 cells/mL]	MAPE	%RMSE
LSTM	CHO	2.66	3.55	7.2%	9.6%
LSTM zero-shot	iPSC	1.96	2.55	27.9%	36.3%
Zero-shot Transfer	iPSC	0.79	1.22	11.3%	17.4%
Fine-tuned Transfer Learning	iPSC	0.92	1.22	13.1%	17.4%
CHO Cross-check (after fine-tuning)	CHO	2.31	3.3	6.3%	9.0%

Table 5: Performance of each model used in main study.

Across bioreactor volumes, however, model performance varied notably (figure 81, 82, 85), reflecting scale-specific data representation. As the model was fine-tuned primarily on smaller-scale data, it may not have fully generalised to the distinct flow and mass transfer regimes of the larger setup. Moreover, the absence of key scale-dependent features, such as k_La and some reactor geometry ratios limited the model's ability to capture underlying mechanistic differences, contributing to residual errors. Including these physical parameters in future iterations would likely strengthen cross-scale fidelity.

Representative growth trajectories confirmed accurate prediction of lag and exponential phases across iPSC runs, though minor over- or underestimation persisted depending on scale. The limited number of iPSC experiments constrained the model's ability to fully specialise to the target domain. Nevertheless, given the strong zero-shot generalisation, expanding the small-scale iPSC dataset would likely reduce prediction uncertainty and enhance scalability.

Finally, the experimental results underscored the suitability of the causal autoregressive LSTM design. This architecture supports stepwise forecasting, using VCD from the previous time point as input, an approach consistent with real-time bioprocess monitoring. The stacked recurrent structure effectively captured both short-term responses to process fluctuations and long-term growth dynamics, while sequence masking naturally accommodated missing or irregular measurements as an essential advantage given the practical constraints of biological data collection.

The results demonstrate that the CHO-trained LSTM framework can be effectively adapted to a distinct mammalian cell system through conservative transfer learning. While performance remains constrained by dataset size and missing scale descriptors, the approach successfully captures conserved bioprocess dynamics, validating its potential as a scalable predictive tool for bioprocess modelling.

6. Conclusion

This study aimed to develop a predictive modelling framework capable of transferring knowledge across mammalian cell lines and bioreactor scales. A multi-scale dataset was constructed from publicly available literature, focusing on the CHO cell line, the industry benchmark for large-scale biomanufacturing.

Using this dataset, an LSTM-based RNN was trained to predict viable cell density trajectories across process scales from 15 mL to 5000 L. The model successfully captured general mammalian growth dynamics and reproduced cell proliferation behaviour across clones and reactor scales. Its stacked LSTM architecture, combined with layer normalization and dropout, effectively modelled the complex, non-linear nature of cell growth, confirming suitability for transfer learning.

Fine-tuning the CHO-trained model on a small iPSC dataset achieved successful cross-domain transfer, with accurate generalisation to unseen scales. This demonstrates that learned temporal features can bridge distinct mammalian systems and that meaningful predictions can be achieved without large target-domain datasets, supporting the goal of reducing dependence on extensive experimental data.

Beyond technical performance, these findings have significant scientific and industrial relevance, aligning with the Pharma 4.0 vision of data-driven bioprocessing. The model offers a predictive framework for scale-up, supporting planning and decision-making before costly experiments. By identifying transferable process relationships, it enhances process understanding, reproducibility, and the efficiency of early-stage iPSC development.

This study represents a novel contribution to the field of bioprocess modelling by demonstrating the feasibility of applying cross-cell-line transfer learning between mammalian systems using an LSTM-based framework. While existing predictive models are largely restricted to a single system, this study successfully transferred temporal features learned from CHO to iPSC processes. By combining deep learning with engineering-based feature selection, it bridges a critical knowledge gap in bioprocess scale-up and provides a foundation for data-efficient modelling of emerging cell systems in regenerative medicine and biomanufacturing.

The positive outcomes of this study are accompanied by several limitations and challenges that could be addressed through further research. The analysis relied on publicly available datasets, which often lack high-quality, well-annotated data. The available records are dominated by

successful experimental runs, reducing variability and narrowing the design space for model training. Nevertheless, the compiled dataset is extensive, covering numerous scale-up cases and including less successful runs, thereby increasing representativeness and practical relevance.

The developed model was also sensitive to preprocessing parameters such as sequence length and padding values. Standardising time-series lengths inevitably reduced temporal continuity, yet the chosen configuration effectively balanced run comparability and trajectory representation.

A further challenge concerned feature availability. Several biologically relevant variables were underreported or absent across studies. Although the model performed well with available data, incorporating additional process descriptors would likely enhance predictive and explanatory power. Therefore, establishing standardised, open-access datasets and promoting data-sharing initiatives across academia and industry would significantly enhance reproducibility and knowledge exchange, facilitating the development of more efficient bioprocess scale-up strategies.

Another key observation concerns the dependence of transfer learning performance on domain similarity. While both CHO and iPSC systems represent mammalian cell lines, they differ substantially in physiology and process behaviour. iPSCs, unlike CHO cells, tend to form aggregates that influence mixing efficiency, shear stress exposure, mass transfer, growth kinetics, oxygen uptake rate, and kLa relationships. These differences highlight the potential of integrating hybrid modelling approaches that combine mechanistic with data-driven layers. Prior research has demonstrated that such hybrid models can substantially improve prediction accuracy and interpretability compared with purely data-driven methods (Polak et al., 2024).

Model interpretability remains another important consideration. LSTM-based architectures are often perceived as black-box models, limiting insight into how predictions are formed. Integrating explainability tools, such as SHAP analysis, could help identify key feature contributions and guide variable selection in future studies.

Future research should focus on expanding datasets to include a wider range of CHO and iPSC process scales and operating conditions, incorporating mechanistic variables, such as shear stress, energy dissipation rate, mixing time, and testing alternative RNN architectures alongside other modelling approaches.

Summarising, this research demonstrated that data-driven models, grounded in engineering principles, can accelerate process understanding and enable predictive, resource-efficient scale-up of complex mammalian systems. These results validate the proof-of-concept hypothesis, demonstrating that machine learning can effectively capture transferable process dynamics and significantly reduce experimental burden, offering a scalable and data-efficient pathway for predictive bioprocess scale-up in data-scarce environments. The approach exemplifies a forward-looking perspective aligned with Biopharma 4.0, where data-centric process development fosters innovation, efficiency, and sustainability across modern biomanufacturing.

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Appendices.

A. Ethical Consideration

The first stage of the project focused on collecting secondary data from publicly available repositories and scientific literature to compile structured, multi-scale datasets for both CHO and iPSC bioprocesses. These datasets were aligned to ensure compatibility of input features, enabling effective model transfer and cross-system comparison. The datasets contained no identifiable personal or institutional information, eliminating any confidentiality risks.

The research adhered to recognised ethical principles, emphasising integrity, accuracy, transparency, and respect for intellectual property, thereby maintaining compliance with academic and professional standards.

Data integrity, transparency, and reproducibility were prioritised throughout the study. Each stage of preprocessing, model development, and evaluation was documented to ensure that the results could be independently validated and replicated.

B. List of Abbreviations

Adam – Adaptive Moment Estimation (optimizer)

ANN – Artificial Neural Network

ANNs – Artificial Neural Networks

BiLSTM - Bidirectional Long Short-Term Memory

CatBoost – Gradient boosting library

CHO – Chinese Hamster Ovary (cells)

DO – Dissolved Oxygen

EDA – Exploratory Data Analysis

FFNN – Feedforward Neural Network

GRU – Gated Recurrent Unit

HEK293T – Human Embryonic Kidney 293T cells

IoT – Internet of Things

iPSC – induced Pluripotent Stem Cell

kLa – Volumetric Oxygen Transfer Coefficient

LSTM – Long Short-Term Memory (network)

MAE – Mean Absolute Error

MAPE – Mean Absolute Percentage Error

ML – Machine Learning

MSE – Mean Squared Error

Np – Impeller Power Number

ODEs – Ordinary Differential Equations

P/V – Power per Unit Volume

PAT – Process Analytical Technology

PCA – Principal Component Analysis

pCO₂ – Partial Pressure of Carbon Dioxide

pO₂ – Partial Pressure of Oxygen

Re – Reynolds Number

ReLU – Rectified Linear Unit (activation function)

RF – Random Forest

RL – Reinforcement Learning

RMSE – Root Mean Squared Error

RNN – Recurrent Neural Network

SHAP – SHapley Additive exPlanations

STR - Stirred-Tank Reactor

SVR – Support Vector Regression

TL – Transfer Learning

VCD – Viable Cell Density

vvm – Volume of Gas per Volume of Liquid per Minute (volumetric gas flow rate)

XGB – Extreme Gradient Boosting

XGBoost – Extreme Gradient Boosting (library)

μ (mu) – Dynamic Viscosity

ρ (rho) – Fluid Density

C. Definitions

Clone

A clone refers to a genetically distinct mammalian cell line originating from a single progenitor cell. Each clone is characterised by specific growth kinetics, metabolic activity, and productivity profiles.

iPSC (Induced Pluripotent Stem) Cells

iPSCs are somatic cells that have been reprogrammed into a pluripotent state, allowing them to self-renew and differentiate into multiple cell types. They are increasingly used in regenerative medicine, disease modelling, and emerging biomanufacturing applications. In this study, iPSCs serve as the target system for model adaptation, representing a more complex and less characterised mammalian cell type.

Process

In this study, a process refers to a complete cell cultivation workflow representing the same biological system operated across multiple bioreactor scales. Each process captures the progression of cell growth under a consistent set of operating principles, such as feeding strategy and seeding density, while varying scale-dependent parameters.

STR (Stirred-Tank Reactor)

An STR is a type of bioreactor commonly used for cultivating mammalian, microbial, or stem cell cultures under controlled conditions. In this study, STRs are the primary cultivation systems across scales, providing time-resolved bioprocess data used for modelling VCD and assessing scale-dependent effects.

D. Code Repository Overview

GitHub link: https://github.com/pmatkowska94/MSC_Data_Science/tree/main

Pilot study:

- (1) Pilot_EDA.ipynb – code for EDA on the source (CHO) domain.
- (2) Pilot_Normalize+RNN.ipynb – code for preprocessing and the first baseline RNN model on the CHO dataset (1st stage).
- (3) Pilot_Normalize+RNN_improved.ipynb – code for preprocessing and improved model refinement (2nd stage).
- (4) Pilot_knowledge_transfer.ipynb – code for the pilot study implementing transfer learning from CHO to iPSC (3rd stage).
- (5) Pilot_EDA_iPSC.ipynb – code for EDA on the target (iPSC) domain.

The main study expands this workflow to the full dataset and formal transfer learning framework:

- (6) CHO_EDA.ipynb – main study EDA on the CHO dataset.
- (7) CHO_preprocessing.ipynb – preprocessing pipeline for CHO data.
- (8) CHO_Augmentation+Scaling.ipynb – data augmentation and feature scaling applied to the CHO dataset.
- (9) CHO_seq+RNN.ipynb – sequence generation and RNN model development on the source domain.
- (10) iPSC_EDA+preprocessing.ipynb – EDA and preprocessing for the target (iPSC) dataset.
- (11) iPSC_transfer_knowledge_zero_shot.ipynb – zero-shot model evaluation on the iPSC dataset.
- (12) iPSC_transfer_knowledge_tl1.ipynb – first transfer learning attempt on the iPSC dataset.
- (13) Copy_of_iPSC_transfer_knowledge_tl1.ipynb – second transfer learning attempt with reduced LR.
- (14) CHO_seq+RNN_model_cross_validation.ipynb – cross-validation of transfer learning models.
- (15) RMSE,_MAE_Comparison_TL.ipynb – visuals.

